

Verification as a Costed Action: When Checking Beats Containing Under Attack–Fault Ambiguity

Chiheb Mejrissi*

Department of Computer Science, Example University, 1 Example Street, Example City, Country

Abstract

Order flow alone cannot always tell whether a trading-gateway anomaly comes from a compromised account or a benign fault. A check such as asking the account holder to confirm through a separate channel reads authorization; if its outcome is conditionally independent of order flow given the cause, it is not bound by what order flow can reveal, composes with any passive posterior and, when order flow is uninformative, lowers the cost of ambiguity exactly when the prior lies in a verification interval. The check has a sensitivity, a specificity, a fixed cost and a delay during which a compromise keeps doing harm. The interval is that of the classical two-threshold leave / verify / contain rule; because harm during the delay accrues only under compromise, it vanishes at a closed-form critical delay, a time budget for the check. The interval does not depend on the prior; the prior sets how much of a population’s posterior mass it reaches. In a companion study’s cells where passive classification does no better than the best constant policy, an exploratory test, specified after that study’s cost table was examined, supports the prediction that targeted verification beats that policy over the region the rule predicts. The support is fragile: restricted to points where a saving is predicted, a post-hoc reading, one cell falls below the threshold, and on the study’s provisional cells the test weakened the prediction. Where passive classification succeeds, the classifier is sometimes cheaper, which suggests, but does not establish, miscalibrated posteriors.

Keywords: costly verification, value of information, threshold decision rule,

*Corresponding author.

Email address: `chiheb.mejrissi@example.org` (Chiheb Mejrissi)

1. Introduction

An operator of an electronic trading gateway sees an account submit an unusual pattern of orders. The pattern may come from an adversary who has taken over the account or the session, or from a benign malfunction of the client’s own systems. The operator can *contain*—suspend the session, pull the orders—or *leave* trading running. Containing a fault disrupts a legitimate client for nothing; leaving a compromise lets the adversary keep trading. When the two causes produce order flow that looks the same, no amount of order-flow analysis settles the matter.

The separate channel. A companion study, Paper 1 [1], makes this precise. In its linear-Gaussian order-flow model, when the confounding condition holds, order-flow observation cannot separate a compromise from a fault; the results of Paper 1 used here are its Propositions 1 and 2 (Appendix B). By Paper 1’s Proposition 1, an operator who sees only order flow can then do no better than act on the prior, and pays a cost of ambiguity, over an operator who knows the cause, equal to the expected regret of the better of the two terminal actions taken at the prior (for our loss table, Corollary 1(i)). Verification reads another channel, *authorization*, and, if its outcome is conditionally independent of order flow given the cause (Assumption 2), it is not subject to that limit; Section 3.4 develops the argument.

When a check pays. A check is not free. It costs analyst time [2, 3] and the friction of a re-authentication challenge for the client [4], and it takes time during which a real adversary keeps trading. We model verification as an action with sensitivity s , specificity t , a fixed cost c_v and a delay τ during which a true compromise causes damage at rate r_K , on top of a general loss table (C_K, C_F, H_K, H_F) for the two terminal actions. The passive operator contains when the belief in a compromise exceeds a threshold q^* (Lemma 1). The value of a check is a tent peaking at q^* (Lemma 3), and the check pays exactly where the tent exceeds its cost. The result is the classical two-threshold rule of Pauker and Kassirer [5], restated in this setting: leave, verify or contain, with thresholds $q_1 < q^* < q_2$ in closed form (Theorem 1).

We state it because every other result is read against q_1 and q_2 , not as a contribution (Section 2).

What the setting adds is the shape of the cost. Because harm during the delay accrues only if the cause is a compromise, the expected cost of checking, $c_v + q r_K \tau$, grows with the belief. As the delay grows, the interval (q_1, q_2) shrinks and vanishes at a critical delay τ^* (Theorem 2). Its use here is operational: τ^* is a time budget for a verification procedure. A check that takes longer than τ^* should never be run, at any belief.

What the prior changes, and what it does not. The verification interval is a set of beliefs; it does not depend on the prior probability of compromise. What depends on the prior is how much of the posterior mass of a population of cases falls inside it, and hence how much targeted verification saves on that population. We define these per-population quantities; the saving is at most pa and so vanishes with the prior, and we show that the mass in the interval vanishes too unless the check is both free and never flags a fault and the order flows of compromise and fault overlap (Lemma 4). A low prior therefore does not by itself make verification more valuable; whether a check helps in a given population is an empirical question about where that population’s posteriors fall relative to q_1 .

Contributions.

1. *The separate channel.* If the outcome of a check that reads authorization is conditionally independent of order flow given the cause (Assumption 2, a modelling assumption, not a result), the check composes with any passive posterior (Lemma 2) and lowers the confounded cost of ambiguity exactly when the prior lies in the verification interval (Corollary 1). The confounded case is then the worst case for the passive and the verify-first operator alike (Corollary 2, a direct consequence stated for completeness).
2. *The cause-borne delay.* A closed-form critical delay τ^* at which the verification interval vanishes, with nested intervals below it and the manner of vanishing characterised (Theorem 2), under a cost of delay borne only by a true compromise. The direction of the bound depends on how the delay is charged: τ^* is a lower bound on the critical delay when the cost of leaving a compromise already includes part of the delay harm (Corollary 3), and an upper bound when a fault also costs during the check (Corollary 4).

3. *What the prior does.* The interval does not depend on the prior, but the posterior mass that reaches it vanishes as compromise becomes rare, except for a free check that never flags a fault, where its limit is positive exactly when the order-flow laws of compromise and fault are not mutually singular (Lemma 4(iii)–(iv)).
4. *Exploratory evidence* on the companion study’s market designs and priors (Section 5), with the grid of (s, t, c_v, τ) , the plausible set, the five policies and the criteria fixed before any quantity of the primary test was computed. Both outcomes are exploratory. In the cells where passive classification fails, the primary test supports the prediction that targeted verification beats the best constant policy over the region the two-threshold rule predicts; the support is not robust, since a post-hoc reading restricted to the points where a saving is predicted weakens it in one cell. Across all cells, the policy comparison weakens the expectation that targeted verification is never beaten: in some of the cells where passive classification succeeds it is more costly than the passive classifier at some plausible grid points, which suggests (but does not establish) miscalibrated corrected posteriors there. No pre-registered confirmation has been run.

Three further results are stated as supporting extensions, not as headline results (Section 4.9), and the paper does not rest on them. The interval and the critical delay carry over from a binary check to a check with any output, and then depend on the check only through its total variation (Theorem 3); this follows in one line from known results. Proposition 2 states when targeted verification on a miscalibrated score still beats the passive rule, a candidate mechanism for the policy comparison’s outcome that this paper does not test. Proposition 3 replaces the conditional independence of Assumption 2 by lower bounds on the check’s accuracy, a monotonicity argument.

Section 2 positions the paper. Section 3 sets out the model and the separate-channel assumption, and Section 4 derives the rule, the critical delay, the per-population quantities and the three extensions; proofs are in Appendix A. Section 5 specifies the empirical design and reports its exploratory results. Section 6 discusses scope and limitations, and Section 7 concludes.

2. Related work

Threshold analysis and the value of information. The leave / verify / contain rule of Section 4 is the threshold approach of Pauker and Kassirer [5]

transposed to attack-fault triage. There, a clinician’s probability of disease is split by a *testing threshold* and a *test-treatment threshold* into three regions: withhold treatment, test and follow the result, treat without testing. The thresholds follow from the test’s sensitivity and specificity, its cost or risk, and the benefit and harm of treatment. Our q_1 and q_2 play the same roles, with contain in place of treat and a general loss table (C_K, C_F, H_K, H_F) in place of the clinical utilities. The underlying calculation—a single experiment is worth running exactly when its expected value of sample information exceeds its cost, and for two actions and two states that value is piecewise linear and vanishes wherever no outcome can change the action—is classical Bayesian decision theory [6]. The height of that value at the threshold is also known: the largest proportional reduction of expected regret that a test achieves is its Youden index $s + t - 1$, reached at the treatment threshold [7], and the Youden cut-off of a continuous test is cost-optimal exactly when the pre-test probability equals the threshold [8]. For an experiment with any output, the value of information is convex in the belief [9], and for two actions it is governed at the threshold by the total variation distance between the outcome laws, which is the best Youden index of any binary reading of the output [10]. Fixing a test’s sensitivity and specificity when its output is continuous narrows the range of beliefs over which testing pays [11]. Outside medicine, three-way decisions derive accept / defer / reject thresholds from a loss table; when the loss of deferring differs by class, the expected cost of deferring is affine in the belief, and the deferral region vanishes when the two thresholds cross [12]. Deferral there carries no information, so it gives a critical delay for waiting without a test, not for a check with a sensitivity and a specificity.

Costs of delay that depend on the state. The feature that distinguishes our setting is a cost of checking that falls on one cause: an adversary keeps trading while the check runs, while a fault is taken not to get worse (Assumption 4, relaxed in Corollary 4). Costs of delay that depend on the belief about the underlying state are a central object of time-critical decision analysis, where the expected cost of deferring an action is computed from a distribution over states and a time-dependent utility [13, 14]. In medical decision making, Schroeder et al. [15] use the threshold framework to trade the accuracy of a slower send-out test against the clinical benefit lost while waiting for its result; this is the closest precedent we know for the delay term $qr_K\tau$ of our model. At the population level, the value of further research is likewise eroded

by the decisions made while it is pending [16]. In operations, the value of diagnostic information is weighed against the delay that acquiring it imposes on the cases waiting, and belief-threshold diagnosis under congestion charges the delay to the queue rather than to one cause [17]; decision makers set this trade-off poorly [18]. Given the tent, the critical delay τ^* of Theorem 2 follows in one step, and its extension to a check with any output (Theorem 3) follows in one line from the total-variation form of the peak. Neither the rule, nor the tent shape of the value of information (Lemma 3), nor the critical delay is new as decision theory; what is new here is the channel and the cause-borne delay: their use for attack-fault triage, where the cost of delay falls only on a compromise, and the separate-channel argument of Section 3.4.

Costly and active information acquisition. Costly observation actions in a partially observed decision process go back to Monahan [19]. Choosing which observations to buy before acting is studied as Bayesian active learning and adaptive diagnosis, where tests are selected sequentially to identify a hypothesis at least cost [20], and as optimal stopping with costly sequential information, where the reversibility of the terminal decision shapes when to stop [21]. The value of information has also been characterised for stopping problems in which the decision maker pays for signals before an irreversible action [22]. Our problem is the one-shot special case with two causes, two terminal actions and a single available check. Its sequential analogue is the Bayesian sequential probability ratio test, whose continuation interval of beliefs shrinks as the cost of sampling rises [23]; there the cost of an observation does not depend on the cause. We keep it one-shot on purpose: it gives the thresholds and the critical delay in closed form, and the sequential version is a separate problem.

Intrusion response and alert verification. Intrusion response has long been cast as a cost-sensitive decision, in which the damage an attack would cause is weighed against the cost of responding [24], and later as a game or a partially observed control problem [25, 26, 27]. Alert verification checks whether an intrusion attempt actually succeeded before an alert is raised [28]; it is a technical filter, with no cost, delay or threshold attached. More recently, investigating an alert has been modelled as costly information acquisition that improves the defender’s belief [29, 30]; deferring uncertain alerts to an analyst is learned in the triage of security operations centres [31]; and a two-threshold rule escalates high scores, closes low ones and defers the rest

[32]. There the thresholds come from error-rate budgets or are learned; the cost of the check does not grow with the belief, and there is no critical delay. Separating cyberattacks from physical faults has been studied through active detection [33] and, recently, by adding cyberspace logs to physical measurements, an analysis that in industry is manual and slow [34], and attacks on transportation systems have been detected through the redundancy between physical and social data [35]. These are the closest precedents we know for reading a second channel; what it does not give is a rule for when reading that channel is worth its cost and delay. Our contribution sits between these lines: a verify-then-respond decision layer with an explicit accuracy, cost and cause-borne delay, justified by the fact that the check reads a channel the observation-side analysis does not cover.

Prior shift and cost-sensitive classification. The empirical design of Section 5 uses classifier scores as passive posteriors. Scores trained at one class balance are moved to another prior by the standard correction [36], and a calibrated posterior is turned into a cost-minimising decision by thresholding it at the cost ratio [37], a rule also used with instance-dependent costs in fraud detection [38]. A cost-ratio threshold is optimal only for calibrated scores [39]; the passive threshold q^* of Lemma 1 is that threshold for our loss table. Miscalibration lowers the net benefit of a risk model and can make it harmful at some thresholds [40], and thresholds that minimise the worst-case regret over miscalibrated predictors have been derived for two actions [41]; Proposition 2 is the three-action counterpart for the verification band.

3. Model

An operator watches an anomaly on a trading gateway and must decide what caused it. Two causes are possible: a *compromise* K , in which someone the account holder did not authorize submits orders through the holder’s valid credential or session (a stolen credential, a hijacked session), and a *fault* F , in which the orders are the holder’s own and the same symptoms come from a benign malfunction. The operator can *contain* (suspend the account or session, pull the orders) or *leave* (let trading continue). Before committing to either, the operator may also *verify*: query a source of evidence that the order flow does not carry. This section sets out the decision problem; Section 4 solves it.

3.1. Loss table and passive risk

The operator’s belief that the cause is a compromise is $q \in [0, 1]$. Without verification, the operator picks the terminal action with the lower expected loss.

Definition 1 (Loss table, passive risk). There are two causes, compromise K (the orders are submitted through a valid credential or session by someone the holder of that credential or session did not authorize: a stolen credential, a hijacked session) and fault F (the orders are the holder’s own, distorted by a benign malfunction), and two terminal actions, *contain* and *leave*. Containing costs C_K under K and C_F under F ; leaving costs H_K under K and H_F under F . For a belief $q = P(K) \in [0, 1]$ the *passive risk* is

$$R(q) = \min\{q C_K + (1 - q) C_F, q H_K + (1 - q) H_F\}.$$

Write $a = H_K - C_K$ and $b = C_F - H_F$.

The four entries of the loss table are kept general. Containing a compromise (C_K) costs the disruption of a legitimate client plus the operational cost of the response; containing a fault (C_F) costs the same disruption with nothing gained; leaving a compromise (H_K) costs the harm the adversary goes on to do; leaving a fault (H_F) costs whatever the malfunction costs if it is not interrupted, counted from the terminal decision on; what a malfunction costs during the check itself is the rate r_F of Corollary 4, zero under Assumption 4. The quantity a is what containment saves when the cause is a compromise, and b is what it wastes when the cause is a fault. Paper 1’s loss table [1] (its Definition D6) is the special case $H_F = 0$, with Paper 1’s containment cost, disruption and harm in the roles of C_K , C_F and H_K ; the cost table on which Paper 1 classifies its cells uses, in addition, a measured cost of leaving a fault, which differs by design and regime (Section 5.2).

Nothing in the model depends on the prior probability of compromise except through the belief q . The prior of a single case enters in Section 4.6 (Definition 4 and Corollary 1); the distribution of passive posteriors over a population of cases enters in Section 4.8 (Definition 5).

3.2. Verification

Verification is a costed action with an uncertain outcome. On a trading gateway it is, for example, a call from the desk to the client to confirm the activity, or a step-up authentication that asks the submitter for a factor bound to the account holder. A check that the credential or session is valid

is not a verification in this sense: a stolen credential or a hijacked session is valid.

Definition 2 (Verification action). Verification returns an outcome $Z \in \{+, -\}$ with sensitivity $s = P(Z = + \mid K)$ and specificity $t = P(Z = - \mid F)$. It costs c_v and takes a delay τ ; during the delay a true compromise causes harm at rate r_K . From belief q ,

$$P(+) = qs + (1 - q)(1 - t), \quad P(-) = q(1 - s) + (1 - q)t,$$

$$q^+ = \frac{qs}{P(+)}, \quad q^- = \frac{q(1 - s)}{P(-)}$$

(a posterior whose outcome has probability zero is never used; see Definition 3).

The sensitivity s is the probability that verification flags a compromise that is present; the specificity t is the probability that it clears a fault. The cost c_v is paid whatever the cause: analyst time, the call, the friction imposed on the client. The delay τ is the time the check takes, and the damage rate r_K is the harm per unit of that time done by an adversary who is really present (a rate, distinct from the cost H_K of leaving a compromise). We assume that a fault accrues no additional cost while the operator is checking, whereas a compromise does (Assumption 4); Corollary 4 drops this assumption.

Definition 3 (Verify-first risk, value of information, verification region). The *verify-first risk* is

$$R_v(q) = c_v + q r_K \tau + P(+) R(q^+) + P(-) R(q^-),$$

where $P(z) R(q^z)$ is read as $\min\{P(z, K) C_K + P(z, F) C_F, P(z, K) H_K + P(z, F) H_F\}$ with $P(z, K) = P(z \mid K) q$ and $P(z, F) = P(z \mid F) (1 - q)$ (equal to the product when $P(z) > 0$, and zero when $P(z) = 0$). The *value of information* is $V(q) = R(q) - [P(+) R(q^+) + P(-) R(q^-)]$, the *verification cost* is $\chi(q) = c_v + q r_K \tau$, and the *verification region* is

$$S = \{q \in [0, 1] : R_v(q) < R(q)\} = \{q \in [0, 1] : V(q) > \chi(q)\}.$$

Ties are resolved in favour of the passive action: the operator verifies only when verifying is strictly cheaper.

The verify-first risk adds two terms to the expected passive risk at the updated belief: the cost of the check and the harm expected during the delay. The value of information V is the familiar reduction in terminal risk that the

outcome buys [6]; the verification cost χ collects what the check itself costs. Because harm during the delay accrues only under K , χ grows with the belief: checking is more expensive precisely when a compromise is more likely. The verification region S is the set of beliefs at which the check pays for itself.

3.3. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$; that is, $a > 0$ and $b > 0$: leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it.

This excludes the cases in which one terminal action dominates the other whatever the cause, and the case $a, b < 0$, which is the same problem with the names of the two actions exchanged (Remark 1(b)). On a gateway it says that an unchecked adversary does more damage than a response costs, and that suspending a healthy client costs more than letting a benign fault run. Remark 1 in Section 4 records what happens when it fails.

Assumption 2 (Separate channel). Verification reads authorization, not order flow: it asks whether the holder of the credential or session authorized the orders, through a channel the person submitting them does not control (for example an out-of-band confirmation with the account holder, or a step-up authentication with a factor bound to the holder). A check of whether the credential or session is valid is not such a check, since a stolen credential or a hijacked session is valid under K . Its outcome Z is conditionally independent of the order-flow observation Y given the cause: $P(Z, Y | c) = P(Z | c) P(Y | c)$ for $c \in \{K, F\}$, and (s, t) do not depend on Y .

Assumption 2 is the conceptual core of the paper and is discussed in Section 3.4.

Assumption 3 (Informative one-shot verification). $s, t \in [0, 1]$ with $s + t > 1$; $c_v \geq 0$, $\tau \geq 0$, $r_K \geq 0$. Verification is available once; after its outcome the operator takes a terminal action (contain or leave) and cannot verify again.

The condition $s + t > 1$ says that a positive outcome is more likely under a compromise than under a fault; a check that fails it carries no information (Remark 1). The one-shot restriction matches a triage step in which one check is run and the operator then acts. Repeated or sequential checking is a stopping problem and is outside the scope of this paper.

Assumption 4 (Delay accounting). The harm $r_K \tau$ accrued by a true compromise during the verification delay is *additional* to the terminal cost

(C_K or H_K) incurred afterwards, whichever terminal action is taken; the terminal costs C_K, C_F, H_K, H_F and the verification characteristics s, t do not depend on τ ; a fault accrues no additional cost during the delay (its damage rate is $r_F = 0$); and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 4 fixes how the delay is charged. The harm accrued during the check is paid in addition to the terminal cost, whatever the operator then does; this is the conservative accounting for verification, since it charges the delay in full even when the terminal cost H_K might already include part of it. Any double counting is therefore against verification only: under any accounting in which H_K already includes part or all of that harm, the critical delay τ^* of Theorem 2 is a lower bound (Corollary 3). The check does not change the loss table or its own accuracy, a fault costs nothing extra while it is checked (relaxed in Corollary 4), and the operator does not update on further order flow while waiting for the outcome.

3.4. When verification is not bound by the order-flow limit

In its linear-Gaussian order-flow model, when the confounding condition holds, Paper 1 [1] shows that order-flow observation cannot separate a compromise from a fault, and that a passive operator who sees only order flow then pays a cost of ambiguity that no classifier can remove. That limit is a statement about the law of the order-flow observation Y under the two causes; it constrains every decision rule that is a function of Y alone. It does not constrain evidence that is not a function of Y .

Verification is such evidence. Such a check reads the authorization channel: whether the entity submitting the orders is the one entitled to submit them. Assumption 2 states that, given the cause, the outcome of that check is independent of what the order flow shows, and that its accuracy (s, t) does not depend on the order flow. Operationally, whether the account holder confirms the orders, or the submitter passes a challenge bound to the holder, does not depend on how the orders look. The sensitivity s is the probability that the holder disavows the orders, or that the submitter fails the challenge, under a compromise; the specificity t is the probability that the holder confirms them under a fault. The check does not detect a holder who trades maliciously on their own account, nor an adversary who also controls the holder’s second factor; the model leaves both out of K . Under this assumption verification composes with whatever the passive analysis concludes:

Lemma 2 in Section 4 shows that the joint posterior is the verification update applied to the order-flow posterior. That the characteristics (s, t) are the same whatever the order-flow law is not a finding of the lemma but Assumption 2 itself; what the lemma adds is that, given the assumption, the two updates compose, including when the order-flow laws under K and F coincide. That is the case in which Paper 1’s passive operator can do no better than act on the prior; Corollary 1 in Section 4 states when verification brings the cost of ambiguity strictly below Paper 1’s value there.

Assumption 2 is a modelling choice, not a consequence of the setting; Section 6.2 states what is lost when it fails.

4. The two-threshold rule and the critical delay

This section derives when verification pays. The passive rule has a single threshold (Lemma 1). Verification composes with any order-flow posterior (Lemma 2), and its value is a tent peaking at that threshold (Lemma 3). Comparing the tent with the cost of the check gives the classical leave / verify / contain rule with two thresholds, restated in this setting (Theorem 1), and a critical delay beyond which the check never pays (Theorem 2). Lemmas 1 and 3, Theorem 1 and Remark 1 are background, stated because every other result is read against them. Corollary 1 applies these results to the confounded case, Corollary 2 records that this case is the worst case, Corollaries 3 and 4 show how the critical delay depends on the accounting of the delay, and Remark 1 records the degenerate parameters. Lemma 4 and Proposition 1 then turn from a single belief to a population of cases with a given prior, which is where the prior enters. Section 4.9 closes with three supporting extensions: a check with any output (Definition 6, Theorem 3), a miscalibrated score (Proposition 2) and a relaxed separate channel (Proposition 3). Proofs are in Appendix A.

4.1. The passive threshold

Lemma 1 (Passive threshold). Under Assumption 1, let $q^* = b/(a + b) \in (0, 1)$. Leaving is strictly optimal for $q < q^*$, containing is strictly optimal for $q > q^*$, and

$$R(q) = q H_K + (1 - q) H_F - (a + b) (q - q^*)^+,$$

so R is continuous, concave and piecewise linear on $[0, 1]$ with a single kink at q^* .

The operator who cannot verify contains exactly when the belief in a compromise exceeds q^* : the share of the total stakes $a + b$ that a wrong containment wastes. A response that is costly to a healthy client (large b) pushes q^* up; an adversary who does much damage if left alone (large a) pulls it down. At q^* the two terminal actions are equally good, and the passive risk has its only kink.

4.2. Verification composes with the passive posterior

Lemma 2 (Verification composes with any passive posterior).

Under Assumption 2, let $q(y) = P(K \mid Y = y)$ be the passive (order-flow) posterior. Then $P(K \mid Y = y, Z = z)$ equals the verification update of Definition 2 applied to $q = q(y)$: it is $q(y)^+$ if $z = +$ and $q(y)^-$ if $z = -$. By Assumption 2, the characteristics (s, t) do not depend on the order-flow law; hence V , χ and S of Definition 3, which are built from q , (s, t) and the constants alone, are the same whatever the order-flow law, including when the order-flow laws under K and F coincide, and the order-flow law enters only through the belief $q(y)$ at which they are evaluated.

Lemma 2 separates the two stages of the decision. The operator first reduces the order-flow evidence to a belief $q(y)$, with whatever classifier or model is available, and then applies the verification analysis below to that belief, with the same (s, t) , which Assumption 2 posits rather than Lemma 2 derives. The verification rule need not be re-derived for each classifier, and it remains meaningful when the order flow carries no information about the cause (Section 3.4). All results below are stated for a belief q and hold at $q = q(y)$.

4.3. The value of information is a tent

Lemma 3 (The value of information is a tent). Under Assumptions 1 and 3,

$$V(q) = \max\{0, \min\{G^+(q), G^-(q)\}\},$$

where

$$G^+(q) = q s a - (1 - q)(1 - t) b, \quad G^-(q) = (1 - q) t b - q(1 - s) a.$$

Consequently, with

$$q_L = \frac{(1 - t) b}{s a + (1 - t) b}, \quad q_U = \frac{t b}{t b + (1 - s) a},$$

(i) $0 \leq q_L < q^* < q_U \leq 1$; (ii) $V(q) > 0$ if and only if $q \in (q_L, q_U)$, equivalently if and only if $q^- < q^* < q^+$ (the two posteriors fall strictly on opposite sides of q^*); (iii) V is affine and strictly increasing on $[q_L, q^*]$, affine and strictly decreasing on $[q^*, q_U]$, and zero outside (q_L, q_U) ; (iv) V attains its maximum uniquely at q^* , with

$$V(q^*) = (s + t - 1) \frac{a b}{a + b} = (s + t - 1) a q^* > 0.$$

Lemma 3 is the standard value-of-information calculation for two actions and two states [6]; that its peak at the threshold scales with the Youden index is known from regret graphs [7]. A check is worth something only when its outcome can change the terminal action: a positive outcome must move the belief above q^* and a negative outcome below it. Outside the interval (q_L, q_U) neither outcome can overturn the passive decision, and the check is worthless there. That interval widens as the check becomes more accurate: a perfect check ($s = t = 1$) has $q_L = 0$ and $q_U = 1$, and has value at every belief strictly between 0 and 1. The value rises linearly from q_L to its peak at q^* and falls linearly to q_U , with slopes that differ in general, so the tent is not symmetric about q^* (Figure 1). The height of the peak is the product of the Youden index $s + t - 1$ and the quantity $a q^* = b(1 - q^*)$, which is the expected regret at the threshold, relative to an operator who knows the cause, of either terminal action.

4.4. The two-threshold rule

Theorem 1 (Two-threshold leave / verify / contain rule). Under Assumptions 1, 3 and 4, the verification region S is an open interval (possibly empty). It is non-empty if and only if

$$V(q^*) > c_v + q^* r_K \tau,$$

and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$, where

$$q_1 = \frac{c_v + (1 - t) b}{s a + (1 - t) b - r_K \tau}, \quad q_2 = \frac{t b - c_v}{t b + (1 - s) a + r_K \tau}$$

(both denominators are positive when $S \neq \emptyset$). If $S \neq \emptyset$, the optimal one-shot policy is: *leave* if $q \leq q_1$, *verify* if $q_1 < q < q_2$ and then contain on $Z = +$ and leave on $Z = -$, *contain* if $q \geq q_2$. If $S = \emptyset$, write $q_1 = q_2 = q^*$; the optimal policy is the passive rule of Lemma 1, *leave* if $q < q^*$ and *contain* if

$q > q^*$, and at $q = q^*$, where both terminal actions are optimal, the operator leaves by convention. When $c_v = 0$ and $\tau r_K = 0$, $S = (q_L, q_U)$.

Theorem 1 is the classical threshold rule of Pauker and Kassirer [5], restated in this setting, with q_1 and q_2 in the roles of the testing and test-treatment thresholds (Section 2); the only feature the setting adds is a cost of the check that grows with the belief, through the delay term $q r_K \tau$.

For an operator the rule reads as follows. Below q_1 the check costs more than it is expected to save, and the operator leaves. Above q_2 a compromise is likely enough that the operator contains without checking. In between, the operator verifies and follows the outcome (Figure 1). The verification interval always contains the passive threshold q^* when it is non-empty, and it lies inside the interval (q_L, q_U) on which the check has any value at all. Its two ends move inward as the check becomes more expensive or, for $\tau < \tau^*$, slower: c_v raises the lower end and lowers the upper one, and the delay term $r_K \tau$ enters both denominators.

4.5. The critical delay

Theorem 2 (Critical delay). Under Assumptions 1, 3 and 4, write $S(\tau)$ for the verification region at delay τ , all other parameters fixed. (i) *Nesting.* $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$. If $r_K > 0$ and $V(q^*) > c_v$, then $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$ (part (ii)), and on $[0, \tau^*)$ the end point $q_2(\tau)$ of Theorem 1 is strictly decreasing in τ , and $q_1(\tau)$ is strictly increasing in τ unless $c_v = 0$ and $t = 1$, in which case $q_1(\tau) = 0$ on $[0, \tau^*)$. For $\tau \geq \tau^*$, $S(\tau) = \emptyset$ and the convention $q_1 = q_2 = q^*$ of Theorem 1 applies. (ii) *Critical delay.* If $r_K > 0$ and $V(q^*) > c_v$, let

$$\tau^* = \frac{V(q^*) - c_v}{q^* r_K} = \frac{(s + t - 1) a - c_v / q^*}{r_K} > 0.$$

Then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau^*$. (iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $r_K = 0$, $S(\tau)$ does not depend on τ . (iv) *How the interval vanishes.* Suppose $r_K > 0$ and $V(q^*) > c_v$ (the hypotheses of (ii)). If $c_v > 0$ or $t < 1$, then $q_1 \rightarrow q^*$ and $q_2 \rightarrow q^*$ as $\tau \uparrow \tau^*$: the interval shrinks continuously to the point q^* . If $c_v = 0$ and $t = 1$, then $S(\tau) = (0, q_2(\tau))$ for $\tau < \tau^*$ with $q_2(\tau) \downarrow q^*$, and $S(\tau^*) = \emptyset$: the interval vanishes discontinuously.

Because the value of information is a tent peaking at q^* and the cost of the check is affine and increasing in the belief, the check pays at some belief

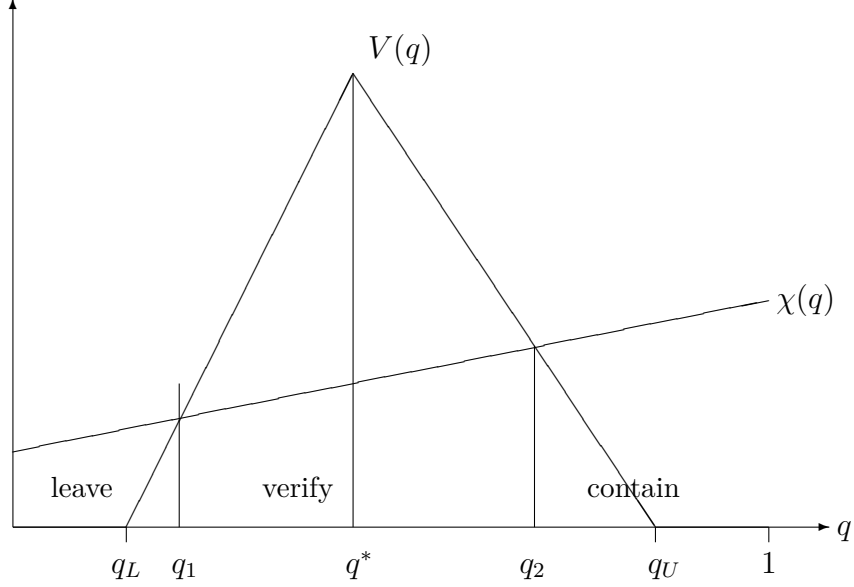


Figure 1: Schematic of Lemma 3 and Theorem 1 (not to scale; no parameter values; critical delays for the checks of the empirical study are in Table 7, and a worked example with Paper 1’s costs is in Section 4.5). The value of information V is a tent on (q_L, q_U) with its peak at the passive threshold q^* ; the verification cost χ increases with the belief because harm during the delay accrues only under compromise. Verification pays on (q_1, q_2) , where the tent lies above the cost line. Raising the delay τ tilts the cost line upward and shrinks the interval (Theorem 2).

if and only if it pays at q^* , and solving that inequality for the delay gives τ^* (precedents in Section 2).

For an operator, τ^* is a time budget for the check. This reading assumes that a fault costs nothing during the check (Assumption 4); if it does, the budget is the shorter τ_F^* of Corollary 4. A verification procedure that takes longer than τ^* should never be run, at any belief: the operator should act on the passive rule instead. Below τ^* the verification interval shrinks as the check slows down, and the operator verifies over a narrower band of beliefs around q^* . The budget grows with the accuracy of the check (through $s + t - 1$) and shrinks with the damage rate r_K and with the fixed cost c_v . If the fixed cost alone is at least the peak value of information, $c_v \geq V(q^*)$, no delay is short enough: the check is never worth running (Theorem 2(iii)). In the limiting case of a free check that never flags a fault ($c_v = 0$ and $t = 1$), the interval reaches down to zero belief and disappears all at once

at τ^* ; otherwise it closes continuously onto q^* , and beyond τ^* the rule is the passive one (at $q = q^*$ both terminal actions are optimal; the operator leaves by convention).

In the units of the empirical study (Section 5.3), the critical delay is $r_K \tau^*/a = s + t - 1 - x$, with x the fixed cost of the check as a share of the peak value of a perfect, free and instantaneous check; for a sense of scale, take Paper 1's cost matrix: $a = 826,562$, $b = 473,958$, so $q^* = 0.364$. A check with $s = 0.9$, $t = 0.9$ and fixed cost $c_v = 0.01 ab/(a + b) = 3,012$ (the point $x = 0.01$, $y = 0$ of the grid of Section 5.3) pays for beliefs between $q_1 = 0.064$ and $q_2 = 0.832$ when it is instantaneous, and its critical delay is $\tau^* = 0.790 a/r_K$: the check must return before a compromise has done 0.790 times the damage that containing it averts. Table 7 gives $r_K \tau^*/a$ for every check of the grid. An absolute time budget needs a documented damage rate r_K , which the paper does not supply.

4.6. Comparison with the passive cost of ambiguity

The results above are stated for a belief. To compare verification with the passive analysis of Paper 1 [1], the decision problem is stated over policies that act on the order-flow observation.

Definition 4 (Decision problem with verification; verify-first cost of ambiguity). The cause M is K with prior probability $p = P(K) \in (0, 1)$ and F otherwise. Given M , the order-flow observation Y has law P_M^Y , and the verification outcome Z has the law of Definition 2, jointly as in Assumption 2. Write $L(\text{contain}, K) = C_K$, $L(\text{contain}, F) = C_F$, $L(\text{leave}, K) = H_K$, $L(\text{leave}, F) = H_F$ (Definition 1). A *verify-first policy* is a quadruple of measurable maps $\delta, \pi_0, \pi_+, \pi_-$ from observations y to $[0, 1]$: with probability $\delta(y)$ the operator verifies and then contains with probability $\pi_Z(y)$; otherwise it contains with probability $\pi_0(y)$. Its expected cost is

$$\begin{aligned} J(\delta, \pi) = \mathbb{E} \bigg[& (1 - \delta(Y)) \left(\pi_0(Y) L(\text{contain}, M) \right. \\ & \left. + (1 - \pi_0(Y)) L(\text{leave}, M) \right) \\ & + \delta(Y) \left(c_v + \mathbb{1}_{[M=K]} r_K \tau + \pi_Z(Y) L(\text{contain}, M) \right. \\ & \left. \left. + (1 - \pi_Z(Y)) L(\text{leave}, M) \right) \right]. \end{aligned}$$

A *passive policy* is a verify-first policy with $\delta \equiv 0$. Let R_Y^* and $R_{Y,v}^*$ be the infima of J over passive and over verify-first policies, $R_{\text{oracle}} = p \min\{C_K, H_K\} +$

$(1 - p) \min\{C_F, H_F\}$ the expected cost of an operator who observes M , and

$$\text{CoA}^* = R_Y^* - R_{\text{oracle}}, \quad \text{CoA}_v^* = R_{Y,v}^* - R_{\text{oracle}}$$

the passive and the verify-first cost of ambiguity.

Corollary 1 (Verification lowers the confounded cost of ambiguity). Under Assumptions 1–4, in the setting of Definition 4, suppose the order-flow laws coincide, $P_K^Y = P_F^Y$ (the confounded case). Then: (i) $\text{CoA}^* = \min\{p a, (1 - p) b\} > 0$; (ii) the verify-first cost of ambiguity is

$$\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\},$$

$$E(p) = c_v + p r_K \tau + p(1 - s) a + (1 - p)(1 - t) b,$$

equivalently $\text{CoA}_v^* = \text{CoA}^* - (V(p) - \chi(p))^+$, and it is attained by the policy of Theorem 1 applied at the belief $q = p$, whatever Y ; (iii) $\text{CoA}_v^* < \text{CoA}^*$ if and only if $p \in S$, that is, if and only if $V(q^*) > c_v + q^* r_K \tau$ and $q_1 < p < q_2$ (Theorem 1); if $r_K > 0$ and $V(q^*) > c_v$, this holds if and only if $\tau < \tau^*$ and $q_1(\tau) < p < q_2(\tau)$ (Theorem 2), and if $V(q^*) \leq c_v$ it holds for no p and no τ ; (iv) over $p \in (0, 1)$, the reduction $\text{CoA}^* - \text{CoA}_v^*$ is largest at $p = q^*$, where it equals $(V(q^*) - c_v - q^* r_K \tau)^+$; (v) $\text{CoA}_v^* = 0$ if and only if $s = t = 1$, $c_v = 0$ and $r_K \tau = 0$. Under the special case $H_F = 0$ of Paper 1’s Definition D6, CoA^* in (i) is the cost of ambiguity of Paper 1, Proposition 1, so (iii) states when verification brings the cost of ambiguity strictly below that value.

In the confounded case the order flow carries no information about the cause, and the passive cost of ambiguity in part (i), stated for the general loss table, reduces to the value of Paper 1’s Proposition 1 when $H_F = 0$. Corollary 1 states when verification brings it strictly lower: exactly when the prior lies in the verification region of Theorem 1, so, for a harmful compromise and a check whose fixed cost is below the peak value of information, only when the check is faster than τ^* . The reduction is largest when the prior sits at the passive threshold, and the cost of ambiguity is removed entirely only by a perfect and free check that accrues no harm during its delay. The optimal verify-first policy ignores the order flow, which here carries no information, and applies the rule of Theorem 1 to the prior.

Corollary 2 (Uniform ceiling: the confounded case is the worst case). Under Assumptions 1–4, in the setting of Definition 4, let $E(p)$ be as in Corollary 1. For every pair of order-flow laws P_K^Y, P_F^Y and every $p \in (0, 1)$: (i) the *verify-everything* policy ($\delta \equiv 1$, $\pi_+ \equiv 1$, $\pi_- \equiv 0$) has expected cost

$R_{\text{oracle}} + E(p)$, whatever the order-flow laws; (ii) $\text{CoA}^* \leq \min\{p a, (1-p) b\}$ and

$$\text{CoA}_v^* \leq \min\{\text{CoA}^*, E(p)\} \leq \min\{p a, (1-p) b, E(p)\};$$

(iii) every inequality in (ii) is an equality in the confounded case $P_K^Y = P_F^Y$ (Corollary 1), so that, over all pairs of order-flow laws, the largest passive cost of ambiguity is $\min\{p a, (1-p) b\}$ and the largest verify-first cost of ambiguity is $\min\{p a, (1-p) b, E(p)\}$, both attained in the confounded case; (iv) the largest verify-first cost of ambiguity is strictly below the largest passive one if and only if $p \in S$, and the gap is $(V(p) - \chi(p))^+$. Under the special case $H_F = 0$ of Paper 1's Definition D6 and the hypotheses of Paper 1, Proposition 1, $\min\{p a, (1-p) b\}$ is the cost of ambiguity of Paper 1, Proposition 1, and the passive bound in (ii) is the upper bound of Paper 1, Proposition 2 (total-variation extension of Proposition 1): the value of Paper 1's Proposition 1 is the worst case of the passive cost of ambiguity over all order-flow models, and verification caps the cost of ambiguity at $E(p)$ for every order-flow model.

Corollary 2 is a direct consequence of Corollary 1 and the cost of verifying every case, stated for completeness: the confounded case is the worst case for the passive and for the verify-first operator alike, and whatever the order-flow model, verification caps the cost of ambiguity at $E(p)$, the cost of ambiguity of the policy that verifies every case.

Corollary 3 (Delay accounting: τ^* is a lower bound). For $\theta \in [0, 1]$, *Assumption 4_θ* is Assumption 4 except that the harm $r_K \tau$ accrued by a true compromise during the delay is added to the terminal cost in full if the terminal action is *contain* and in the share θ if it is *leave*. Assumption 4_1 is Assumption 4; Assumption 4_0 is the accounting in which H_K already includes the harm a compromise left in place causes during the delay. Under Assumption 4_θ the verify-first risk is

$$R_v^\theta(q) = c_v + \sum_{z \in \{+, -\}} \min\left\{P(z, K) (C_K + r_K \tau) + P(z, F) C_F, \right. \\ \left. P(z, K) (H_K + \theta r_K \tau) + P(z, F) H_F\right\},$$

with $P(z, K)$, $P(z, F)$ as in Definition 3, and the verification region is $S_\theta(\tau) = \{q \in [0, 1] : R_v^\theta(q) < R(q)\}$. Let $\sigma_\theta = s + \theta(1-s) \in [s, 1]$. Under Assumptions 1, 3 and 4_θ : (i) $S_\theta(\tau) = \{q \in [0, 1] : V(q) > c_v + \sigma_\theta q r_K \tau\}$, the verification

region of Definition 3 with r_K replaced by $\sigma_\theta r_K$; for $q \in S_\theta(\tau)$ the optimal action after verification is to contain on $Z = +$ and leave on $Z = -$. Theorems 1 and 2 therefore hold under Assumption 4 $_\theta$ with r_K replaced by $\sigma_\theta r_K$. (ii) For $0 \leq \theta \leq \theta' \leq 1$ and every $\tau \geq 0$, $S(\tau) = S_1(\tau) \subseteq S_{\theta'}(\tau) \subseteq S_\theta(\tau) \subseteq S_0(\tau)$. (iii) If $r_K > 0$ and $V(q^*) > c_v$, then $S_\theta(\tau) \neq \emptyset$ if and only if $\tau < \tau_\theta^*$, where

$$\tau_\theta^* = \frac{\tau^*}{\sigma_\theta}, \quad \tau^* \leq \tau_\theta^* \leq \frac{\tau^*}{s},$$

with τ^* of Theorem 2. So τ^* is a lower bound on the critical delay under every accounting 4 $_\theta$, attained if and only if $\theta = 1$ or $s = 1$.

Corollary 3 answers the objection that the harm done during the delay is counted twice when H_K already includes it: charging that harm in full, as Assumption 4 does, only shortens the critical delay, so τ^* is a lower bound on the critical delay under every such accounting.

Corollary 4 (A fault that accrues cost during the delay). For $r_F \geq 0$, Assumption 4 $_F$ is Assumption 4 except that a fault also accrues cost at rate r_F during the verification delay, additional to the terminal cost whichever terminal action is taken (Assumption 4 is the case $r_F = 0$). Under Assumption 4 $_F$ the verify-first risk is $R_v(q) + (1 - q)r_F\tau$, so the verification region is that of Definition 3 with χ replaced by

$$\chi_F(q) = c_v + qr_K\tau + (1 - q)r_F\tau = (c_v + r_F\tau) + q(r_K - r_F)\tau.$$

Under Assumptions 1, 3 and 4 $_F$: (i) S is an open interval, non-empty if and only if $V(q^*) > c_v + \bar{r}\tau$ with $\bar{r} = q^*r_K + (1 - q^*)r_F$, and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$,

$$q_1 = \frac{c_v + r_F\tau + (1 - t)b}{sa + (1 - t)b - (r_K - r_F)\tau}, \quad q_2 = \frac{tb - c_v - r_F\tau}{tb + (1 - s)a + (r_K - r_F)\tau},$$

both denominators positive; the optimal policy is that of Theorem 1 with these end points. (ii) $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $\bar{r} > 0$ and $V(q^*) > c_v$, then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau_F^*$, where

$$\tau_F^* = \frac{V(q^*) - c_v}{q^*r_K + (1 - q^*)r_F}.$$

(iii) If moreover $r_K > 0$, then $\tau_F^* = \tau^* q^* r_K / \bar{r} \leq \tau^*$, with equality if and only if $r_F = 0$: the critical delay τ^* of Theorem 2 is an upper bound on the critical delay when faults accrue cost during the delay.

If a fault also costs during the check, the verification region is still an interval and vanishes at a critical delay $\tau_F^* \leq \tau^*$; so τ^* is the time budget when a fault costs nothing during the check and an upper bound on it otherwise, while Corollary 3 shows it is a lower bound when H_K already includes the delay harm.

4.7. Degenerate parameters

Remark 1 (Degenerate parameters). (a) If $s+t = 1$, then $q^+ = q^- = q$ whenever defined, $V \equiv 0$, and $S = \emptyset$ for all $c_v \geq 0$, $\tau \geq 0$. (b) Assumption 1 fails exactly when $a \leq 0$ or $b \leq 0$, that is, in one of two cases. (b1) *Weak signs mixed or zero:* $a \leq 0 \leq b$ or $b \leq 0 \leq a$ (so $q^* = b/(a+b) \notin (0,1)$ or is undefined). Then one terminal action is weakly optimal at every belief, R is affine, $V \equiv 0$, and $S = \emptyset$. (b2) *Both negative:* $a < 0$ and $b < 0$. Then $q^* = b/(a+b) \in (0,1)$, containing is strictly optimal for $q < q^*$ and leaving for $q > q^*$. Exchanging the names of the two terminal actions gives a problem with the same R , V , χ and S that satisfies Assumption 1 with (a,b) replaced by $(-a,-b)$ and the same q^* ; every result below then holds for that problem, i.e. for the original one with (a,b) replaced by $(-a,-b)$ and *contain* and *leave* exchanged (in Theorem 1: contain if $q \leq q_1$, verify if $q_1 < q < q_2$ and then leave on $Z = +$ and contain on $Z = -$, leave if $q \geq q_2$). (c) The requirement $c_v \geq 0$ in Assumption 3 is needed for Theorem 1: if $c_v < 0$, S can contain beliefs near 0 where $V = 0$ and can then be disconnected.

Parts (a) and (b) say that the results above need a check that is informative and a decision that is not settled in advance. Part (c) explains why Assumption 3 requires a non-negative cost: a check that pays the operator to run it would be worth running at beliefs where its outcome cannot change the action, and the verification region could then fail to be an interval.

4.8. Populations of cases: what the prior changes

The verification region $S = (q_1, q_2)$ is a set of beliefs. Its end points depend on the loss table and on (s, t, c_v, τ, r_K) , and not on the prior probability of compromise. An operator facing a population of cases—for instance all the anomalies raised in one simulated market at one base rate of compromise—sees instead a distribution of passive posteriors, and the prior moves that distribution. The following quantities measure how much of it falls in the verification region and what verifying there saves.

Definition 5 (Verification mass, saving and recovered share of a cell). An *order-flow model* c is a pair of order-flow laws P_K^c, P_F^c on a

measurable space, with densities f_K^c, f_F^c with respect to a σ -finite measure ν dominating both (e.g. $\nu = P_K^c + P_F^c$). A *cell* (c, p) is an order-flow model c together with a prior $p \in (0, 1)$; in Paper 1's grid, c is a market design in one regime and p is a prior of the grid. In the setting of Definition 4 with $P_K^Y = P_K^c$, $P_F^Y = P_F^c$ and prior $p \in (0, 1)$, write $P_p = p P_K^c + (1 - p) P_F^c$ for the law of Y and

$$q_p(y) = \frac{p f_K^c(y)}{p f_K^c(y) + (1 - p) f_F^c(y)}$$

for the passive posterior (defined where the denominator is positive, a set of full P_p -measure). The *verification mass*, the *saving* and the *recovered share* of the cell (c, p) are

$$m_c(p) = P_p(q_p(Y) \in S), \quad D_c(p) = \mathbb{E}_p[(V - \chi)^+(q_p(Y))],$$

$$\rho_c(p) = \frac{D_c(p)}{\text{CoA}_c^*(p)},$$

with S, V, χ as in Definition 3 and $\text{CoA}_c^*(p)$ the passive cost of ambiguity of Definition 4 for this cell; $\rho_c(p)$ is defined when $\text{CoA}_c^*(p) > 0$. Where $f_K^c, f_F^c > 0$, $\text{logit } q_p(y) = \text{logit } p + \Lambda_c(y)$ with $\Lambda_c = \log(f_K^c/f_F^c)$.

Lemma 4 (Saving identity and low-prior limit). Under Assumptions 1–4, in the setting of Definition 5: (i) *Saving.* $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p) = R_Y^* - R_{Y,v}^*$, where $\text{CoA}_{v,c}^*(p)$ is the verify-first cost of ambiguity of Definition 4 for this cell; it is attained by the policy that verifies iff $q_p(Y) \in S$, then contains on $Z = +$ and leaves on $Z = -$, and otherwise takes the passive action of Lemma 1 at $q_p(Y)$. Moreover $0 \leq D_c(p) \leq \min\{\text{CoA}_c^*(p), V(q^*) m_c(p)\}$, so $0 \leq \rho_c(p) \leq 1$ where defined. (ii) *The saving vanishes at low priors.* For every order-flow model c , $D_c(p) \rightarrow 0$ as $p \rightarrow 0$. (iii) *The mass vanishes at low priors when $q_1 > 0$.* If $S = \emptyset$, then $m_c \equiv 0$. If $S = (q_1, q_2) \neq \emptyset$ and $q_1 > 0$ (equivalently, $c_v > 0$ or $t < 1$), then $m_c(p) \rightarrow 0$ as $p \rightarrow 0$. (iv) *The exception.* If $S \neq \emptyset$, $c_v = 0$ and $t = 1$ (so $q_1 = 0$), then $m_c(p) \rightarrow P_F^c(f_K^c(Y) > 0)$ as $p \rightarrow 0$. This limit is positive if and only if P_K^c and P_F^c are not mutually singular, and zero when they are (compromise and fault perfectly separable by order flow); the saving still vanishes, by (ii). No condition on the order-flow model is needed beyond Definition 5: the densities are finite, and $f_F^c > 0$ holds P_F^c -almost surely by construction.

Part (ii) is immediate: by part (i) and Corollary 2(ii), $D_c(p) \leq \text{CoA}_c^*(p) \leq p a$. What is not immediate is part (iii): the mass in the interval also vanishes, except in the case of part (iv).

Part (i) identifies the saving of targeted verification with the reduction in the cost of ambiguity, and bounds it by the peak value of information times the mass in the interval. Parts (ii) and (iii) state what happens when compromise becomes rare. As the prior falls, the passive posteriors move towards zero and, in the limit, below the lower threshold q_1 whenever the check has a fixed cost or can flag a fault, so the mass in the interval and the saving both vanish. A low prior therefore does not make verification more valuable on its own: what matters is whether the upper tail of the posteriors still reaches above q_1 . Part (iv) is the only exception for the mass: for a free check that never flags a fault $q_1 = 0$ by the formula of Theorem 1, so the interval reaches down to zero belief, and the mass tends to $P_F^c(f_K^c(Y) > 0)$, which is positive exactly when the order-flow laws of compromise and fault are not mutually singular, although the saving still vanishes by part (ii).

Lemma 4 is the reason the empirical question of Section 5 is posed per population. Whether targeted verification saves anything in a given population depends on where that population's posteriors fall relative to q_1 , which the theory does not decide. For a Gaussian order-flow model the quantities of Definition 5 have closed forms.

Proposition 1 (Closed forms for a Gaussian order-flow model).

Under Assumptions 1–4, let the order-flow model c of Definition 5 be Gaussian: $Y \mid M \sim \mathcal{N}(\mu_M, \Sigma_Y)$, $M \in \{K, F\}$, with common covariance $\Sigma_Y \succ 0$ and whitened separation $d = \sqrt{(\mu_K - \mu_F)^\top \Sigma_Y^{-1} (\mu_K - \mu_F)} > 0$. Suppose $S = (q_1, q_2) \neq \emptyset$. Write $\ell = \text{logit } p$, $\ell_1 = \text{logit } q_1$, $\ell^* = \text{logit } q^* = \log(b/a)$, $\ell_2 = \text{logit } q_2$ (with $\text{logit } 0 = -\infty$, $\text{logit } 1 = +\infty$, $\Phi(\pm\infty) = 1, 0$ accordingly), and for $-\infty \leq u < v \leq +\infty$

$$\begin{aligned}\Pi_K(u, v) &= \Phi\left(\frac{v - \ell}{d} - \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} - \frac{d}{2}\right), \\ \Pi_F(u, v) &= \Phi\left(\frac{v - \ell}{d} + \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} + \frac{d}{2}\right).\end{aligned}$$

Then

$$\begin{aligned}m_c(p) &= p \Pi_K(\ell_1, \ell_2) + (1 - p) \Pi_F(\ell_1, \ell_2), \\ D_c(p) &= p(\alpha_1 - \beta_1) \Pi_K(\ell_1, \ell^*) - (1 - p) \beta_1 \Pi_F(\ell_1, \ell^*) \\ &\quad + p(\beta_2 - \alpha_2) \Pi_K(\ell^*, \ell_2) + (1 - p) \beta_2 \Pi_F(\ell^*, \ell_2),\end{aligned}$$

with $\alpha_1 = s a + (1 - t) b - r_K \tau$, $\beta_1 = c_v + (1 - t) b$, $\alpha_2 = t b + (1 - s) a + r_K \tau$, $\beta_2 = t b - c_v$ (so that $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$, Theorem 1). As $d \rightarrow 0$,

$m_c(p) \rightarrow \mathbb{I}[p \in S]$ for $p \notin \{q_1, q_2\}$ and $D_c(p) \rightarrow (V - \chi)^+(p)$ for every p , the confounded values of Corollary 1.

Proposition 1 shows how the mass and the saving depend on the separation d and on the prior through the logit shift ℓ , and it recovers the confounded values of Corollary 1 as $d \rightarrow 0$.

The order-flow models of Paper 1 [1], to which Remark 2 and Section 5 refer, are simulated with ABIDES, an agent-based discrete-event simulator of a limit order book market [42]. In Paper 1 a *market design* is one way of injecting a compromise and a fault into the simulated market, and each design is run in two *regimes*: in the *summary-matched* regime the two injections are parameterised so that summary statistics of their order flow match, and in the *contrast* regime they are not. A design in one regime is an order-flow model c in the sense of Definition 5.

Remark 2 (The Gaussian model is a calibration model). The hypotheses of Proposition 1 are a Gaussian order-flow model of the form of Paper 1’s Definition D5 with any separation $d > 0$; they involve no value of d and no property of Paper 1’s cells. Paper 1’s ABIDES order-flow models were never shown to be Gaussian, so the Gaussian model is an *equivalent-Gaussian calibration model*, not Paper 1’s cells, and Proposition 1 describes those cells only insofar as they are Gaussian with that separation. A separation d backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d/2)$ is a calibration heuristic only; wherever the closed forms of Proposition 1 are evaluated at such a d , d is also swept directly over a grid. Every statement about Paper 1’s cells, including where targeted verification beats the best constant policy in them, rests on quantities computed on each cell’s corrected classifier posteriors (Definition 5 applies to any order-flow model with finite densities), with no Gaussian assumption, and not on Proposition 1.

The benchmark $\text{Acc}^* = \Phi(d/2)$ of Remark 2 is the accuracy of the Bayes classifier of the Gaussian model at equal priors, $p = 1/2$. At a prior p with $\ell = \text{logit } p$, the Bayes accuracy of the same model is $p\Phi(d/2 + \ell/d) + (1 - p)\Phi(d/2 - \ell/d)$, which exceeds $\Phi(d/2)$ whenever $p \neq 1/2$; a separation backed out through $\Phi(d/2)$ from an accuracy measured at another prior therefore overstates d . This is a further reason why such a d is a calibration heuristic only and why d is also swept directly over a grid.

4.9. Extensions: any check, a miscalibrated score, a relaxed channel

The results above are stated for a binary check with fixed characteristics (s, t) , a calibrated belief, and a check whose accuracy does not depend on

the order flow. This subsection relaxes each of the three in turn. The three results are supporting extensions; each follows from known results or from a short argument, and none is a headline result of the paper. No other result of the paper depends on any of the three.

A check with any output. Many checks return more than a yes or a no: a score, a graded confidence, a multi-level outcome. Fixing the sensitivity and the specificity of such a check at one cut-off narrows the range of beliefs over which it pays [11].

Definition 6 (A check with an arbitrary output). A *check* returns an outcome Z in a measurable space $(\mathcal{Z}, \mathcal{A})$, with law P_K^Z under K and P_F^Z under F , and densities g_K, g_F with respect to $\mu = P_K^Z + P_F^Z$; it costs c_v and takes a delay τ during which a true compromise causes harm at rate r_K , as in Definition 2. Its *total variation* is $\text{TV}_Z = \frac{1}{2} \int |g_K - g_F| d\mu = \sup_{A \in \mathcal{A}} (P_K^Z(A) - P_F^Z(A))$. For a belief q , the *verify-first risk* is $R_{v,Z}(q) = c_v + q r_K \tau + R_Z(q)$ with

$$R_Z(q) = \int \min \{ q g_K C_K + (1 - q) g_F C_F, \quad q g_K H_K + (1 - q) g_F H_F \} d\mu,$$

the value of information is $V_Z(q) = R(q) - R_Z(q)$, and the verification region is $S_Z = \{q \in [0, 1] : V_Z(q) > \chi(q)\}$, with $\chi(q) = c_v + q r_K \tau$ as in Definition 3. The check of Definition 2 is the case $\mathcal{Z} = \{+, -\}$, $P_K^Z(\{+\}) = s$, $P_F^Z(\{-\}) = t$. A *binary reading* of a check is the check $\mathbb{I}[Z \in A]$ for some $A \in \mathcal{A}$, with sensitivity $s_A = P_K^Z(A)$ and specificity $t_A = P_F^Z(\mathcal{Z} \setminus A)$; V_A, S_A denote its value of information and verification region.

Theorem 3 (Any check: the interval, and a critical delay set by total variation). Under Assumption 1, let a check be as in Definition 6 with $\text{TV}_Z > 0$, $c_v \geq 0$, $\tau \geq 0$, $r_K \geq 0$, available once, and let Assumption 4 hold with the laws (P_K^Z, P_F^Z) in place of (s, t) . Then: (i) V_Z is continuous on $[0, 1]$, $V_Z \geq 0$, $V_Z(0) = V_Z(1) = 0$, and V_Z is convex on $[0, q^*]$ and on $[q^*, 1]$; explicitly

$$V_Z(q) = \int (q a g_K - (1 - q) b g_F)^+ d\mu - (q a - (1 - q) b)^+.$$

(ii) S_Z is an open interval (possibly empty). It is non-empty if and only if $V_Z(q^*) > c_v + q^* r_K \tau$, and then $S_Z = (q_1, q_2)$ with $0 \leq q_1 < q^* < q_2 \leq 1$. An optimal one-shot policy is: take the passive action of Lemma 1 if $q \notin S_Z$; if $q \in S_Z$, verify and then contain if and only if $q a g_K(Z) > (1 - q) b g_F(Z)$.

(iii) $V_Z(q^*) = \text{TV}_Z \frac{ab}{a+b} = \text{TV}_Z a q^*$. (iv) $\tau < \tau'$ implies $S_Z(\tau') \subseteq S_Z(\tau)$. If $V_Z(q^*) \leq c_v$, then $S_Z(\tau) = \emptyset$ for every $\tau \geq 0$. If $r_K = 0$, the delay costs nothing, $S_Z(\tau) = S_Z(0)$ for every $\tau \geq 0$, and $S_Z(\tau)$ is non-empty for every $\tau \geq 0$ if and only if $V_Z(q^*) > c_v$, i.e. $\text{TV}_Z a q^* > c_v$; there is then no critical delay. If $r_K > 0$ and $V_Z(q^*) > c_v$, then $S_Z(\tau) \neq \emptyset$ if and only if $\tau < \tau_Z^*$, where

$$\tau_Z^* = \frac{\text{TV}_Z a - c_v/q^*}{r_K}.$$

So whether a verification interval exists, and the critical delay, depend on the check only through TV_Z . (v) For the binary check of Definition 2, $\text{TV}_Z = |s+t-1|$; when $s+t > 1$ (Assumption 3), $\text{TV}_Z = s+t-1$ and (iii)–(iv) are Lemma 3(iv) and Theorem 2(ii). For every binary reading $\mathbb{K}[Z \in A]$, $V_A \leq V_Z$ pointwise, so $S_A \subseteq S_Z$; its Youden index $s_A + t_A - 1 = P_K^Z(A) - P_F^Z(A)$ is at most TV_Z , with equality for the likelihood-ratio reading $A^* = \{g_K > g_F\}$. Hence, when $r_K > 0$ and $V_A(q^*) > c_v$, the critical delay τ_A^* of the reading satisfies $\tau_A^* \leq \tau_Z^* = \tau_{A^*}^*$.

Parts (i) and (iii) are known: the value of an experiment is convex in the belief [9], and its value at the threshold is the total variation distance of the outcome laws times the regret at the threshold, the arbitrary-output form of the Youden-index result of [7] through Le Cam’s identity between total variation and the best binary test [10]; that the Youden cut-off is optimal at the threshold is also known [8]. Given (iii), the critical delay τ_Z^* of part (iv) follows in one line, exactly as τ^* follows from Lemma 3(iv) in Theorem 2: the cost of the check is affine and increasing in the belief, so the check pays somewhere if and only if it pays at q^* . For a binary check, parts (iii) and (iv) specialise to the value formula of Lemma 3(iv) and the critical delay of Theorem 2(ii). For an operator, the consequence is a way to compare checks. Whether a check can pay at all, and its time budget, depend on the check only through one number, its total variation, which is its best Youden index over all ways of reading its output; a binary reading of the output never has a longer time budget than the check read in full (part (v)).

A miscalibrated score. The rule of Theorem 1 is applied in practice to a score, such as a prior-shift-corrected classifier posterior, that need not be calibrated. Proposition 2 states exactly when verifying on such a score still beats the passive rule applied to the same score.

Proposition 2 (Plug-in rules on a miscalibrated score). Under Assumptions 1, 3 and 4, let $S = (q_1, q_2) \neq \emptyset$ (Theorem 1), and let $\hat{Q}$ be a $[0, 1]$ -

valued score computed from the order-flow observation (for example a prior-shift-corrected classifier posterior), jointly distributed with the cause M , with the verification outcome Z conditionally independent of $\hat{Q}$ given M (as under Assumption 2). Write $\eta(\hat{q}) = P(M = K \mid \hat{Q} = \hat{q})$ for its calibration map (a version of the conditional probability), and $B_1 = (q_1, q^*]$, $B_2 = (q^*, q_2)$. The *passive plug-in rule* leaves if $\hat{Q} \leq q^*$ and contains otherwise; the *targeted plug-in rule* verifies if $\hat{Q} \in S$, then contains on $Z = +$ and leaves on $Z = -$, and otherwise acts as the passive plug-in rule. Let

$$D_1 = s a + (1 - t) b - r_K \tau, \quad D_2 = t b + (1 - s) a + r_K \tau,$$

the denominators of q_1 and q_2 in Theorem 1, both positive. Then: (i) conditional on $\hat{Q} = \hat{q}$, the expected cost of the passive plug-in rule minus that of the targeted one is $D_1(\eta(\hat{q}) - q_1)$ if $\hat{q} \in B_1$, $D_2(q_2 - \eta(\hat{q}))$ if $\hat{q} \in B_2$, and 0 otherwise; (ii) hence at a score $\hat{q} \in S$ the targeted rule is strictly cheaper than the passive rule if and only if $\eta(\hat{q}) > q_1$ when $\hat{q} \leq q^*$, and if and only if $\eta(\hat{q}) < q_2$ when $\hat{q} > q^*$: a score in B_1 may overstate the probability of compromise by less than $\hat{q} - q_1$, and a score in B_2 may understate it by less than $q_2 - \hat{q}$, and verifying there still costs less than the passive rule; (iii) (*band-frequency identity*) the expected cost of the passive plug-in rule minus that of the targeted one is

$$D_1 [P(M = K, \hat{Q} \in B_1) - q_1 P(\hat{Q} \in B_1)] + D_2 [q_2 P(\hat{Q} \in B_2) - P(M = K, \hat{Q} \in B_2)];$$

(iv) if $\hat{Q}$ is calibrated ($\eta(\hat{Q}) = \hat{Q}$ almost surely), this difference is $\mathbb{E}[(V - \chi)^+(\hat{Q})] \geq 0$, the saving of Definition 5 and Lemma 4(i).

Part (ii) gives a tolerance: in the lower half of the band a score may overstate the probability of compromise by up to its distance to q_1 , not to q^* , before verifying there loses to the passive rule; in the upper half, symmetrically, up to its distance to q_2 . Part (iii) turns this into a population statement that needs only the true causes and the scores, not the calibration map. For two actions, the harm miscalibration does to decisions has been shown in simulation [40] and bounded by a worst-case regret threshold [41]; Proposition 2 is the exact counterpart for the three actions of the verification band. Proposition 2 names a candidate mechanism for the outcome of the policy comparison of Section 5.8, where targeted verification is more costly than the passive classifier in some cells; whether that mechanism is at work there is not tested in this paper.

A relaxed separate channel. Assumption 2 asks that the outcome of the check be conditionally independent of the order flow given the cause. An adversary may, however, defeat an out-of-band confirmation more often when the orders look more suspicious, so the accuracy of the check may depend on the order flow. Proposition 3 addresses this case.

Proposition 3 (Robust verification region under a relaxed separate channel). Under Assumptions 1, 3 and 4, replace Assumption 2 by the following: given the order-flow observation $Y = y$, the verification outcome has sensitivity $s(y) = P(Z = + \mid M = K, Y = y)$ and specificity $t(y) = P(Z = - \mid M = F, Y = y)$, measurable in y , with $s(y) \geq s_{lo}$ and $t(y) \geq t_{lo}$ for every y , where $s_{lo}, t_{lo} \in [0, 1]$ and $s_{lo} + t_{lo} > 1$. Write $V_{s,t}$ and $S_{s,t}$ for the value of information and the verification region of Definition 3 with characteristics (s, t) , and $q(y) = P(K \mid Y = y)$ for the *calibrated* posterior: the true conditional probability of compromise given the order-flow observation, not a classifier’s score (a miscalibrated score is the case of Proposition 2). Then: (i) $V_{s,t}(q)$ is nondecreasing in s and in t for every $q \in [0, 1]$; hence $S_{s_{lo}, t_{lo}} \subseteq S_{s(y), t(y)}$ for every y ; (ii) $P(K \mid Y = y, Z = z)$ is the update of Definition 2 applied to $q(y)$ with characteristics $(s(y), t(y))$; (iii) the *robust policy*, which verifies if and only if $q(Y) \in S_{s_{lo}, t_{lo}}$, then contains on $Z = +$ and leaves on $Z = -$, and otherwise takes the passive action of Lemma 1 at $q(Y)$, has, conditional on $Y = y$, an expected cost below that of the passive rule by $(V_{s(y), t(y)} - \chi)(q(y)) \geq (V_{s_{lo}, t_{lo}} - \chi)(q(y)) > 0$ if $q(y) \in S_{s_{lo}, t_{lo}}$, and equal to it otherwise. It is never worse than the passive rule at any y , and it uses only the lower bounds (s_{lo}, t_{lo}) , not $s(\cdot)$ and $t(\cdot)$.

Proposition 3 is a monotonicity argument: the value of information does not decrease when the check becomes more accurate, so a region computed from lower bounds on the accuracy is contained in the region at every observation. What the results need from the separate channel is therefore not independence but a lower bound on the check’s accuracy that holds whatever the order flow; the price is a smaller region than the one the true accuracy would give. Proposition 3 assumes a calibrated posterior; a miscalibrated score is the case of Proposition 2.

5. Exploratory empirical study on Paper 1’s cells

This section specifies an empirical study on the order-flow models and priors of Paper 1 [1]. The study is exploratory: its primary test was specified after Paper 1’s cost table for the same cells had been examined, so what-

ever it finds is exploratory evidence, not the outcome of a pre-registered test. The check of the interval was specified before that and is a check of the computation. The policy comparison’s criterion was also stated before any cost existed, but its operative reading as a per-point test was confirmed only after provisional results existed, so it too is exploratory. The grids, plausible set, policies, metrics and outcome criteria below were fixed before any quantity of the primary test was computed, and were not changed afterwards. The cells were not fixed by this study: they are Paper 1’s, and they changed once, when Paper 1 refitted its classifiers (Section 5.6). The theorems of Section 4 are proved and are not tested here; where the study computes a quantity that a theorem gives in closed form, a mismatch is an error of computation, not evidence.

5.1. Data and passive posteriors

The order-flow data are Paper 1’s simulated order-flow features for each of its 5 market designs (V1, Study 4, V2, V3, V4; full names in Table 8), in both the summary-matched and the contrast regime (Section 4.8). On each design and regime, Paper 1’s classifier of compromise against fault is refitted exactly as Paper 1 fits it, on Paper 1’s primary training split, and applied to the held-out test instances, which are balanced between the two causes: $n_K = 300$ compromises and $n_F = 300$ faults in every design and regime (Table 8). Fits that Paper 1’s procedure leaves non-converged are kept as they are, because the cells studied are defined by Paper 1’s own pipeline; convergence is recorded per design and regime (Table 8).

The classifier’s scores are estimated at the class balance of the training set, with training share $\pi_{\text{tr}} = 0.5$ of compromises. They are moved to a prior p by the standard prior-shift correction [36]: a score r becomes the corrected posterior

$$q_p(r) = \frac{(p/\pi_{\text{tr}}) r}{(p/\pi_{\text{tr}}) r + ((1-p)/(1-\pi_{\text{tr}}))(1-r)}.$$

The corrected posteriors are the passive beliefs $q(y)$ to which Lemma 2 applies the verification analysis.

5.2. Cells and the failure cells

A cell is a design, a regime and a prior (Definition 5): the design in one regime is the order-flow model c , and p ranges over the 5 priors of Paper 1’s grid, $p \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$. Every quantity of Definition 5 is an

expectation under $P_p = p P_K^c + (1 - p) P_F^c$. On the balanced test set this law is obtained by weighting each compromise instance by p/n_K and each fault instance by $(1 - p)/n_F$, with n_K and n_F the numbers of test instances of each cause; quantiles of the posterior are lower quantiles of this weighted empirical distribution. Without the weights, the test set would describe the posteriors under the training balance, which is the prior the correction removes.

The *failure cells* are the cells in which Paper 1 reports that its classifier does not beat the best constant policy, always contain or always leave. They are read from Paper 1’s reported cost table, with Paper 1’s best constant policy for each cell, and are not re-derived here; they are listed, cell by cell, in Table 3.

The primary loss table is Paper 1’s cost triple with $H_F = 0$ (its Definition D6), in the form of Section 3.1: Paper 1’s containment cost, disruption and harm in the roles of C_K , C_F and H_K . This fixes a , b and the passive threshold $q^* = b/(a + b)$ of Lemma 1, which are the same in every cell. The measured- H_F table from which Paper 1’s cells are classified is the second arm (Section 5.7). Both arms were fixed before any result, and both are reported for every criterion.

5.3. Verification grid in model units

The model involves the delay and the damage rate only through their product $r_K \tau$ (Definition 2, Assumption 4), so the grid has four axes, written in units of the model. Let $V_{\max} = a q^* = ab/(a + b)$, the peak value of a perfect, free and instantaneous check (Lemma 3), which no check can exceed at any belief. The axes are the sensitivity s , the specificity t ,

$$x = \frac{c_v}{V_{\max}} \quad \text{and} \quad y = \frac{r_K \tau}{a},$$

the cost of a check as a share of the most any check can save, and the harm a true compromise accrues during the delay as a share of the harm cost difference a . In these units the condition of Theorem 1 for a non-empty verification interval reads $x + y < s + t - 1$, and Theorem 2 gives the critical delay as $r_K \tau^*/a = s + t - 1 - x$, so critical delays are reported in units of a/r_K . Whether the interval exists at a grid point therefore does not depend on the monetary scale of a cell. The same monetary values of c_v and $r_K \tau$ are used in every cell.

Sensitivity and specificity range over the full product of a common set of levels, every pair of which satisfies Assumption 3. The cost axis x and the

delay axis y each range over a set of levels starting at a near-free check and at no delay respectively, and extending far enough that, for every pair (s, t) , the grid has points on both sides of the boundary $x + y = s + t - 1$ at which the interval vanishes (Theorem 2(ii)). The levels of every axis are listed with the results (Section 5.8).

Plausible set. A check is trivially useful at low beliefs when its lower threshold is near zero. By the formula for q_1 in Theorem 1, $q_1 = 0$ when the fixed cost c_v is zero and the specificity t is one, and a near-free check that almost never flags a fault pushes q_1 close to zero; the limiting case is Lemma 4(iv), in which the mass in the interval does not vanish at low priors. A grid point is therefore *implausible* if and only if its specificity is the highest level of the grid and its cost is one of the two lowest levels of x , whatever its sensitivity and its delay; every other grid point is *plausible*. The sensitivity is not restricted, because it enters q_1 only through the factor sa in the denominator, which the grid bounds away from zero. The delay is not restricted, because the delay cost $qr_K\tau$ is scaled by the belief and cannot make a check near-free at low posteriors. Costly or slow checks are not implausible: they only shrink or empty the interval, which the criteria below count against the prediction.

5.4. Policies compared

Five policies are compared on every cell and grid point: always contain; always leave; the passive classifier, which contains when the corrected posterior exceeds q^* (Lemma 1); verify everything and follow the outcome; and targeted verification by the rule of Theorem 1 applied to the corrected posterior. Verification outcomes are not simulated. For a test instance with true cause M and posterior q , the cost of each policy is its exact expectation over the outcome Z , with $P(Z = + | K) = s$ and $P(Z = + | F) = 1 - t$ (Definition 2), plus the fixed cost and the harm during the delay where the policy verifies (Assumption 4). The only randomness left in the study is the resampling of test instances for the confidence intervals.

5.5. Metrics

Per cell and grid point the study computes:

- q_1 , q_2 and τ^* in closed form (Theorems 1 and 2), cross-checked against a numerical search of the rule;

- the location of the corrected posteriors: their quantiles under each cause, the shares below q_1 , inside (q_1, q_2) and above q_2 , and the distance of the median corrected logit posterior from logit q^* ;
- the verification mass $m_c(p)$ of Definition 5, estimated as the P_p -weighted share of test instances whose corrected posterior lies in (q_1, q_2) ;
- the saving $\widehat{D}_c(p)$ of Definition 5, estimated in two ways: the *plug-in saving* $\widehat{D}_c(p)$, the P_p -weighted mean of $(V - \chi)^+$ at the corrected posteriors, which is Definition 5 evaluated as if the corrected posteriors were the true ones; and the *realised saving* $\widehat{D}_c(p)$, the P_p -weighted cost of the passive rule minus that of targeted verification, computed from the true causes of the test instances. By Lemma 4(i) the two agree in expectation when the corrected posteriors are calibrated;
- the recovered share $\rho_c(p)$, estimated as $\widehat{D}_c(p)$ divided by the passive cost of ambiguity, itself estimated as the P_p -weighted cost of the passive rule minus the cost of an operator who knows the cause; it is not reported where that cost of ambiguity is indistinguishable from zero. Unlike $\rho_c(p)$ of Definition 5, which lies in $[0, 1]$ by Lemma 4(i), this realised share is not constrained: it is negative where the realised saving is negative, which, beyond sampling error, points to miscalibrated corrected posteriors, whereas the plug-in saving $\widehat{D}_c(p)$ is non-negative by construction, as Lemma 4(i) requires of $D_c(p)$;
- for every policy, its expected cost (Table 9) and, beside it, its false-positive and false-negative rates.

False-positive and false-negative rates are reported beside every cost because, as in Paper 1, the prior-shift correction trades one for the other. Confidence intervals are bootstrap percentile intervals over test instances, resampled within each cause so that the class sizes, and hence the weights of Section 5.2, are exact in every resample. The confidence level and the number of resamples are listed with the results.

5.6. Primary exploratory test: targeted verification in the failure cells

The primary test asks, in each failure cell, over which region of (s, t, c_v, τ) targeted verification beats the cell's best constant policy, and whether that region is the one predicted from Theorem 1 and the cell's corrected posteriors.

Contrast and predicted saving. Write $\eta(c, g)$ for the expected cost of targeted verification minus that of the best constant policy, in cell (c, p) at grid point g , with bootstrap confidence interval $[L, U]$. When the best constant policy is always leave, Definition 1 and Lemma 1 give that leaving costs $(a+b)(q-q^*)^+$ more than the passive rule at belief q , and Lemma 4(i) gives that targeted verification costs $(V-\chi)^+(q)$ less than it. The model’s prediction for the tested contrast is therefore $-\Gamma(c, g)$, with predicted saving

$$\Gamma(c, g) = \mathbb{E}_p[(a+b)(q-q^*)^+ + (V-\chi)^+(q)],$$

the expectation taken over the corrected posteriors under P_p . When the best constant policy is always contain, $(q-q^*)^+$ is replaced by $(q^*-q)^+$. The two terms are reported separately; a positive first term in a cell where the passive rule does not beat the best constant policy indicates miscalibrated corrected posteriors and is reported as such.

Predicted region. By Theorem 1 and Lemma 4(i), a saving over always leaving can come only from posterior mass above q_1 . Let $Q_\alpha = \inf\{u : P_p(q \leq u) \geq \alpha\}$ be the α -quantile of the corrected posterior q under P_p , and fix two quantile levels $\alpha_{\text{in}} < \alpha_{\text{out}}$ in the upper tail. A grid point is

- *predicted in* if $q_1 < Q_{\alpha_{\text{in}}}$ and $P_p(q_1 < q < q_2) > 0$;
- *predicted out* if $q_1 \geq Q_{\alpha_{\text{out}}}$, or if all of the mass above q_1 lies at or above q_2 ;
- in the *band* otherwise.

The prediction speaks of the upper tail of the posteriors without naming a level. Outside the band, the position of q_1 relative to Q_α is the same at every level $\alpha \in [\alpha_{\text{in}}, \alpha_{\text{out}}]$, so the classification does not depend on the choice of level; band points are exactly those whose prediction would change with it, so they are counted and reported, with the quantile at an intermediate level, and enter no share. The two levels are set from the size of the test set ($n_K = 300$, $n_F = 300$), not from any result. At α_{in} the tail above q_1 holds enough weighted test instances for the bootstrap interval to resolve a saving of the predicted order, so a point is declared in only where the test can see the predicted saving; at α_{out} it holds too few instances for any interval to resolve one, and the model puts almost no mass where verification acts, so a point is declared out only there. The predicted region uses the point estimates on the full test set.

Agreement. At a predicted-in point the empirical result *agrees* if $U < 0$, *disagrees* if $U \geq 0$ and $L > -\Gamma$, and is *unresolved* if the interval contains both 0 and $-\Gamma$, which is a failure of power rather than a disagreement. At a predicted-out point it agrees if $U \geq 0$ and disagrees if $U < 0$, a saving the model does not predict. A cell's agreement share is the number of agreeing points divided by the number of agreeing and disagreeing points, over plausible grid points only; implausible points are classified and reported, and enter no share. A cell *agrees* if its share is at least $\bar{\sigma} = 1 - 2(1 - \gamma)$, with γ the confidence level of the intervals. If the prediction holds and the posteriors are calibrated, a point disagrees only through the error of its interval, at a rate of at most $1 - \gamma$; $\bar{\sigma}$ allows twice that rate, for the correlation of neighbouring grid points and residual miscalibration of the corrected posteriors. A cell is *unresolved* if more than half of its plausible predicted-in points are unresolved. Shares are decided per failure cell, because the predicted region is computed from each cell's own posteriors; a pooled share is reported and decides nothing. The values of α_{in} , α_{out} and γ are listed with the results.

Power. The size of the test set is fixed by Paper 1's design, so no sample size was chosen for a target power, and no power analysis in that sense was run. Power is assessed instead point by point, against the effect the model itself predicts: at a predicted-in point the interval either separates the predicted saving Γ from zero or contains both 0 and $-\Gamma$, and in the second case the point is unresolved, not a disagreement. The outcome below uses this in both directions: a cell with a majority of unresolved predicted-in points makes the outcome *inconclusive at this power* rather than a support, and a refutation requires that the test had power where the model predicts a saving.

Outcome. The outcome is one of four, applied in this order.

1. *Refutes* if no plausible point of any failure cell has $U < 0$, and the test has power: at least one failure cell has plausible predicted-in points, and in every such cell at least half of them disagree. A cell with no predicted-in points predicts an empty region and cannot refute.
2. *Weakens* if some failure cell has an agreement share below $\bar{\sigma}$, or if $U < 0$ only at implausible points.
3. *Inconclusive at this power* if no cell weakens or refutes but some cell is unresolved.
4. *Supports* if every failure cell agrees, no cell is unresolved, and $U < 0$ at one plausible point of one failure cell at least.

A cell with no decided point is listed as such; it counts as unresolved if it has predicted-in points and otherwise enters no criterion. The outcome is decided with Paper 1’s cost matrix as above; a version with the measured cost of leaving a fault (Section 5.7) is reported beside it and decides nothing.

How the tested claim was arrived at. The claim tested here replaced an earlier formulation, which stated that targeted verification acts where the passive posteriors straddle Paper 1’s threshold q^* , and that, because q^* is low with Paper 1’s loss tables, the verification mass and the recovered share are largest in the low-prior cells where Paper 1’s classifier fails. That formulation rested on a premise: q^* lies closer, on the logit scale, to the priors of the failure cells than to those of the success cells. The premise was checked on Paper 1’s cost table before any quantity of this study was computed, and it failed. The claim was then replaced by the one tested here, and the test was specified; this is why the test is exploratory. The test was then run twice with the same rules: first on a provisional version of Paper 1’s cells, then on Paper 1’s final cells, after Paper 1 refitted its classifiers for its own reasons. The first run was labelled provisional in its own output, because the refit was known to be coming and could change which cells fail; the re-run on the final cells used the same rules unchanged. On the provisional cells the outcome was *weakens*: in some failure cells the agreement share fell below the threshold. Those cells are not failure cells on the final basis.

5.7. Secondary and descriptive analyses

Policy comparison. If the corrected posteriors are calibrated, targeted verification is, in expectation, at least as cheap as every other policy on every cell (Lemma 4(i) and the definitions of the constant policies). The comparison is read as weakening this if some other policy beats targeted verification beyond sampling error, that is, if the confidence interval of targeted verification minus that policy lies above zero at some grid point of some cell, which would indicate miscalibrated posteriors; and as refuting it if targeted verification never beats the best constant policy in the failure cells. Differences of the order of floating-point rounding, where two policies take the same action on every instance, are treated as ties.

Mass, saving and recovered share. The quantities $m_c(p)$, $\widehat{D}_c(p)$ beside $\widetilde{D}_c(p)$, and $\rho_c(p)$ are reported for every cell, with confidence intervals. They are descriptive and decide no claim.

Sensitivity analyses. The passive threshold is recomputed with the measured cost of leaving a fault in place of $H_F = 0$: the value of H_F that Paper 1’s cost table reports for each design and regime, the execution cost of the fault’s own measured order volume at the pace at which it actually ran, so that the recomputed threshold differs by design and regime (Table 10), with the same monetary values of c_v and $r_K\tau$, so that the sensitivity analysis changes q^* and not the check. The fault rate r_F of Corollary 4 is not measured; the sensitivity analysis keeps $r_F = 0$. Where a cell’s scores are close to Gaussian, the closed forms of Proposition 1 may be evaluated as a cross-check, labelled as a calibration model (Remark 2), with the separation also swept directly over a grid; the cross-check decides nothing.

5.8. Results

The primary test, its sensitivity analysis and the policy comparison are exploratory; the check of the interval is a check of the computation, and the secondary analyses decide nothing. The design values are listed in Table 1, the outcome of each criterion in Table 2, the test instances and fits in Table 8, and the per-cell counts, costs and quantities in Tables 3–10, with the expected cost of every policy in Table 9. Negative outcomes are reported beside the positive one, in the order of Section 5.6.

Interval and critical delay (not evidence). The computed verification intervals agree with the closed forms of Theorem 1 at every grid point, shrink in the delay and are empty exactly at and beyond the critical delay of Theorem 2 (Table 7); since these are theorems, this is a check of the computation, not evidence for them.

Primary exploratory test. On the 6 failure cells of Paper 1, the outcome of the primary test is *supports*, and it is exploratory: the test was specified after Paper 1’s cost table had been examined, so it is not a pre-registered confirmation. Every failure cell reaches the minimum agreement share $\bar{\sigma} = 0.9$ over the 528 plausible grid points (smallest share 0.989), no cell is unresolved, and in every failure cell targeted verification beats always leaving, the best constant policy there, at plausible grid points (Table 3). That share counts predicted-out points, where the tail above q_1 is too thin for a saving to be resolved, so they can rarely disagree. Restricted to predicted-in points, a post-hoc reading that the criterion did not state, the agreement share lies between 0.861 and 1.000 across the failure cells and falls below $\bar{\sigma}$ in 1 of them,

the failure cell of V2 in the summary-matched regime at the lowest prior p (Table 3, columns In agree and In share, beside the pre-stated share). Under that reading the outcome would be *weakens*; the criterion as stated decides *supports*. The agreement is not perfect: each failure cell keeps a few disagreeing points (Table 3), and in the contrast regime of V1 many points (341) fall in the band between the two quantile levels, where the prediction depends on a level the claim does not name, and so enter no share. The failure cells are those of Paper 1 on its final basis; they differ from the cells of an earlier, provisional basis on which the same test had weakened the claim, and the cells responsible for that earlier outcome are no longer failure cells, so the present outcome is not a replication on the same cells. With the measured cost of leaving a fault in place of $H_F = 0$, the outcome is the same (smallest share 0.989); this sensitivity analysis decides nothing, and agrees. The choice of loss table changes nothing that decides: under both tables the failure cells are the same (whether the passive classifier beats the best constant policy agrees in all 50 cells), and every outcome word is the same, including that of the post-hoc predicted-in reading. No pre-registered confirmation of this outcome has been run (Section 6.1).

Policy comparison. The policy comparison, also exploratory, *weakens* the expectation that targeted verification is at least as cheap as every other policy on every cell. Targeted verification is never significantly more costly than always containing, always leaving or verifying everything, in any cell, nor than the passive classifier in any failure cell, and in every failure cell it beats the best constant policy; but in 13 of the 44 cells where Paper 1’s classifier succeeds it is significantly more costly than the passive classifier at between 1 and 315 of the 528 plausible grid points, depending on the cell (Table 4, column vs passive). The criterion counts a single grid point, and it has no correction for the many points and cells tested. Points of one cell share the same instances, so their tests are not independent, and no chance rate can be computed exactly. No correction for multiplicity was applied. As a descriptive guide, with no error rate attached, in 8 of these 13 cells the count exceeds the fraction $1 - \gamma$ of the cell’s plausible points ($\gamma = 0.95$). This split is descriptive: it is not a test, and it changes with the loss table (under the measured cost of leaving a fault, a partly different set of cells). The miscalibration of the corrected posteriors is a hypothesis these cells suggest, not a finding of this study. These differences are not rounding ties, which are treated as ties, and not a failure of the fits, which all converged; in

those cells, the realised saving over the passive rule is negative while the saving predicted from the corrected posteriors is positive, which the criterion named in advance as the sign of miscalibrated corrected posteriors; no test of miscalibration was run. A cost-ratio threshold, and hence the plug-in saving, is optimal only for calibrated scores [39], and Proposition 2 states exactly where miscalibration makes targeted verification lose to the passive rule; whether it explains these cells is not tested here.

Secondary analyses (decide nothing). The mass in the verification interval is larger in the failure cells than in the success cells at most plausible grid points but not at all of them, and the recovered share is not larger there at most points, so the ordering expected under the earlier formulation of the claim (Section 5.6) does not hold; nor does its premise. Both are reported for completeness and decide nothing (Tables 5 and 6; sensitivity analyses in Table 10).

6. Discussion

6.1. What the empirical study shows and does not show

The study of Section 5 does not test the theorems of Section 4, which are proved. What it can test is whether, on the classifier posteriors of a particular set of simulated market designs, the region where targeted verification beats the best constant policy is the region that Theorem 1 predicts from those posteriors. Because the test was specified after Paper 1’s cost table for the same cells had been examined, any outcome of it is exploratory; a confirmation would call for a pre-registered test on fresh data, and a disagreement would point first to the calibration of the corrected posteriors (Section 6.4) before it bears on the model. Both outcomes of the study are exploratory. In the failure cells of Paper 1 the primary test *supports* the prediction; the support is not robust, since a post-hoc reading restricted to predicted-in points, which the criterion did not state, reads *weakens* in one cell, and on Paper 1’s provisional cells the same test weakened the prediction (Section 5.8). The policy comparison *weakens* the expectation that targeted verification is never beaten: in some of the cells where Paper 1’s classifier succeeds, the passive classifier is cheaper at some plausible grid points, which suggests (but does not establish) miscalibrated corrected posteriors in those cells. A confirmation of the primary test needs a pre-registered test on fresh instances, decided on the predicted-in points under both loss tables. Such

Table 1: Design values of the study, fixed before any quantity of the primary test was computed.

Quantity	Value
Designs (Paper 1)	Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4)
Regimes	contrast, summary-matched
Priors p	0.05, 0.10, 0.20, 0.30, 0.50
Training share of compromises π_{tr}	0.5
Test instances n_K, n_F	300, 300 (every design and regime)
Sensitivity s	0.7, 0.8, 0.9, 0.99
Specificity t	0.7, 0.8, 0.9, 0.99
Cost of a check $x = c_v/V_{\max}$	0.001, 0.01, 0.05, 0.1, 0.2, 0.5
Delay $y = r_K \tau / a$	0, 0.01, 0.05, 0.1, 0.25, 0.5
Grid points (plausible / implausible)	576 (528 / 48)
Implausible points	$t = 0.99$ and $x \leq 0.01$
Quantile levels α_{in} , band, α_{out}	0.9, 0.95, 0.99
Confidence level γ	0.95
Agreement threshold $\bar{\sigma} = 1 - 2(1 - \gamma)$	0.9
Bootstrap resamples (within cause)	2000
Passive threshold q^* (Paper 1's cost matrix, $H_F = 0$)	0.364

Table 2: Outcome of each criterion of Section 5, as decided by the pre-stated rules on Paper 1's final cells. Beside it: for the primary test and its sensitivity analysis, the outcome under the post-hoc reading restricted to predicted-in points, which the criterion did not state and which decides nothing (Table 3, columns In agree and In share); and the outcome of the same rules on the provisional version of Paper 1's cells, superseded when Paper 1 refitted its classifiers.

Criterion	Outcome	Predicted- in reading (post hoc)	Provisional basis	Exploratory	Decides
Interval and critical delay: closed form against numerical search	cross-check passed	–	–	no	nothing (a theorem; a mismatch is an error of computation)
Primary exploratory test (Paper 1's cost matrix)	supports	weakens	weakens	yes	the outcome of the primary test (exploratory)
Primary test, measured cost of leaving a fault (sensitivity)	supports	weakens	weakens	yes	nothing (sensitivity)
Policy comparison	weakens	–	weakens	yes	the policy-comparison criterion

a test was designed; under its pre-registered power check, no sample size in the costed range reached the required power in one of the failure cells, so by its own rule its outcome is *inconclusive at this power* and it was not run. It adds no evidence either way. No pre-registered confirmation of either outcome exists. The verification parameters are a grid in model units, not measurements of a particular gateway check, so the study says where a check would pay if it had given characteristics, not whether any existing check has them.

6.2. When the separate channel fails

Every result of Section 4 that involves order flow rests on Assumption 2. If the outcome of the check depends on the order flow given the cause, the verification update is no longer a function of the passive posterior alone and Lemma 2 does not hold. The assumption is a modelling choice about what the check reads, and its fit to a particular verification procedure has to be argued for that procedure. Proposition 3 weakens what has to be argued: if the check’s sensitivity and specificity may depend on the order flow but stay above known lower bounds whatever the order flow, verifying on the region computed from those bounds is never worse than the passive rule. What it does not cover is a check whose accuracy can fall to chance, for instance an adversary who also controls the holder’s second factor.

6.3. One check, then act

The rule is derived for a single check followed by a terminal action (Assumption 3), with the delay charged in full on top of the terminal cost and no order flow used while the check runs (Assumption 4). A fault that costs during the check is covered by Corollary 4, which shortens the critical delay; the fault rate r_F of Corollary 4 is not measured, and the sensitivity analysis keeps $r_F = 0$. Repeated checks, graded responses and evidence that keeps arriving during the delay turn the problem into a sequential one, which this paper does not treat.

6.4. Posteriors and calibration

The saving identity of Lemma 4 holds for the true posterior. Applied to classifier scores it holds only as far as the corrected scores are calibrated, which is why the empirical design reports the plug-in saving $\widetilde{D}_c(p)$ beside the realised saving $\widehat{D}_c(p)$ (Section 5.5); a gap between them beyond sampling error points to miscalibration, not to a failure of the lemma. For a

miscalibrated score, Proposition 2 replaces the lemma: at each score, targeted verification beats the passive rule exactly when the true frequency of compromise at that score lies above q_1 (scores below q^*) or below q_2 (scores above it), and on a population its saving is a weighted difference between the frequency of compromise in each half of the band and the band’s end point. It names recalibration of the posterior inside the band as the remedy; whether recalibration removes the losses of the policy comparison is an empirical question this paper leaves open.

7. Conclusion

Under the assumption that a check reading authorization is conditionally independent of order flow given the cause, verification is not bound by what order flow can reveal, and it composes with any passive posterior. Transposing the threshold approach, a check pays exactly on an interval of beliefs around the passive containment threshold, and because a true compromise keeps doing harm while the check runs, that interval vanishes at a critical delay in closed form. The interval does not depend on the prior; the saving it brings to a population of cases does. The interval and the critical delay carry over to a check with any output, through its total variation, and the separate-channel assumption can be weakened to lower bounds on the check’s accuracy.

Section 5 specifies an exploratory test of where targeted verification beats the best constant policy in the cells of Paper 1 where passive classification does no better, with the grids, plausible set, policies and outcome criteria fixed before any quantity of the primary test was computed. Both of its outcomes are exploratory. The primary test supports the prediction that targeted verification beats that policy over the region the two-threshold rule predicts, but not robustly: a post-hoc reading restricted to the points where a saving is predicted weakens it in one cell. The policy comparison weakens the expectation that targeted verification is never beaten, since in some of the cells where passive classification succeeds the passive classifier is cheaper at some plausible grid points, which suggests (but does not establish) miscalibrated corrected posteriors there. No pre-registered confirmation has been run.

Data and code availability

The code that computes the thresholds, the critical delay and the empirical test of Section 5.8, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The data of the empirical test are the posteriors of the companion study [1], whose deposit they share.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Proofs

This appendix proves the results of Section 4, in the order in which they are stated there.

Proof of Lemma 1. Let $L_c(q) = qC_K + (1 - q)C_F$ (contain) and $L_\ell(q) = qH_K + (1 - q)H_F$ (leave). Their difference is affine:

$$\begin{aligned}\Delta(q) &:= L_\ell(q) - L_c(q) = q(H_K - C_K) - (1 - q)(C_F - H_F) \\ &= qa - (1 - q)b = (a + b)(q - q^*),\end{aligned}$$

with $q^* = b/(a + b)$. By Assumption 1, $a, b > 0$, so $a + b > 0$ and $q^* \in (0, 1)$. Hence $\Delta < 0$ (leave strictly cheaper) for $q < q^*$ and $\Delta > 0$ (contain strictly cheaper) for $q > q^*$. Since $\min\{L_\ell, L_c\} = L_\ell - \max\{0, \Delta\}$,

$$R(q) = L_\ell(q) - (a + b)(q - q^*)^+.$$

R is the minimum of two affine functions, hence continuous and concave; it is piecewise linear with slopes $H_K - H_F$ on $[0, q^*]$ and $C_K - C_F$ on $[q^*, 1]$, which differ by $-(a + b) \neq 0$, so there is exactly one kink, at q^* . $\square$

Proof of Lemma 2. Fix y with $P(Y = y) > 0$ (for densities, replace probabilities of Y by densities; the argument is unchanged), write $q = q(y)$, and let z be an outcome with $P(Z = z | Y = y) > 0$. By Bayes' rule and Assumption 2,

$$P(K | y, z) = \frac{P(z | K) P(y | K) P(K)}{\sum_{c \in \{K, F\}} P(z | c) P(y | c) P(c)}.$$

Divide numerator and denominator by $P(y) = \sum_c P(y | c) P(c)$ and use $P(y | c) P(c) / P(y) = P(c | y)$:

$$P(K | y, z) = \frac{P(z | K) q}{P(z | K) q + P(z | F) (1 - q)}.$$

For $z = +$ this is $qs/(qs + (1 - q)(1 - t)) = q^+$; for $z = -$ it is $q(1 - s)/(q(1 - s) + (1 - q)t) = q^-$. Likewise $P(Z = z | Y = y) = P(z | K) q + P(z | F) (1 - q)$, which is $P(+)$ or $P(-)$ of Definition 2 at $q = q(y)$. So, conditionally on $Y = y$, verification acts on the belief $q(y)$ exactly as in Definition 2 with the same (s, t) , and every quantity built from Definitions 2–3 (R_v, V, χ, S) is evaluated at $q(y)$. These are functions of q , of (s, t) and of the constants $C_K, C_F, H_K, H_F, c_v, r_K, \tau$ only, and (s, t) do not depend on the order-flow law

by Assumption 2; so they are the same functions and set whatever that law is. Nothing in the argument uses the law of Y beyond $q(y)$; in particular it holds when $P(Y | K) = P(Y | F)$, in which case $q(y)$ equals the prior. Without Assumption 2 the factorisation in the first display fails and the update would depend on y through $P(z | y, c)$. $\square$

Proof of Lemma 3. Step 1 (homogeneous form). For an outcome z write $\alpha_z = P(z | K)q$ and $\beta_z = P(z | F)(1 - q)$, so $P(z) = \alpha_z + \beta_z$ and, by Definition 3, $P(z)R(q^z) = \min\{\alpha_z C_K + \beta_z C_F, \alpha_z H_K + \beta_z H_F\}$. The difference of the two arguments is $(\alpha_z H_K + \beta_z H_F) - (\alpha_z C_K + \beta_z C_F) = \alpha_z a - \beta_z b$, so

$$P(z)R(q^z) = (\alpha_z H_K + \beta_z H_F) - \max\{0, \alpha_z a - \beta_z b\}.$$

Here $\alpha_+ a - \beta_+ b = qsa - (1 - q)(1 - t)b = G^+(q)$ and $\alpha_- a - \beta_- b = q(1 - s)a - (1 - q)tb = -G^-(q)$. Summing over z and using $\alpha_+ + \alpha_- = q$, $\beta_+ + \beta_- = 1 - q$, with $L_\ell(q) = qH_K + (1 - q)H_F$:

$$P(+)R(q^+) + P(-)R(q^-) = L_\ell(q) - \max\{0, G^+(q)\} - \max\{0, -G^-(q)\}.$$

By the proof of Lemma 1, $R(q) = L_\ell(q) - \max\{0, \Delta(q)\}$ with $\Delta(q) = qa - (1 - q)b$, and summing the two outcome terms gives $G^+(q) - G^-(q) = \Delta(q)$. Therefore

$$V(q) = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}. \quad (1)$$

Step 2 (zeros and the value at q^).* G^+ is affine with slope $sa + (1 - t)b > 0$ (as $a, b > 0$ and $s > 1 - t \geq 0$); G^- is affine with slope $-(tb + (1 - s)a) < 0$ (as $t > 1 - s \geq 0$). Their unique zeros are q_L and q_U of the statement, and the formulas give $0 \leq q_L$ and $q_U \leq 1$. At q^* , $\Delta(q^*) = 0$, so $G^+(q^*) = G^-(q^*) = q^*sa - (1 - q^*)(1 - t)b$. Since $q^*a = (1 - q^*)b = ab/(a + b)$, this equals $(s + t - 1)ab/(a + b)$, which is > 0 by Assumption 3. As G^+ is increasing with $G^+(q^*) > 0$, $q_L < q^*$; as G^- is decreasing with $G^-(q^*) > 0$, $q^* < q_U$. This proves (i).

Step 3 (the tent). Evaluate (1) on three ranges.

- $q \leq q_L$: $G^+ \leq 0$ and (as $q < q_U$) $G^- > 0$, so $G^+ - G^- < 0$; every term of (1) vanishes and $V = 0$.
- $q \geq q_U$: $G^- \leq 0$ and (as $q > q_L$) $G^+ > 0$, so $G^+ - G^- > 0$ and $V = G^+ - G^- - (G^+ - G^-) = 0$.

- $q_L < q < q_U$: $G^+ > 0$ and $G^- > 0$, so $V = G^+ - \max\{0, G^+ - G^-\} = \min\{G^+, G^-\} > 0$.

Hence $V = \max\{0, \min\{G^+, G^-\}\}$ on $[0, 1]$, and $V > 0$ exactly on (q_L, q_U) . On that interval, $G^+ < G^-$ iff $\Delta < 0$ iff $q < q^*$. So $V = G^+$ (affine, strictly increasing) on $[q_L, q^*]$ and $V = G^-$ (affine, strictly decreasing) on $[q^*, q_U]$; V is continuous, and its unique maximiser is q^* , with $V(q^*) = (s+t-1)ab/(a+b) = (s+t-1)aq^*$. This proves (iii) and (iv).

Step 4 (posterior form of (ii)). For $q \in (0, 1)$, $P(+)>0$ and $P(-)>0$ because $s>0$ and $t>0$. Using $(a+b)q^* = b$,

$$(a+b)P(+)(q^+ - q^*) = (a+b)qs - bP(+) = qsa - (1-q)(1-t)b = G^+(q),$$

and in the same way $(a+b)P(-)(q^* - q^-) = G^-(q)$. So $G^+(q) > 0 \iff q^+ > q^*$ and $G^-(q) > 0 \iff q^- < q^*$; hence $q \in (q_L, q_U) \iff q^- < q^* < q^+$. At $q \in \{0, 1\}$, $V = 0$, and every defined posterior equals q , so the posteriors cannot straddle $q^* \in (0, 1)$: both sides of (ii) are false. $\square$

Proof of Theorem 1. Assumption 4 is what makes Definition 3 the correct verify-first risk: under K (probability q) verification adds $c_v + r_K\tau$ to the terminal cost, under F it adds c_v , the terminal costs and (s, t) are unchanged by the delay, and the terminal action is chosen optimally on the posterior, giving $c_v + qr_K\tau + \sum_z P(z)R(q^z)$. So $R_v < R$ iff $V > \chi$, with $\chi(q) = c_v + qr_K\tau$ affine and nondecreasing.

Step 1 ($S \subseteq (q_L, q_U)$). By Assumption 3, $\chi(q) \geq 0$ on $[0, 1]$. By Lemma 3, $V = 0$ outside (q_L, q_U) , so $V > \chi$ fails there.

Step 2 (S is an interval). Let $\varphi(q) = \min\{G^+(q), G^-(q)\} - \chi(q)$ on $[0, 1]$. It is the minimum of two affine functions minus an affine function, hence concave and continuous. On (q_L, q_U) , $V = \min\{G^+, G^-\}$ (Lemma 3), and outside it $\min\{G^+, G^-\} \leq 0$ (since $G^+ \leq 0$ for $q \leq q_L$ and $G^- \leq 0$ for $q \geq q_U$), so $\varphi \leq -\chi \leq 0$ there. Hence $S = \{q : \varphi(q) > 0\}$, a strict superlevel set of a continuous concave function: an open (relative to $[0, 1]$) interval, possibly empty. Since $\varphi(0) = \min\{-(1-t)b, tb\} - c_v \leq 0$ and $\varphi(1) = \min\{sa, -(1-s)a\} - c_v - r_K\tau \leq 0$, the endpoints are excluded and S is an open interval in $(0, 1)$.

Step 3 (non-emptiness criterion). On $[q^*, q_U]$, $\varphi = G^- - \chi$ is strictly decreasing (G^- strictly decreasing, χ nondecreasing). On $[q_L, q^*]$, $\varphi = G^+ - \chi$ is affine with slope $\sigma = sa + (1-t)b - r_K\tau$. If $\sigma \leq 0$, φ is nonincreasing on $[q_L, q_U]$, so $\sup_{(q_L, q_U)} \varphi \leq \varphi(q_L) = -\chi(q_L) \leq 0$ and $S = \emptyset$. If $\sigma > 0$,

φ is strictly increasing then strictly decreasing, so $\max_{[q_L, q_U]} \varphi = \varphi(q^*) = V(q^*) - \chi(q^*)$. Either way, $S \neq \emptyset$ implies $\varphi(q^*) > 0$ (in the first case S is empty; in the second the maximum is at q^*). Conversely $\varphi(q^*) > 0$ gives $q^* \in S$. Hence $S \neq \emptyset \iff V(q^*) > c_v + q^* r_K \tau$, and then $q^* \in S$. Also, $S \neq \emptyset$ forces $\sigma > 0$: if $\sigma \leq 0$ we showed $S = \emptyset$.

Step 4 (thresholds). Assume $S \neq \emptyset$. Then $\sigma > 0$, and the left endpoint q_1 of S is the zero of the affine function $G^+ - \chi = \sigma q - ((1-t)b + c_v)$ on $[q_L, q^*)$ if $\varphi(q_L) < 0$, or q_L itself if $\varphi(q_L) = 0$; in both cases $q_1 = (c_v + (1-t)b)/\sigma$ (when $\varphi(q_L) = 0$, $\chi(q_L) = 0$, so $c_v = 0$ and $q_L r_K \tau = 0$, and the formula returns q_L). The right endpoint q_2 is the zero of $G^- - \chi = tb - c_v - q(tb + (1-s)a + r_K \tau)$ on $(q^*, q_U]$, i.e. $q_2 = (tb - c_v)/(tb + (1-s)a + r_K \tau)$; its denominator is $\geq tb > 0$. As $G^+(q_1) = \chi(q_1) \geq 0$ we get $q_1 \geq q_L$, and as $G^-(q_2) = \chi(q_2) \geq 0$ we get $q_2 \leq q_U$; $q_1 < q^* < q_2$ because $q^* \in S$ and S is open.

Step 5 (policy). For $q \notin S$ the optimal action is the passive one of Lemma 1; since $q_1 < q^* < q_2$, this is leave on $q \leq q_1$ and contain on $q \geq q_2$ when $S \neq \emptyset$. When $S = \emptyset$ it is leave for $q < q^*$ and contain for $q > q^*$; at $q = q^*$ both terminal actions have cost $R(q^*)$ (Lemma 1), so the convention (leave) is optimal. For $q \in S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so by Lemma 1 the optimal terminal action after verification is contain on $Z = +$ and leave on $Z = -$.

Step 6 ($c_v = 0, r_K \tau = 0$). Then $\chi \equiv 0$ and $S = \{V > 0\} = (q_L, q_U)$ by Lemma 3(ii); the threshold formulas reduce to q_L and q_U . $\square$

Proof of Theorem 2. Throughout, only τ varies; V, q^*, q_L, q_U do not depend on τ (Assumption 4: s, t and the loss table are delay-free).

(i) *Nesting.* For $q \in [0, 1]$, $\chi_\tau(q) = c_v + q r_K \tau$ is nondecreasing in τ because $q r_K \geq 0$. So $V(q) > \chi_{\tau'}(q)$ implies $V(q) > \chi_\tau(q)$ for $\tau < \tau'$, i.e. $S(\tau') \subseteq S(\tau)$. Now let $r_K > 0$ and $V(q^*) > c_v$. By (ii) below, whose proof does not use (i), $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$; so on $[0, \tau^*)$ the end points $q_1(\tau), q_2(\tau)$ are given by the formulas of Theorem 1 with positive denominators, and for $\tau \geq \tau^*$ the region is empty and Theorem 1's convention $q_1 = q_2 = q^*$ applies, so no monotonicity is claimed there. On $[0, \tau^*)$: $q_2 = (tb - c_v)/(tb + (1-s)a + r_K \tau)$ has a positive numerator (because $q_2 > q^* > 0$ and the denominator is positive) and a denominator strictly increasing in τ , so it is strictly decreasing. $q_1 = (c_v + (1-t)b)/(sa + (1-t)b - r_K \tau)$ has a positive denominator strictly decreasing in τ and a nonnegative numerator, zero iff $c_v = 0$ and $t = 1$; so q_1 is strictly increasing on $[0, \tau^*)$ unless $c_v = 0$ and $t = 1$, where $q_1 = 0$ on $[0, \tau^*)$.

(ii) *Critical delay.* By Theorem 1, $S(\tau) \neq \emptyset \iff V(q^*) > c_v + q^* r_K \tau$. With $r_K > 0$ and $q^* > 0$, this is $\tau < (V(q^*) - c_v)/(q^* r_K) = \tau^*$, which is > 0 when $V(q^*) > c_v$. By Lemma 3(iv), $V(q^*)/q^* = (s + t - 1)a$, giving the second expression for τ^* .

(iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $V(q^*) \leq c_v + q^* r_K \tau$ for every $\tau \geq 0$, so $S(\tau) = \emptyset$ by Theorem 1. If $r_K = 0$, χ does not involve τ , so neither does S .

(iv) *How the interval vanishes.* By hypothesis $r_K > 0$ and $V(q^*) > c_v$, so τ^* is defined by (ii); let $\tau \uparrow \tau^*$. The denominator of q_2 is continuous and positive at τ^* , so $q_2(\tau) \rightarrow q_2(\tau^*)$, the zero of $G^- - \chi_{\tau^*}$. At τ^* , $G^-(q^*) = V(q^*) = \chi_{\tau^*}(q^*)$, so that zero is q^* (it is unique: $G^- - \chi_{\tau^*}$ is strictly decreasing). For q_1 : at τ^* the left-piece function $G^+ - \chi_{\tau^*}$ vanishes at q^* and equals $-\chi_{\tau^*}(q_L) = -(c_v + q_L r_K \tau^*)$ at q_L . If $c_v > 0$ or $t < 1$ (the latter giving $q_L > 0$), this is < 0 , so the slope $\sigma(\tau^*) = sa + (1 - t)b - r_K \tau^*$ is > 0 , the formula for q_1 is continuous at τ^* , and $q_1(\tau) \rightarrow$ the zero of $G^+ - \chi_{\tau^*}$, which is q^* . If $c_v = 0$ and $t = 1$, then $q_L = 0$, $G^+(q) = qsa$ and $\chi_{\tau}(q) = qr_K \tau$, and $\tau^* = sa/r_K$ (as $s + t - 1 = s$). For $\tau < \tau^*$, $G^+ - \chi_{\tau} = q(sa - r_K \tau) > 0$ on $(0, q^*]$, so $S(\tau) = (0, q_2(\tau))$ with $q_2(\tau) \downarrow q^*$ by (i) and the limit above; at τ^* , $G^+ - \chi_{\tau^*} \equiv 0$ on $[0, q^*]$ and $S(\tau^*) = \emptyset$ by (ii). So the interval drops from $(0, q^*)$ (in the limit) to $\emptyset$. $\square$

Proof of Corollary 1. Reduction to a constant belief. By Assumption 2 the joint law of (M, Y, Z) is $P(M = m) P_m^Y(dy) P(Z = z | m)$. Let $\mu = P_K^Y = P_F^Y$. Then Y has law μ and is independent of (M, Z) , and $P(M = m, Z = z) = P(m) P(z | m)$. Write $P(z, K) = P(z | K) p$ and $P(z, F) = P(z | F) (1 - p)$ as in Definition 3. Conditioning on $Y = y$ in Definition 4 gives $J(\delta, \pi) = \int \ell(y) \mu(dy)$ with

$$\begin{aligned} \ell(y) = & (1 - \delta(y)) \left[\pi_0(y) (pC_K + (1 - p)C_F) \right. \\ & \left. + (1 - \pi_0(y)) (pH_K + (1 - p)H_F) \right] \\ & + \delta(y) \left[c_v + pr_K \tau + \sum_{z \in \{+, -\}} \left(\pi_z(y) (P(z, K)C_K + P(z, F)C_F) \right. \right. \\ & \left. \left. + (1 - \pi_z(y)) (P(z, K)H_K + P(z, F)H_F) \right) \right]. \end{aligned}$$

Here c_v and $r_K \tau$ are paid only on verification, the latter only under K (Assumption 4). Each expression of the form $\pi x + (1 - \pi) x'$ with $\pi \in [0, 1]$ is at least $\min\{x, x'\}$. So the first bracket is at least $R(p)$ (Definition 1). The

second is at least $c_v + pr_K\tau + \sum_z \min\{P(z, K)C_K + P(z, F)C_F, P(z, K)H_K + P(z, F)H_F\} = R_v(p)$, by the reading of Definition 3. Hence, for every y ,

$$\ell(y) \geq (1 - \delta(y))R(p) + \delta(y)R_v(p) \geq \min\{R(p), R_v(p)\},$$

and $J \geq \min\{R(p), R_v(p)\}$. The constant policy that verifies iff $R_v(p) < R(p)$ and takes the minimizing terminal action in each case attains equality. So

$$R_{Y,v}^* = \min\{R(p), R_v(p)\}, \quad R_Y^* = R(p)$$

(the second with $\delta \equiv 0$). The attaining policy depends on p only, not on y . When $p \in S$ it is the verify branch of Theorem 1 at $q = p$: contain on $Z = +$ and leave on $Z = -$, because $p^- < q^* < p^+$ by Lemma 3(ii). Otherwise it is the passive rule of Lemma 1.

(i) By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = pC_K + (1 - p)H_F$. Subtracting it from both terms of $R(p)$ gives $\text{CoA}^* = \min\{(1 - p)(C_F - H_F), p(H_K - C_K)\} = \min\{pa, (1 - p)b\}$. This is > 0 since $a, b > 0$ and $0 < p < 1$.

(ii) By Definition 3, $R_v(p) = \chi(p) + R(p) - V(p)$. So

$$\text{CoA}_v^* = \min\{R(p), R_v(p)\} - R_{\text{oracle}} = \text{CoA}^* - (V(p) - \chi(p))^+.$$

With $G^\pm$ as in Lemma 3,

$$\begin{aligned} \text{CoA}^* - E(p) &= \min\{pa - p(1 - s)a - (1 - p)(1 - t)b, \\ &\quad (1 - p)b - p(1 - s)a - (1 - p)(1 - t)b\} - \chi(p) \\ &= \min\{G^+(p), G^-(p)\} - \chi(p). \end{aligned}$$

By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$, and $\chi(p) \geq 0$ by Assumption 3. Suppose $\min\{G^+, G^-\} > \chi$. Then $V = \min\{G^+, G^-\}$ and $(V - \chi)^+ = \min\{G^+, G^-\} - \chi$. Otherwise $V \leq \max\{0, \chi\} = \chi$, and $(V - \chi)^+ = 0 = (\min\{G^+, G^-\} - \chi)^+$. In both cases $(V - \chi)^+ = (\text{CoA}^* - E)^+$, so $\text{CoA}_v^* = \text{CoA}^* - (\text{CoA}^* - E)^+ = \min\{\text{CoA}^*, E(p)\}$. (Direct check: $E(p)$ is the cost of the verify branch above minus R_{oracle} . That is c_v , plus $r_K\tau$ under K , a miss a with probability $1 - s$ under K , and a false containment b with probability $1 - t$ under F .)

(iii) By (ii), $\text{CoA}_v^* < \text{CoA}^* \iff V(p) > \chi(p) \iff p \in S$ (Definition 3). By Theorem 1, $S \neq \emptyset \iff V(q^*) > c_v + q^*r_K\tau$, and then $S = (q_1, q_2)$. If $r_K > 0$ and $V(q^*) > c_v$, Theorem 2(ii) turns the non-emptiness condition into $\tau < \tau^*$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for all $\tau \geq 0$ by Theorem 2(iii).

(iv) The reduction is $g(p) = (V(p) - \chi(p))^+$, and $V - \chi \leq -\chi \leq 0$ outside (q_L, q_U) . On $[q^*, q_U]$, $V - \chi = G^- - \chi$ is strictly decreasing (Lemma 3(iii); χ is nondecreasing). On $[q_L, q^*]$, $V - \chi$ is affine. It is ≤ 0 at q_L and equals $V(q^*) - \chi(q^*)$ at q^* . So its maximum over $[q_L, q^*]$ is attained at an endpoint, and $\max_p g(p) = (V(q^*) - \chi(q^*))^+ = (V(q^*) - c_v - q^* r_K \tau)^+$, attained at q^* . When this maximum is positive, g is strictly increasing on $[q_L, q^*]$ (an affine function going from ≤ 0 to a positive value) wherever it is positive, and strictly decreasing on $[q^*, q_U]$. So the maximizer is unique.

(v) Since $\text{CoA}^* > 0$ by (i), $\text{CoA}_v^* = 0$ iff $E(p) = 0$. Each of the four terms of $E(p)$ is nonnegative (Assumption 3). With $p \in (0, 1)$ and $a, b > 0$, all four vanish iff $c_v = 0$, $r_K \tau = 0$, $s = 1$ and $t = 1$. $\square$

Proof of Corollary 2. Fix order-flow laws P_K^Y, P_F^Y and $p \in (0, 1)$. By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = p C_K + (1 - p) H_F$.

(i) *Verify-everything.* With $\delta \equiv 1$, $\pi_+ \equiv 1$, $\pi_- \equiv 0$, the integrand of J in Definition 4 does not depend on Y : the operator pays c_v , pays $r_K \tau$ under K (Assumption 4), contains on $Z = +$ and leaves on $Z = -$. By Assumption 2, Z given M has the law of Definition 2 whatever Y , so

$$J = c_v + p r_K \tau + p (s C_K + (1 - s) H_K) + (1 - p) ((1 - t) C_F + t H_F).$$

Subtracting R_{oracle} and using $H_K - C_K = a$, $C_F - H_F = b$,

$$J - R_{\text{oracle}} = c_v + p r_K \tau + p (1 - s) a + (1 - p) (1 - t) b = E(p).$$

(ii) *Bounds.* The constant passive policies $\pi_0 \equiv 1$ (always contain) and $\pi_0 \equiv 0$ (always leave) cost $p C_K + (1 - p) C_F$ and $p H_K + (1 - p) H_F$, that is $R_{\text{oracle}} + (1 - p) b$ and $R_{\text{oracle}} + p a$. So $R_Y^* \leq R_{\text{oracle}} + \min\{p a, (1 - p) b\}$, which is the first bound. Every passive policy is a verify-first policy ($\delta \equiv 0$), so $R_{Y,v}^* \leq R_Y^*$; by (i), $R_{Y,v}^* \leq R_{\text{oracle}} + E(p)$. Hence $\text{CoA}_v^* \leq \min\{\text{CoA}^*, E(p)\}$, and the first bound gives the second inequality.

(iii) *Equality in the confounded case.* If $P_K^Y = P_F^Y$, Corollary 1(i) gives $\text{CoA}^* = \min\{p a, (1 - p) b\}$ and Corollary 1(ii) gives $\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\}$, so every inequality in (ii) is an equality. Together with (ii), which holds for every pair of order-flow laws, the suprema over pairs of order-flow laws of CoA^* and of CoA_v^* are $\min\{p a, (1 - p) b\}$ and $\min\{p a, (1 - p) b, E(p)\}$, and both are attained by any confounded pair (e.g. $P_K^Y = P_F^Y$ a point mass).

(iv) *Gap.* Write $m = \min\{p a, (1 - p) b\}$. The gap between the two suprema is $m - \min\{m, E(p)\} = (m - E(p))^+$. In the confounded case

$m = \text{CoA}^*$ and Corollary 1(ii) gives $\min\{m, E(p)\} = m - (V(p) - \chi(p))^+$, so the gap is $(V(p) - \chi(p))^+$. It is positive if and only if $V(p) > \chi(p)$, that is (Definition 3) if and only if $p \in S$. $\square$

Proof of Corollary 3. Fix $q \in [0, 1]$, $\tau \geq 0$ and $\theta \in [0, 1]$. Assumption 3 gives $s + t > 1$ with $s, t \leq 1$, so $s > 0$ and $t > 0$; hence $\sigma_\theta \in [s, 1] \subset (0, 1]$.

Four plans. Each minimum in $R_v^\theta(q)$ is over the two terminal actions after outcome z , so $R_v^\theta(q)$ is the least of the costs of the four plans (action after $Z = +$, action after $Z = -$); randomizing between actions cannot do better, since the cost is linear in the probabilities. Write $L = qH_K + (1 - q)H_F$ and $C = qC_K + (1 - q)C_F$, so $R(q) = \min\{L, C\}$, and let $G^\pm$ be as in Lemma 3. For each plan, the saving $R(q) - \text{cost}$ is:

- *contain, contain:* cost $c_v + C + q r_K \tau$, saving $\leq -c_v - q r_K \tau \leq 0$;
- *leave, leave:* cost $c_v + L + \theta q r_K \tau$, saving ≤ 0 ;
- *leave after +, contain after -:* the terminal part is $T' = P(-, K)C_K + P(-, F)C_F + P(+, K)H_K + P(+, F)H_F$, with $L - T' = q(1 - s)a - (1 - q)tb = -G^-(q)$ and $C - T' = -qsa + (1 - q)(1 - t)b = -G^+(q)$; the saving is at most $\min\{-G^+(q), -G^-(q)\}$. Now $G^+(q) = q(sa + (1 - t)b) - (1 - t)b$ is increasing and vanishes at q_L , and $G^-(q) = tb - q(tb + (1 - s)a)$ is decreasing and vanishes at q_U (both slopes are non-zero since $a, b, s, t > 0$); as $q_L < q_U$ (Lemma 3(i)), at every q one of $G^+(q) \geq 0$, $G^-(q) \geq 0$ holds, so this saving is ≤ 0 ;
- *contain after +, leave after -:* the terminal part is $T = P(+, K)C_K + P(+, F)C_F + P(-, K)H_K + P(-, F)H_F$, with $L - T = qsa - (1 - q)(1 - t)b = G^+(q)$ and $C - T = (1 - q)tb - q(1 - s)a = G^-(q)$; the delay part is $P(+, K)r_K\tau + \theta P(-, K)r_K\tau = \sigma_\theta q r_K \tau$. The saving is $\min\{G^+(q), G^-(q)\} - c_v - \sigma_\theta q r_K \tau$.

(i) $q \in S_\theta(\tau)$ if and only if some plan has a strictly positive saving. The first three never do, so $q \in S_\theta(\tau)$ if and only if $\min\{G^+(q), G^-(q)\} > c_v + \sigma_\theta q r_K \tau$, and then the fourth plan is the unique optimal one. The right-hand side is ≥ 0 ($c_v \geq 0$, $r_K \geq 0$, Assumption 3), so by Lemma 3 ($V = \max\{0, \min\{G^+, G^-\}\}$) the condition is $V(q) > c_v + \sigma_\theta q r_K \tau$. This is the condition $V(q) > \chi(q)$ of Definition 3 with r_K replaced by $r'_K = \sigma_\theta r_K \geq 0$. The proofs of Theorems 1 and 2 use Assumption 4 only to

write the verification cost as $c_v + q r_K \tau$ with $r_K \geq 0$, and the optimal post-verification action there is the fourth plan, as here; so both theorems hold under Assumption 4 $_\theta$ with r'_K in place of r_K .

(ii) σ_θ is nondecreasing in θ (its derivative is $1 - s \geq 0$), with $\sigma_1 = 1$. For fixed q and τ , $c_v + \sigma_\theta q r_K \tau$ is nondecreasing in θ because $q r_K \tau \geq 0$; by (i), $S_{\theta'}(\tau) \subseteq S_\theta(\tau)$ for $\theta \leq \theta'$. At $\theta = 1$ the condition is that of Definition 3, so $S_1(\tau) = S(\tau)$.

(iii) Let $r_K > 0$ and $V(q^*) > c_v$. Then $r'_K = \sigma_\theta r_K > 0$, and Theorem 2(ii) with r'_K (valid by (i)) gives $S_\theta(\tau) \neq \emptyset$ if and only if $\tau < (V(q^*) - c_v)/(q^* \sigma_\theta r_K) = \tau^*/\sigma_\theta$. Since $s \leq \sigma_\theta \leq 1$, $\tau^* \leq \tau^*/\sigma_\theta \leq \tau^*/s$, and $\tau_\theta^* = \tau^*$ if and only if $\sigma_\theta = 1$, that is $(1 - \theta)(1 - s) = 0$: $\theta = 1$ or $s = 1$. $\square$

Proof of Corollary 4. Verify-first risk. Under Assumption 4 $_F$, verifying adds $c_v + r_K \tau$ to the terminal cost under K and $c_v + r_F \tau$ under F ; neither depends on the terminal action, so the optimal action after each outcome is the one of Definition 3, and the verify-first risk is $c_v + q r_K \tau + (1 - q) r_F \tau + P(+R(q^+) + P(-)R(q^-) = R_v(q) + (1 - q) r_F \tau$. Hence verifying is strictly cheaper iff $V(q) > \chi_F(q)$. χ_F is affine and $\chi_F \geq 0$ on $[0, 1]$, since $\chi_F(0) = c_v + r_F \tau \geq 0$ and $\chi_F(1) = c_v + r_K \tau \geq 0$ (Assumption 3, $r_F \geq 0$).

(i) As in Steps 1–2 of the proof of Theorem 1, which use only that the verification cost is affine and ≥ 0 : $S \subseteq (q_L, q_U)$, and $S = \{q : \varphi(q) > 0\}$ with $\varphi = \min\{G^+, G^-\} - \chi_F$ concave and continuous, so S is an open interval in $(0, 1)$, possibly empty. On $[q_L, q^*]$, $\varphi = G^+ - \chi_F$ is affine with slope $\sigma_1 = sa + (1 - t)b - (r_K - r_F)\tau$; on $[q^*, q_U]$, $\varphi = G^- - \chi_F$ is affine with slope $-\sigma_2$, $\sigma_2 = tb + (1 - s)a + (r_K - r_F)\tau$. At the ends, $\varphi(q_L) = -\chi_F(q_L) \leq 0$ and $\varphi(q_U) = -\chi_F(q_U) \leq 0$. If $\sigma_1 \leq 0$, then φ is nonincreasing on $[q_L, q^*]$ and, being concave, also on $[q^*, q_U]$, so $\varphi \leq \varphi(q_L) \leq 0$ on $[q_L, q_U]$ and $S = \emptyset$. If $\sigma_2 \leq 0$, symmetrically φ is nondecreasing on $[q_L, q_U]$, $\varphi \leq \varphi(q_U) \leq 0$ and $S = \emptyset$. If $\sigma_1, \sigma_2 > 0$, φ is maximal at q^* . So in every case $S \neq \emptyset$ iff $\varphi(q^*) > 0$, i.e. iff $V(q^*) > \chi_F(q^*) = c_v + \bar{r} \tau$, and then $\sigma_1, \sigma_2 > 0$ and $q^* \in S$. The left end q_1 is the unique zero of $G^+ - \chi_F = \sigma_1 q - ((1 - t)b + c_v + r_F \tau)$ in $[q_L, q^*)$ (it lies there because $\varphi(q_L) \leq 0 < \varphi(q^*)$), which is the stated formula; the right end q_2 is the unique zero of $G^- - \chi_F = tb - c_v - r_F \tau - \sigma_2 q$ in $(q^*, q_U]$, the stated formula. $q_1 < q^* < q_2$ because $q^* \in S$ and S is open. The policy follows as in Step 5 of the proof of Theorem 1: outside S the passive rule of Lemma 1; on $S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so contain on $Z = +$ and leave on $Z = -$.

(ii) For fixed q , $\chi_F(q) = c_v + \tau(q r_K + (1 - q) r_F)$ is nondecreasing in τ ,

so $V(q) > \chi_F(q)$ at τ' implies it at $\tau < \tau'$: $S(\tau') \subseteq S(\tau)$. By (i), $S(\tau) \neq \emptyset$ iff $V(q^*) - c_v > \bar{r}\tau$. If $V(q^*) \leq c_v$ this fails for every $\tau \geq 0$ ($\bar{r} \geq 0$). If $\bar{r} > 0$ and $V(q^*) > c_v$, it is $\tau < (V(q^*) - c_v)/\bar{r} = \tau_F^*$.

(iii) If $r_K > 0$, then $\tau^* = (V(q^*) - c_v)/(q^*r_K)$ (Theorem 2(ii)), so $\tau_F^* = \tau^* q^* r_K / \bar{r}$. Since $\bar{r} - q^* r_K = (1 - q^*) r_F \geq 0$ with $1 - q^* > 0$ (Lemma 1), $\tau_F^* \leq \tau^*$, with equality iff $r_F = 0$. $\square$

Proof of Remark 1. Step 1 of the proof of Lemma 3 used only Definitions 1–3 and holds without Assumptions 1 and 3: $V = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}$, with $G^+ - G^- = \Delta = qa - (1 - q)b$.

(a) If $s + t = 1$, then $P(z | K) = P(z | F)$ for both outcomes, so $q^z = q$ whenever $P(z) > 0$. Then $\sum_z P(z)R(q^z) = R(q)$ (the $P(z) = 0$ term is zero by Definition 3), so $V \equiv 0$. As $\chi \geq 0$, $V > \chi$ never holds and $S = \emptyset$.

(b1) If $a \leq 0 \leq b$, then $\Delta(q) = qa - (1 - q)b \leq 0$ (the difference of the proof of Lemma 1) on $[0, 1]$, so leaving is weakly optimal everywhere and $R = L_\ell$ is affine. Each outcome term satisfies $\alpha_z a - \beta_z b \leq 0$ (as $\alpha_z, \beta_z \geq 0$), so $\sum_z P(z)R(q^z) = \sum_z (\alpha_z H_K + \beta_z H_F) = L_\ell(q) = R(q)$ and $V \equiv 0$. The case $b \leq 0 \leq a$ is symmetric ($\Delta \geq 0$, contain weakly optimal everywhere, each $\alpha_z a - \beta_z b \geq 0$, so $\sum_z P(z)R(q^z) = L_c(q) = R(q)$). In both, $S = \emptyset$ since $\chi \geq 0$. These two sign patterns cover every failure of Assumption 1 except $a < 0$ and $b < 0$: if one of a, b is ≤ 0 and the other is ≥ 0 , one of the two patterns holds, and the only remaining way to have $a \leq 0$ or $b \leq 0$ is $a < 0$ and $b < 0$.

(b2) Let $a < 0$ and $b < 0$. Then $a + b < 0$ and $q^* = b/(a + b) \in (0, 1)$, and $\Delta(q) = (a + b)(q - q^*)$ is > 0 (contain strictly cheaper) for $q < q^*$ and < 0 for $q > q^*$. Exchange the names of the two terminal actions: the new costs are $C'_K = H_K$, $C'_F = H_F$, $H'_K = C_K$, $H'_F = C_F$, so $a' = H'_K - C'_K = -a > 0$, $b' = C'_F - H'_F = -b > 0$ and $b'/(a' + b') = q^*$: the renamed problem satisfies Assumption 1. R is the minimum over the two actions (Definition 1) and each term $P(z)R(q^z)$ is the minimum of the two joint costs (Definition 3), so R , V and S do not depend on the names; χ involves no terminal action; Assumptions 2–4 do not refer to the names (Assumption 4 charges $r_K \tau$ whichever terminal action is taken). In Definition 4, renaming maps a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$ to $(\delta, 1 - \pi_0, 1 - \pi_+, 1 - \pi_-)$, a bijection that preserves J , and R_{oracle} is symmetric in the two actions. Hence every result stated under Assumption 1 holds for the renamed problem with $(a', b') = (-a, -b)$; read back in the original names, contain and leave are exchanged in each policy, which gives the rule stated in (b2).

(c) Take $c_v < 0$, $t < 1$ (so $q_L > 0$ under Assumption 1), $r_K > 0$, and $\tau > 0$ small enough that $V(q^*) > q^* r_K \tau$. Choose $c_v \in (-q_L r_K \tau, 0)$. Then $q^* \in S$, because $V(q^*) > q^* r_K \tau > \chi(q^*)$. At $q = 0$, $V(0) = 0 > c_v = \chi(0)$, so $0 \in S$. At $q = q_L$, $V(q_L) = 0 < c_v + q_L r_K \tau = \chi(q_L)$, so $q_L \notin S$. Since $0 < q_L < q^*$, S is not an interval. $\square$

Proof of Lemma 4. Write $f_K = f_K^c$, $f_F = f_F^c$, $f_p = pf_K + (1-p)f_F$ (the ν -density of P_p), $q = q_p$, and $g = (V - \chi)^+$. Let $N = \{y : f_p(y) = 0\}$; then $P_p(N) = 0$, and q is a measurable map from N^c to $[0, 1]$.

Properties of g . By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$ with $G^\pm$ affine, so V is continuous on $[0, 1]$ and $0 \leq V \leq V(q^*)$; $\chi(q) = c_v + qr_K \tau$ is affine and ≥ 0 (Assumption 3). Hence g is continuous, $0 \leq g \leq V \leq V(q^*)$, and $g > 0$ exactly on S (Definition 3). At $q = 0$: $G^+(0) = -(1-t)b \leq 0$, so $V(0) = 0$ and $g(0) = 0$.

(i) *Saving.* Fix a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$. By Assumption 2 the joint law of (M, Y, Z) has density $P(M = m) f_m(y) P(Z = z | m)$ with respect to (counting $\times \nu \times$ counting). Integrating out m and z in Definition 4,

$$J(\delta, \pi) = \int_{N^c} f_p(y) \ell_{q(y)}(y) \nu(dy),$$

where $\ell_q(y)$ is the expression $\ell(y)$ of the proof of Corollary 1 with p replaced by q , since $pf_K(y)/f_p(y) = q(y)$ and $(1-p)f_F(y)/f_p(y) = 1 - q(y)$. The pointwise argument of that proof gives $\ell_{q(y)}(y) \geq (1 - \delta(y)) R(q(y)) + \delta(y) R_v(q(y)) \geq \min\{R, R_v\}(q(y))$, and, for passive policies ($\delta \equiv 0$), $\ell_{q(y)}(y) \geq R(q(y))$. All costs are bounded, so every integral is finite. Equality holds for the following measurable policy: $\delta(y) = \mathbb{1}[q(y) \in S]$ (S is an interval, Theorem 1); $\pi_0(y) = \mathbb{1}[q(y) > q^*]$ (Lemma 1); $\pi_+ \equiv 1$, $\pi_- \equiv 0$. The last two are optimal on $\{q(y) \in S\}$: there $V(q) > \chi(q) \geq 0$, so $G^+(q) > 0$ and $G^-(q) > 0$ (Lemma 3). Since $G^+(q) = P(+, K)a - P(+, F)b$, containing is strictly better than leaving after $Z = +$. Since $G^-(q) = P(-, F)b - P(-, K)a$, leaving is strictly better after $Z = -$. With $\delta \equiv 0$ and the same π_0 , equality holds in the passive bound. Hence

$$R_{Y,v}^* = \mathbb{E}_p[\min\{R, R_v\}(q(Y))], \quad R_Y^* = \mathbb{E}_p[R(q(Y))].$$

By Definition 3, $R - R_v = V - \chi$, so $R - \min\{R, R_v\} = (V - \chi)^+ = g$, and $R_Y^* - R_{Y,v}^* = \mathbb{E}_p[g(q(Y))] = D_c(p)$. Subtracting R_{oracle} from both infima gives $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p)$. For the bounds: $D_c \geq 0$ since $g \geq 0$. $\text{CoA}_{v,c}^* \geq 0$ because, for every policy, the integrand of J is at least

$\min\{L(\text{contain}, M), L(\text{leave}, M)\}$ ($c_v \geq 0$, $r_K \tau \geq 0$), and the expectation of that is R_{oracle} ; so $D_c \leq \text{CoA}_c^*$. Finally $g \leq V(q^*) \mathbb{1}_S$ gives $D_c(p) \leq V(q^*) m_c(p)$.

Pointwise limit under F . Let y satisfy $f_F(y) > 0$ (a set of full P_F^c -measure). Then $y \notin N$ for every $p \in (0, 1)$, and

$$0 \leq q_p(y) \leq \frac{p f_K(y)}{(1-p) f_F(y)} \longrightarrow 0 \quad (p \rightarrow 0),$$

because $f_K(y) < \infty$.

(ii) One line suffices: by (i), $D_c(p) \leq \text{CoA}_c^*(p)$, and $\text{CoA}_c^*(p) \leq p a$, because always leaving is a passive policy with expected cost $p H_K + (1-p) H_F$ while $R_{\text{oracle}} = p C_K + (1-p) H_F$ under Assumption 1 (this is Corollary 2(ii)); hence $0 \leq D_c(p) \leq p a \rightarrow 0$. The pointwise argument below gives the same limit and is kept because (iii) and (iv) use it. Split $D_c(p) = p \mathbb{E}_K[g(q_p(Y))] + (1-p) \mathbb{E}_F[g(q_p(Y))]$, where $\mathbb{E}_m$ is the expectation under P_m^c (the event N is null under both). The first term is at most $p V(q^*) \rightarrow 0$. In the second, $g(q_p(y)) \rightarrow g(0) = 0$ for P_F^c -almost every y , by the pointwise limit and continuity of g . The integrand is bounded by the constant $V(q^*)$, so dominated convergence gives $\mathbb{E}_F[g(q_p(Y))] \rightarrow 0$. Hence $D_c(p) \rightarrow 0$.

(iii) If $S = \emptyset$, then $m_c \equiv 0$. Let $S = (q_1, q_2) \neq \emptyset$. By Theorem 1 the denominator of q_1 is positive, so $q_1 > 0$ iff $c_v + (1-t)b > 0$, i.e. iff $c_v > 0$ or $t < 1$ ($b > 0$, Assumption 1). Suppose $q_1 > 0$. Then $m_c(p) \leq p + (1-p) P_F^c(q_p(Y) > q_1)$. For P_F^c -almost every y , $q_p(y) \rightarrow 0 < q_1$, so $\mathbb{1}_{[q_p(y) > q_1]} \rightarrow 0$. Dominated convergence (bound 1) gives $P_F^c(q_p(Y) > q_1) \rightarrow 0$, hence $m_c(p) \rightarrow 0$.

(iv) Let $S \neq \emptyset$, $c_v = 0$ and $t = 1$. Then $q_1 = 0$ (Theorem 1) and $S = (0, q_2)$ with $q_2 > q^* > 0$. Write $m_c(p) = p P_K^c(q_p(Y) \in S) + (1-p) P_F^c(q_p(Y) \in S)$; the first term is at most $p \rightarrow 0$. Take y with $f_F(y) > 0$. If $f_K(y) = 0$, then $q_p(y) = 0 \notin S$ for every p . If $f_K(y) > 0$, then $q_p(y) > 0$, and $q_p(y) < q_2$ for all small enough p by the pointwise limit, so $q_p(y) \in S$ eventually. Thus $\mathbb{1}_{[q_p(y) \in S]} \rightarrow \mathbb{1}_{[f_K(y) > 0]}$ for P_F^c -almost every y . Dominated convergence gives $P_F^c(q_p(Y) \in S) \rightarrow P_F^c(f_K(Y) > 0)$, and $(1-p) \rightarrow 1$.

When the limit is positive. Let $A = \{y : f_K(y) > 0\}$, so $P_K^c(A) = 1$. If $P_F^c(A) = 0$, then A separates the two laws and they are mutually singular. Conversely, if $P_K^c(B) = 1$ and $P_F^c(B) = 0$ for some measurable B , then $\int_{B^c} f_K d\nu = P_K^c(B^c) = 0$, so $f_K = 0$ ν -almost everywhere on B^c ; since $P_F^c \ll \nu$, $P_F^c(A \cap B^c) = 0$, and $P_F^c(A) \leq P_F^c(B) + P_F^c(A \cap B^c) = 0$. Hence the

limit $P_F^c(A)$ is zero if and only if P_K^c and P_F^c are mutually singular. That the saving still vanishes is (ii). $\square$

Proof of Proposition 1. Law of the log-likelihood ratio. Let $\Delta = \mu_K - \mu_F$ and $\bar{\mu} = (\mu_K + \mu_F)/2$. The Gaussian densities are positive everywhere, and

$$\Lambda(y) = \log \frac{f_K(y)}{f_F(y)} = \Delta^\top \Sigma_Y^{-1}(y - \bar{\mu}).$$

Under $M = K$, $y - \bar{\mu} \sim \mathcal{N}(\Delta/2, \Sigma_Y)$, so $\Lambda \sim \mathcal{N}(d^2/2, d^2)$. Under $M = F$, $\Lambda \sim \mathcal{N}(-d^2/2, d^2)$. By Definition 5, $\text{logit } q_p(Y) = \ell + \Lambda(Y)$, which is strictly increasing in Λ . For $-\infty \leq u < v \leq \infty$, with Λ continuous ($d > 0$),

$$P_K^c(\text{logit } q_p(Y) \in (u, v)) = P\left(\frac{d^2}{2} + dW \in (u - \ell, v - \ell)\right) = \Pi_K(u, v),$$

with $W \sim \mathcal{N}(0, 1)$, and likewise $P_F^c(\cdot) = \Pi_F(u, v)$. Open and half-open intervals have the same probability.

Mass. $q_p(Y) \in S = (q_1, q_2)$ iff $\text{logit } q_p(Y) \in (\ell_1, \ell_2)$ (with the stated conventions when $q_1 = 0$ or $q_2 = 1$; $q_p(Y) \in (0, 1)$ always). Since $P_p = pP_K^c + (1 - p)P_F^c$, this gives m_c .

Saving. On S , $V > \chi \geq 0$, so $V = \min\{G^+, G^-\}$ (Lemma 3). $G^+ - G^- = qa - (1 - q)b$ is ≤ 0 iff $q \leq q^*$. So on $(q_1, q^*]$, $g := (V - \chi)^+ = G^+ - \chi = \alpha_1 q - \beta_1$; on $[q^*, q_2)$, $g = G^- - \chi = \beta_2 - \alpha_2 q$. Outside S , $g = 0$. Expanding $G^\pm$ from Lemma 3 and $\chi = c_v + q r_K \tau$ gives the stated α_i, β_i . Theorem 1 gives $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$. For any Borel $A \subseteq (0, 1)$,

$$\mathbb{E}_p[q_p(Y) \mathbb{1}_A(q_p(Y))] = \int \frac{p f_K}{f_p} \mathbb{1}_A(q_p) f_p d\nu = p P_K^c(q_p(Y) \in A).$$

Take $A_1 = (q_1, q^*]$ and $A_2 = (q^*, q_2)$, and write $P_p = pP_K^c + (1 - p)P_F^c$. Then

$$\mathbb{E}_p[(\alpha_1 q_p - \beta_1) \mathbb{1}_{A_1}(q_p)] = p(\alpha_1 - \beta_1) P_K^c(A_1) - (1 - p) \beta_1 P_F^c(A_1),$$

$$\mathbb{E}_p[(\beta_2 - \alpha_2 q_p) \mathbb{1}_{A_2}(q_p)] = p(\beta_2 - \alpha_2) P_K^c(A_2) + (1 - p) \beta_2 P_F^c(A_2),$$

where $P_m^c(A)$ abbreviates $P_m^c(q_p(Y) \in A)$. Substituting $P_m^c(A_1) = \Pi_m(\ell_1, \ell^*)$ and $P_m^c(A_2) = \Pi_m(\ell^*, \ell_2)$, and summing, gives D_c .

Confounded limit $d \rightarrow 0$. For fixed finite $u < v$ and $\ell \notin \{u, v\}$, $(v - \ell)/d \mp d/2 \rightarrow \pm\infty$ according to the sign of $v - \ell$. So $\Pi_K(u, v), \Pi_F(u, v) \rightarrow \mathbb{1}[u < \ell < v]$, and the same holds with an infinite endpoint. At $\ell = u$ or

$\ell = v$ the corresponding Φ term tends to $\Phi(0) = 1/2$. Hence $m_c(p) \rightarrow \mathbb{K}[\ell_1 < \ell < \ell_2] = \mathbb{K}[p \in S]$ for $p \notin \{q_1, q_2\}$. For D_c , take $p \in (q_1, q^*)$: the limit is $p(\alpha_1 - \beta_1) - (1 - p)\beta_1 = \alpha_1 p - \beta_1 = g(p)$. Take $p \in (q^*, q_2)$: the limit is $\beta_2 - \alpha_2 p = g(p)$. Take $p \notin [q_1, q_2]$: the limit is $0 = g(p)$. At $p = q^*$ each of the two pieces tends to one half of its bracket. The brackets are $\alpha_1 q^* - \beta_1$ and $\beta_2 - \alpha_2 q^*$, and both equal $g(q^*)$ by continuity of $V - \chi$. So the limit is $g(q^*)$. At $p = q_1 > 0$ the limit is $\frac{1}{2}(\alpha_1 q_1 - \beta_1) = 0 = g(q_1)$. At $p = q_2 < 1$ it is $\frac{1}{2}(\beta_2 - \alpha_2 q_2) = 0 = g(q_2)$. These are the confounded values $\mathbb{K}[p \in S]$ and $(V - \chi)^+(p)$ of Corollary 1(ii)–(iii). $\square$

Proof of Theorem 3. Write $L_c(q) = q C_K + (1 - q) C_F$ and $L_\ell(q) = q H_K + (1 - q) H_F$ for the expected costs of containing and leaving, so that $R = \min\{L_c, L_\ell\}$ and $L_\ell - L_c = q a - (1 - q) b$. Pointwise in z , put $A_z(q) = q g_K(z) C_K + (1 - q) g_F(z) C_F$ and $B_z(q) = q g_K(z) H_K + (1 - q) g_F(z) H_F$; then $B_z - A_z = q a g_K(z) - (1 - q) b g_F(z)$ and $\min\{A_z, B_z\} = B_z - (B_z - A_z)^+$.

(i) Since $\int B_z d\mu = L_\ell(q)$, $R_Z(q) = L_\ell(q) - \int (q a g_K - (1 - q) b g_F)^+ d\mu$; and $R(q) = L_\ell(q) - (L_\ell - L_c)^+(q) = L_\ell(q) - (q a - (1 - q) b)^+$. Subtracting gives the explicit formula. For each z , $q \mapsto \min\{A_z(q), B_z(q)\}$ is concave (a minimum of two affine functions), and it is bounded by $(g_K + g_F)(z) \max\{|C_K|, |C_F|, |H_K|, |H_F|\}$, which is μ -integrable; so R_Z is concave and, by dominated convergence, continuous on $[0, 1]$. $V_Z \geq 0$ because $f(\cdot)^+ \geq (f \cdot)^+$ and $\int (q a g_K - (1 - q) b g_F) d\mu = q a - (1 - q) b$. At $q = 0$ the integrand of the explicit formula is $(-b g_F)^+ = 0$ and $(-b)^+ = 0$ (Assumption 1), so $V_Z(0) = 0$; at $q = 1$ both terms equal a , so $V_Z(1) = 0$. On $[0, q^*]$, $R = L_\ell$ (Lemma 1), which is affine, so $V_Z = L_\ell - R_Z$ is convex there; on $[q^*, 1]$, $R = L_c$ and $V_Z = L_c - R_Z$ is convex.

(ii) Let $f = V_Z - \chi$. It is continuous and, since χ is affine, convex on $[0, q^*]$ and on $[q^*, 1]$, with $f(0) = -c_v \leq 0$ and $f(1) = -c_v - r_K \tau \leq 0$. The set $N_1 = \{q \in [0, q^*] : f(q) \leq 0\}$ is a sublevel set of a convex function on an interval, hence an interval; it is closed by continuity and contains 0. If $f(q^*) \leq 0$, N_1 contains 0 and q^* , so $N_1 = [0, q^*]$; likewise $N_2 = \{q \in [q^*, 1] : f(q) \leq 0\} = [q^*, 1]$, and $S_Z = \emptyset$. If $f(q^*) > 0$, then $N_1 = [0, q_1]$ for some $q_1 \in [0, q^*)$ and $N_2 = [q_2, 1]$ for some $q_2 \in (q^*, 1]$, so $S_Z = [0, 1] \setminus (N_1 \cup N_2) = (q_1, q_2)$. For the policy: outside S_Z verifying is not strictly cheaper (ties to the passive action, Definition 3), and the passive action of Lemma 1 attains R . Inside S_Z verifying is strictly cheaper, and after observing z the cost $\min\{A_z, B_z\}$ is attained by containing iff $A_z < B_z$, i.e. iff $q a g_K(z) > (1 - q) b g_F(z)$ (ties to leave); integrating gives $R_Z(q)$,

hence $R_{v,Z}(q)$.

(iii) At $q^* = b/(a+b)$, $q^*a = (1-q^*)b = ab/(a+b)$, so the second term of the explicit formula vanishes and the first is $\frac{ab}{a+b} \int (g_K - g_F)^+ d\mu = \frac{ab}{a+b} \text{TV}_Z$, because $\int (g_K - g_F) d\mu = 0$ gives $\int (g_K - g_F)^+ d\mu = \frac{1}{2} \int |g_K - g_F| d\mu$. Also $ab/(a+b) = a q^*$.

(iv) V_Z does not depend on τ (Assumption 4 as read here), and $\chi_\tau(q) = c_v + q r_K \tau$ is nondecreasing in τ ; so $V_Z > \chi_{\tau'}$ implies $V_Z > \chi_\tau$ for $\tau < \tau'$, which is the nesting. By (ii), $S_Z(\tau) \neq \emptyset$ iff $V_Z(q^*) > c_v + q^* r_K \tau$. If $V_Z(q^*) \leq c_v$ this fails for all $\tau \geq 0$. If $r_K = 0$, $\chi_\tau = c_v$ for every τ , so $S_Z(\tau) = S_Z(0)$, and the condition reads $V_Z(q^*) > c_v$ whatever τ , i.e. $\text{TV}_Z a q^* > c_v$ by (iii): non-empty for every τ iff $V_Z(q^*) > c_v$, empty for every τ otherwise. If $r_K > 0$ and $V_Z(q^*) > c_v$, it is $\tau < (V_Z(q^*) - c_v)/(q^* r_K)$, which by (iii) equals $(\text{TV}_Z a - c_v/q^*)/r_K = \tau_Z^*$.

(v) For the binary check, $\text{TV}_Z = \frac{1}{2}(|s - (1-t)| + |(1-s) - t|) = |s + t - 1|$, which is $s + t - 1$ when $s + t > 1$ (Assumption 3); then (iii) is Lemma 3(iv) and (iv) is Theorem 2(ii). For a binary reading A , $Z_A = \mathbb{K}[Z \in A]$ has laws $(P_K^Z(A), P_K^Z(A^c))$ and $(P_F^Z(A), P_F^Z(A^c))$, so $R_{Z_A}(q) = \min\{\int_A A_z, \int_A B_z\} + \min\{\int_{A^c} A_z, \int_{A^c} B_z\} \geq \int_A \min\{A_z, B_z\} + \int_{A^c} \min\{A_z, B_z\} = R_Z(q)$; hence $V_A \leq V_Z$ and $S_A \subseteq S_Z$ (the cost χ is the same). Its Youden index is $P_K^Z(A) + P_F^Z(A^c) - 1 = P_K^Z(A) - P_F^Z(A) = \int_A (g_K - g_F) d\mu \leq \int (g_K - g_F)^+ d\mu = \text{TV}_Z$, with equality iff A contains $\{g_K > g_F\}$ and no set of positive μ -measure on which $g_K < g_F$, in particular for A^* . If the reading is informative (Youden index > 0), (iii)–(iv) apply to it with TV_{Z_A} equal to its Youden index, giving $\tau_A^* = (\text{TV}_{Z_A} a - c_v/q^*)/r_K \leq \tau_Z^*$, with equality for A^* . $\square$

Proof of Proposition 2. Fix $\hat{q}$ and write $\eta = \eta(\hat{q})$. Conditional on $\hat{Q} = \hat{q}$, the cause is K with probability η , and, because Z is conditionally independent of $\hat{Q}$ given M , $Z = +$ with probability s under K and $1 - t$ under F . Outside S the two rules take the same action, so the difference is 0.

For $\hat{q} \in S$, the targeted rule verifies; by Assumption 4 its conditional expected cost is

$$c_v + \eta r_K \tau + \eta (s C_K + (1-s) H_K) + (1-\eta) ((1-t) C_F + t H_F).$$

If $\hat{q} \in B_1$, the passive plug-in rule leaves, at cost $\eta H_K + (1-\eta) H_F$. The difference is

$$\eta s a - (1-\eta)(1-t)b - c_v - \eta r_K \tau = \eta D_1 - (c_v + (1-t)b) = D_1(\eta - q_1),$$

using $q_1 = (c_v + (1 - t)b)/D_1$ (Theorem 1). If $\hat{q} \in B_2$, the passive plug-in rule contains, at cost $\eta C_K + (1 - \eta) C_F$, and the difference is

$$(1 - \eta)tb - \eta(1 - s)a - c_v - \eta r_K \tau = (tb - c_v) - \eta D_2 = D_2(q_2 - \eta),$$

using $q_2 = (tb - c_v)/D_2$. $D_1, D_2 > 0$ because $S \neq \emptyset$ (Theorem 1). This is (i), and (ii) follows since $D_1, D_2 > 0$.

(iii) Take expectations of (i) over $\hat{Q}$: $\mathbb{E}[\mathbb{K}_{B_1}(\hat{Q}) D_1(\eta(\hat{Q}) - q_1)] = D_1(\mathbb{E}[\mathbb{K}_{B_1}(\hat{Q}) \eta(\hat{Q})] - q_1 P(\hat{Q} \in B_1))$ and $\mathbb{E}[\mathbb{K}_{B_1}(\hat{Q}) \eta(\hat{Q})] = P(M = K, \hat{Q} \in B_1)$ by the definition of conditional probability; likewise on B_2 .

(iv) If $\eta(\hat{q}) = \hat{q}$: on $B_1 \subseteq (q_L, q^*]$ (Theorem 1: $q_L \leq q_1$), $D_1(\hat{q} - q_1) = \hat{q}sa - (1 - \hat{q})(1 - t)b - \chi(\hat{q}) = G^+(\hat{q}) - \chi(\hat{q}) = V(\hat{q}) - \chi(\hat{q})$ by Lemma 3 ($V = G^+$ on $[q_L, q^*]$), which is positive on B_1 ; on $B_2 \subseteq (q^*, q_U)$, $D_2(q_2 - \hat{q}) = G^-(\hat{q}) - \chi(\hat{q}) = V(\hat{q}) - \chi(\hat{q}) > 0$; and outside S , $(V - \chi)^+ = 0$. Taking expectations gives $\mathbb{E}[(V - \chi)^+(\hat{Q})]$, which is $D_c(p)$ of Definition 5 when $\hat{Q} = q_p(Y)$, as in Lemma 4(i). $\square$

Proof of Proposition 3. (i) By Lemma 3, $V_{s,t} = \max\{0, \min\{G^+, G^-\}\}$ with $G^+(q) = qsa - (1 - q)(1 - t)b$ and $G^-(q) = (1 - q)tb - q(1 - s)a$. For fixed $q \in [0, 1]$, $\partial G^+/\partial s = qa \geq 0$, $\partial G^+/\partial t = (1 - q)b \geq 0$, $\partial G^-/\partial s = qa \geq 0$, $\partial G^-/\partial t = (1 - q)b \geq 0$ (Assumption 1). Minimum and maximum preserve monotonicity, so $V_{s,t}(q)$ is nondecreasing in s and in t on $\{s + t > 1\}$, where Lemma 3 applies. Since χ does not depend on (s, t) , $V_{s_{lo}, t_{lo}}(q) > \chi(q)$ implies $V_{s(y), t(y)}(q) > \chi(q)$, i.e. $S_{s_{lo}, t_{lo}} \subseteq S_{s(y), t(y)}$ (note $s(y) + t(y) \geq s_{lo} + t_{lo} > 1$).

(ii) By Bayes' rule, $P(K | y, +) = q(y)s(y)/(q(y)s(y) + (1 - q(y))(1 - t(y)))$ and similarly for $Z = -$, which is Definition 2 at $q = q(y)$ with $(s(y), t(y))$.

(iii) Condition on $Y = y$ and write $\hat{q} = q(y)$. If $q \notin S_{s_{lo}, t_{lo}}$, the robust policy takes the passive action and the difference is 0. If $q \in S_{s_{lo}, t_{lo}}$, then $q \in S_{s(y), t(y)}$ by (i), so by Theorem 1 at characteristics $(s(y), t(y))$ (whose hypotheses hold, and whose posterior is that of (ii)) the post-verification actions "contain on +, leave on -" are optimal, and the conditional expected cost of verifying is $\chi(q) + R(q) - V_{s(y), t(y)}(q)$ (Definition 3). The passive action costs $R(q)$ (Lemma 1). The difference is $(V_{s(y), t(y)} - \chi)(q) \geq (V_{s_{lo}, t_{lo}} - \chi)(q) > 0$ by (i) and $q \in S_{s_{lo}, t_{lo}}$. The policy uses $s(\cdot)$, $t(\cdot)$ only through the event $q(Y) \in S_{s_{lo}, t_{lo}}$, which depends on the lower bounds alone. $\square$

Appendix B. Results used from Paper 1

The five statements below are restated from Paper 1 [1], a working paper, with the symbols of this paper; their proofs are in Paper 1. Throughout this paper, “Paper 1, Proposition 1” and “Paper 1, Proposition 2” refer to the last two of them and “Proposition 1”, “Proposition 2” and “Proposition 3” without that prefix to this paper’s results of Sections 4.8 and 4.9.

Theorem 1 of [1] (Indistinguishability criterion). Under the model and assumptions of Section 3 of [1], in its output-level injection model (its Definition D2: a latent linear-Gaussian state $x_{t+1} = Ax_t + w_t$, observed as $y_t = C\tilde{x}_t + \varepsilon_t$ with $\tilde{x}_t = x_t + \mathbb{I}[t \geq t_0] \theta_M b_M$, where C is the observation matrix, t_0 the onset time, $\theta_M \in \mathbb{R}$ the severity and b_M the injection direction of the cause $M \in \{K, F\}$), a compromise instance (K, θ_K, b_K) and a fault instance (F, θ_F, b_F) with a common onset induce the same law of the observation trajectory $(y_t)_{t \geq 0}$ if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

In this paper only the consequence of $(*)$ is used: the laws of the observation Y under K and under F coincide, $P_K^Y = P_F^Y$ (the confounded case of Definition 4).

Definition D6 of [1] (Loss table). The general two-action, two-cause loss table $L(a, M)$, $a \in \{\text{contain, leave}\}$, $M \in \{K, F\}$, is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave}, K) &= H_K, & L(\text{leave}, F) &= H_F, \end{aligned}$$

with $C_K, C_F, H_K, H_F \in \mathbb{R}$. The table of [1] (its Section 3) is the special case $H_F = 0$, in which its routine containment cost, disruption cost and harm are C_K , C_F and H_K . Its Proposition 1 is stated for that special case.

Definition D5 of [1] (Gaussian discriminability). If $Y \mid M \sim \mathcal{N}(\mu_M, \Sigma_Y)$, $M \in \{K, F\}$, with a common covariance $\Sigma_Y \succ 0$, let $\delta = \mu_K - \mu_F$ and $d = \sqrt{\delta^\top \Sigma_Y^{-1} \delta} \geq 0$ (the whitened mean separation).

Proposition 1 of [1] (Cost of ambiguity is bounded below under confounding). Suppose condition $(*)$ of Theorem 1 of [1] (restated above) holds (the confounded case, in which the laws of the observation Y under K and under F coincide), $0 < p < 1$, $C_F > 0$, and $H_K > C_K$ (harm strictly worse than routine containment), with $H_F = 0$ (Definition D6). Then the

best achievable expected cost among policies mapping the observations to $\{\text{contain, leave}\}$ is

$$R^* = \min\{p C_K + (1-p) C_F, p H_K\},$$

the oracle's expected cost is $R_{\text{oracle}} = p C_K$, and

$$\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1-p) C_F, p(H_K - C_K)\} > 0,$$

that is, $\text{CoA}^* = \min\{p a, (1-p) b\}$ with $a = H_K - C_K$ and $b = C_F - H_F = C_F$.

Proposition 2 of [1] (Total-variation extension of Proposition 1).

Let the loss table and prior be as in Proposition 1 of [1] ($H_F = 0$, $0 < p < 1$, $C_F > 0$, $H_K > C_K$), but do not assume (*): the laws P_K^Y , P_F^Y of the observation are arbitrary, with densities f_K, f_F with respect to a σ -finite measure ν dominating both, and $\text{TV}(P_K^Y, P_F^Y) = \frac{1}{2} \int |f_K - f_F| d\nu$. Set $\bar{a} = p(H_K - C_K) = p a$ and $\bar{b} = (1-p) C_F = (1-p) b$. Then

$$\text{CoA}^* = \int \min\{\bar{a} f_K, \bar{b} f_F\} d\nu = \frac{\bar{a} + \bar{b} - \int |\bar{a} f_K - \bar{b} f_F| d\nu}{2},$$

$$\min(\bar{a}, \bar{b}) (1 - \text{TV}(P_K^Y, P_F^Y)) \leq \text{CoA}^* \leq \min(\bar{a}, \bar{b}),$$

and, if a common surrogate law $\bar{P}$ satisfies $\text{TV}(P_K^Y, \bar{P}) \leq \epsilon_K$ and $\text{TV}(P_F^Y, \bar{P}) \leq \epsilon_F$,

$$\text{CoA}^* \geq \min(\bar{a}, \bar{b}) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

In particular $\text{CoA}^* \leq \min\{p a, (1-p) b\}$ for every pair of order-flow laws, which is the passive bound of Corollary 2(ii).

References

- [1] C. Mejrissi, Attack-fault confounding at the electronic trading gateway: Fragility under state feedback, the cost of ambiguity over the observation horizon, and a retrospective study of summary matching, working paper (2026).
- [2] S. Tariq, M. Baruwat Chhetri, S. Nepal, C. Paris, Alert fatigue in security operations centres: Research challenges and opportunities, ACM Computing Surveys 57 (9) (2025) 224:1–224:38. doi:10.1145/3723158.

- [3] J. Ghadermazi, A. Shah, S. Jajodia, A machine learning and optimization framework for efficient alert management in a cybersecurity operations center, *Digital Threats: Research and Practice* 5 (2) (2024) 1–23. doi:10.1145/3644393.
- [4] S. Wiefling, P. R. Jørgensen, S. Thunem, L. Lo Iacono, Pump up password security! Evaluating and enhancing risk-based authentication on a real-world large-scale online service, *ACM Transactions on Privacy and Security* 26 (1) (2023) 6:1–6:36. doi:10.1145/3546069.
- [5] S. G. Pauker, J. P. Kassirer, The threshold approach to clinical decision making, *New England Journal of Medicine* 302 (20) (1980) 1109–1117. doi:10.1056/NEJM198005153022003.
- [6] H. Raiffa, R. Schlaifer, *Applied Statistical Decision Theory*, Studies in Managerial Economics, Division of Research, Graduate School of Business Administration, Harvard University, Boston, 1961.
- [7] J. Hilden, P. Glasziou, Regret graphs, diagnostic uncertainty and Youden’s index, *Statistics in Medicine* 15 (10) (1996) 969–986. doi:10.1002/(SICI)1097-0258(19960530)15:10<969::AID-SIM211>3.0.CO;2-9.
- [8] N. Smits, A note on Youden’s J and its cost ratio, *BMC Medical Research Methodology* 10 (2010) 89. doi:10.1186/1471-2288-10-89.
- [9] M. H. DeGroot, Uncertainty, information, and sequential experiments, *The Annals of Mathematical Statistics* 33 (2) (1962) 404–419. doi:10.1214/aoms/1177704567.
- [10] L. Le Cam, *Asymptotic Methods in Statistical Decision Theory*, Springer, New York, 1986.
- [11] M. C. Mungan, The undertesting bias in clinical decision making: The role of test dichotomization, *Medical Decision Making* 46 (7) (2026) 819–824. doi:10.1177/0272989X261471178.
- [12] Y. Yao, Three-way decisions with probabilistic rough sets, *Information Sciences* 180 (3) (2010) 341–353. doi:10.1016/j.ins.2009.09.021.

- [13] E. Horvitz, G. Rutledge, Time-dependent utility and action under uncertainty, in: *Proceedings of the Seventh Conference on Uncertainty in Artificial Intelligence (UAI)*, 1991, pp. 151–158.
- [14] E. Horvitz, A. Seiver, Time-critical action: Representations and application, in: *Proceedings of the Thirteenth Conference on Uncertainty in Artificial Intelligence (UAI)*, 1997, pp. 250–257.
- [15] L. F. Schroeder, P. Rebman, P. Kasaie, E. Kenu, J. Zelner, D. W. Dowdy, A generalizable decision-making framework for selecting onsite versus send-out clinical laboratory testing, *Medical Decision Making* 44 (3) (2024) 307–319. doi:10.1177/0272989X241232666.
- [16] S. Eckermann, A. R. Willan, Time and expected value of sample information wait for no patient, *Value in Health* 11 (3) (2008) 522–526.
- [17] S. Alizamir, F. de Véricourt, P. Sun, Diagnostic accuracy under congestion, *Management Science* 59 (1) (2013) 157–171. doi:10.1287/mnsc.1120.1576.
- [18] M. Kremer, F. de Véricourt, Mismanaging diagnostic accuracy under congestion, *Operations Research* 71 (3) (2023) 895–916. doi:10.1287/opre.2022.2292.
- [19] G. E. Monahan, Optimal stopping in a partially observable Markov process with costly information, *Operations Research* 28 (6) (1980) 1319–1334. doi:10.1287/opre.28.6.1319.
- [20] D. Golovin, A. Krause, D. Ray, Near-optimal Bayesian active learning with noisy observations, in: *Advances in Neural Information Processing Systems* 23, 2010, pp. 766–774.
- [21] R. Xu, T. Zariphopoulou, L. Zhang, Decision making under costly sequential information acquisition: The paradigm of reversible and irreversible decisions, *arXiv preprint* (2026). arXiv:2401.00569v4.
- [22] E. Lehrer, T. Wang, The value of information in stopping problems, *Economic Theory* 78 (2) (2024) 619–648. doi:10.1007/s00199-023-01543-8.

- [23] A. Wald, J. Wolfowitz, Optimum character of the sequential probability ratio test, *The Annals of Mathematical Statistics* 19 (3) (1948) 326–339. doi:10.1214/aoms/1177730197.
- [24] W. Lee, W. Fan, M. Miller, S. J. Stolfo, E. Zadok, Toward cost-sensitive modeling for intrusion detection and response, *Journal of Computer Security* 10 (1–2) (2002) 5–22. doi:10.3233/JCS-2002-101-202.
- [25] K. Hammar, R. Stadler, Intrusion prevention through optimal stopping, *IEEE Transactions on Network and Service Management* 19 (3) (2022) 2333–2348. doi:10.1109/TNSM.2022.3176781.
- [26] K. Hammar, R. Stadler, Learning near-optimal intrusion responses against dynamic attackers, *IEEE Transactions on Network and Service Management* 21 (1) (2024) 1158–1177. doi:10.1109/TNSM.2023.3293413.
- [27] K. Hammar, T. Li, R. Stadler, Q. Zhu, Adaptive security response strategies through conjectural online learning, *IEEE Transactions on Information Forensics and Security* 20 (2025) 4055–4070. doi:10.1109/TIFS.2025.3558600.
- [28] C. Kruegel, W. Robertson, Alert verification: Determining the success of intrusion attempts, in: *Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2004)*, Lecture Notes in Informatics, Gesellschaft für Informatik, 2004.
- [29] Y. Kim, G. Dán, Q. Zhu, Human-in-the-loop cyber intrusion detection using active learning, *IEEE Transactions on Information Forensics and Security* 19 (2024) 8658–8672. doi:10.1109/TIFS.2024.3434647.
- [30] Y. Kim, G. Dán, Dynamic alert prioritization for real-time situational awareness: A hidden Markov model framework with active learning, *IEEE Transactions on Dependable and Secure Computing* 22 (6) (2025) 6913–6928. doi:10.1109/TDSC.2025.3592127.
- [31] F. Jalalvand, M. Baruwat Chhetri, S. Nepal, C. Paris, Adaptive alert prioritisation in security operations centres via learning to defer with human feedback, *Information Sciences* 760 (2027) 124128. doi:10.1016/j.ins.2026.124128.

- [32] F. Şahin, N. S. Mert, Dual-threshold conformal deferral for trustworthy security alert triage, *Electronics* 15 (18) (2026) 4084. doi:10.3390/electronics15184084.
- [33] K. Zhang, C. Keliris, T. Parisini, M. M. Polycarpou, Identification of sensor replay attacks and physical faults for cyber-physical systems, *IEEE Control Systems Letters* 6 (2022) 1178–1183. doi:10.1109/LCSYS.2021.3089674.
- [34] M. S. Ahsan, H. Wang, V. R. Motakatla, M. Zhu, P. Liu, Differentiation between faults and cyberattacks through combined analysis of cyberspace logs and physical measurements, *IEEE Transactions on Smart Grid* (2026). doi:10.1109/TSG.2026.3728276.
- [35] T. Roy, S. Sattarzadeh, S. Dey, Cyber-attack detection in socio-technical transportation systems exploiting redundancies between physical and social data, *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 54 (3) (2024) 1477–1488. doi:10.1109/TSMC.2023.3328319.
- [36] M. Saerens, P. Latinne, C. Decaestecker, Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure, *Neural Computation* 14 (1) (2002) 21–41. doi:10.1162/089976602753284446.
- [37] C. Elkan, The foundations of cost-sensitive learning, in: *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.
- [38] S. Höppner, B. Baesens, W. Verbeke, T. Verdonck, Instance-dependent cost-sensitive learning for detecting transfer fraud, *European Journal of Operational Research* 297 (1) (2022) 291–300. doi:10.1016/j.ejor.2021.05.028.
- [39] T. Silva Filho, H. Song, M. Perello-Nieto, R. Santos-Rodriguez, M. Kull, P. Flach, Classifier calibration: A survey on how to assess and improve predicted class probabilities, *Machine Learning* 112 (9) (2023) 3211–3260. doi:10.1007/s10994-023-06336-7.
- [40] B. Van Calster, A. J. Vickers, Calibration of risk prediction models: Impact on decision-analytic performance, *Medical Decision Making* 35 (2) (2015) 162–169. doi:10.1177/0272989X14547233.

- [41] G. N. Rothblum, G. Yona, Decision-making under miscalibration, in: 14th Innovations in Theoretical Computer Science Conference (ITCS 2023), Vol. 251 of Leibniz International Proceedings in Informatics (LIPIcs), Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023, pp. 92:1–92:20. doi:10.4230/LIPIcs.ITCS.2023.92.
- [42] D. Byrd, M. Hybinette, T. H. Balch, ABIDES: Towards high-fidelity multi-agent market simulation, in: Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS), 2020, pp. 11–22. doi:10.1145/3384441.3395986.

Table 3: Primary exploratory test per failure cell, with Paper 1’s cost matrix and, below it, with the measured cost of leaving a fault (sensitivity, decides nothing). Counts of plausible grid points predicted in, out and in the band; agreeing, disagreeing and unresolved points; the agreement share against $\bar{\sigma} = 0.9$ (the pre-stated criterion); beside it, post hoc, the predicted-in points that agree and their share of the decided predicted-in points (–: none decided). Last column: whether targeted verification beats the baseline at some implausible point. Designs as in Paper 1: Original design (V1), Study 5 (V2); C: contrast, SM: summary-matched regime.

Cost matrix	Design	Regime	p	Baseline	In	Out	Band	Agree	Disagree	Unres.	Share	In agree	In share	Agrees	Unres. cell	Impl. $U < 0$
Paper 1	V1	C	0.05	leave	120	67	341	181	2	4	0.989	114	0.983	yes	no	yes
	V2	C	0.05	leave	103	214	211	299	1	17	0.997	85	0.988	yes	no	yes
	V2	SM	0.05	leave	52	417	59	448	5	16	0.989	31	0.861	yes	no	yes
	V2	SM	0.10	leave	162	242	124	365	2	37	0.995	123	0.984	yes	no	yes
	V2	SM	0.20	leave	377	69	82	399	2	45	0.995	330	0.994	yes	no	yes
	V2	SM	0.30	leave	461	67	0	492	0	36	1.000	425	1.000	yes	no	yes
	pooled (decides nothing)							2184	12		0.995					
measured H_F	V1	C	0.05	leave	129	67	332	188	1	7	0.995	121	0.992	yes	no	yes
	V2	C	0.05	leave	103	213	212	297	1	18	0.997	84	0.988	yes	no	yes
	V2	SM	0.05	leave	52	416	60	447	5	16	0.989	31	0.861	yes	no	yes
	V2	SM	0.10	leave	162	242	124	365	3	36	0.992	123	0.976	yes	no	yes
	V2	SM	0.20	leave	377	69	82	397	2	47	0.995	328	0.994	yes	no	yes
	V2	SM	0.30	leave	461	67	0	492	0	36	1.000	425	1.000	yes	no	yes
	pooled (decides nothing)							2186	12		0.995					

Table 4: Policy comparison per cell: number of plausible grid points at which targeted verification is significantly worse / significantly better than each other policy (95% bootstrap intervals, 2000 resamples), with the median false-positive and false-negative rates of targeted verification and of verify-everything over the plausible points. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	vs always contain	vs always leave	vs passive	vs verify all	Targeted FPR, FNR	Verify-all FPR, FNR
V1	C	0.05	yes	0 / 528	0 / 150	0 / 201	0 / 528	0.010, 0.873	0.200, 0.150
V1	C	0.10	no	0 / 528	0 / 286	0 / 306	0 / 508	0.025, 0.637	0.200, 0.150
V1	C	0.20	no	0 / 528	0 / 528	0 / 334	0 / 439	0.087, 0.364	0.200, 0.150
V1	C	0.30	no	0 / 528	0 / 528	0 / 398	0 / 423	0.161, 0.226	0.200, 0.150
V1	C	0.50	no	0 / 528	0 / 528	0 / 307	0 / 439	0.342, 0.088	0.200, 0.150
V1	SM	0.05	no	0 / 528	0 / 528	0 / 84	0 / 528	0.008, 0.345	0.200, 0.150
V1	SM	0.10	no	0 / 528	0 / 528	5 / 162	0 / 528	0.023, 0.246	0.200, 0.150
V1	SM	0.20	no	0 / 528	0 / 528	0 / 318	0 / 527	0.049, 0.177	0.200, 0.150
V1	SM	0.30	no	0 / 528	0 / 528	0 / 382	0 / 526	0.083, 0.137	0.200, 0.150
V1	SM	0.50	no	0 / 528	0 / 528	0 / 277	0 / 520	0.142, 0.088	0.200, 0.150
Study 4	C	0.05	no	0 / 528	0 / 528	0 / 71	0 / 528	0.000, 0.024	0.200, 0.150
Study 4	C	0.10	no	0 / 528	0 / 528	1 / 0	0 / 528	0.000, 0.013	0.200, 0.150
Study 4	C	0.20	no	0 / 528	0 / 528	315 / 0	0 / 528	0.000, 0.005	0.200, 0.150
Study 4	C	0.30	no	0 / 528	0 / 528	175 / 0	0 / 528	0.000, 0.003	0.200, 0.150
Study 4	C	0.50	no	0 / 528	0 / 528	0 / 0	0 / 528	0.001, 0.003	0.200, 0.150
Study 4	SM	0.05	no	0 / 528	0 / 528	4 / 33	0 / 528	0.003, 0.248	0.200, 0.150
Study 4	SM	0.10	no	0 / 528	0 / 528	0 / 228	0 / 528	0.013, 0.175	0.200, 0.150
Study 4	SM	0.20	no	0 / 528	0 / 528	0 / 266	0 / 528	0.035, 0.118	0.200, 0.150
Study 4	SM	0.30	no	0 / 528	0 / 528	0 / 277	0 / 528	0.057, 0.088	0.200, 0.150
Study 4	SM	0.50	no	0 / 528	0 / 528	0 / 282	0 / 528	0.108, 0.056	0.200, 0.150
V2	C	0.05	yes	0 / 528	0 / 109	0 / 383	0 / 516	0.004, 0.904	0.200, 0.150
V2	C	0.10	no	0 / 528	0 / 371	0 / 245	0 / 496	0.013, 0.716	0.200, 0.150
V2	C	0.20	no	0 / 528	0 / 528	0 / 369	0 / 426	0.064, 0.401	0.200, 0.150
V2	C	0.30	no	0 / 528	0 / 528	0 / 350	0 / 399	0.158, 0.249	0.200, 0.150
V2	C	0.50	no	0 / 528	0 / 528	0 / 331	0 / 464	0.338, 0.098	0.200, 0.150
V2	SM	0.05	yes	0 / 528	0 / 31	0 / 31	0 / 506	0.000, 1.000	0.200, 0.150
V2	SM	0.10	yes	0 / 528	0 / 123	0 / 123	0 / 435	0.003, 0.987	0.200, 0.150
V2	SM	0.20	yes	0 / 528	0 / 330	0 / 341	0 / 259	0.051, 0.415	0.200, 0.150
V2	SM	0.30	yes	0 / 528	0 / 425	0 / 443	0 / 103	0.100, 0.200	0.200, 0.150
V2	SM	0.50	no	0 / 367	0 / 528	0 / 359	0 / 217	0.307, 0.047	0.200, 0.150
V3	C	0.05	no	0 / 528	0 / 528	0 / 21	0 / 528	0.000, 0.027	0.200, 0.150
V3	C	0.10	no	0 / 528	0 / 528	8 / 0	0 / 528	0.000, 0.014	0.200, 0.150
V3	C	0.20	no	0 / 528	0 / 528	0 / 259	0 / 528	0.001, 0.004	0.200, 0.150
V3	C	0.30	no	0 / 528	0 / 528	15 / 0	0 / 528	0.001, 0.001	0.200, 0.150
V3	C	0.50	no	0 / 528	0 / 528	64 / 0	0 / 528	0.003, 0.000	0.200, 0.150
V3	SM	0.05	no	0 / 528	0 / 528	0 / 95	0 / 528	0.005, 0.337	0.200, 0.150
V3	SM	0.10	no	0 / 528	0 / 528	0 / 225	0 / 528	0.014, 0.251	0.200, 0.150
V3	SM	0.20	no	0 / 528	0 / 528	0 / 288	0 / 528	0.034, 0.173	0.200, 0.150
V3	SM	0.30	no	0 / 528	0 / 528	0 / 255	0 / 528	0.054, 0.131	0.200, 0.150
V3	SM	0.50	no	0 / 528	0 / 528	0 / 337	0 / 526	0.102, 0.077	0.200, 0.150
V4	C	0.05	no	0 / 528	0 / 314	0 / 211	0 / 528	0.007, 0.719	0.200, 0.150
V4	C	0.10	no	0 / 528	0 / 443	0 / 293	0 / 522	0.030, 0.515	0.200, 0.150
V4	C	0.20	no	0 / 528	0 / 528	0 / 347	0 / 499	0.092, 0.290	0.200, 0.150
V4	C	0.30	no	0 / 528	0 / 528	0 / 459	0 / 498	0.157, 0.192	0.200, 0.150
V4	C	0.50	no	0 / 528	0 / 528	0 / 383	0 / 498	0.323, 0.083	0.200, 0.150
V4	SM	0.05	no	0 / 528	0 / 528	215 / 0	0 / 528	0.000, 0.001	0.200, 0.150
V4	SM	0.10	no	0 / 528	0 / 528	89 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.20	no	0 / 528	0 / 528	31 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.30	no	0 / 528	0 / 528	28 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.50	no	0 / 528	0 / 528	68 / 0	0 / 528	0.000, 0.000	0.200, 0.150

Table 5: Cells: Paper 1’s failure cells and best constant policy; cost of the passive rule (with its false-positive and false-negative rates), of the best constant policy and the passive cost of ambiguity with its interval; the corrected-posterior quantiles at α_{in} , the band level and α_{out} ; the median corrected logit posterior minus logit q^* . Costs in Paper 1’s monetary units. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	Best const.	Passive cost (FPR, FNR)	Best-const. cost	CoA* [CI]	$Q_{\alpha_{\text{in}}}$	Q_{band}	$Q_{\alpha_{\text{out}}}$	Δlogit
V1	C	0.05	yes	leave	68,724 (0.010, 0.960)	65,875	44,178 [39,809, 50,047]	0.106	0.172	0.427	-2.79
V1	C	0.10	no	leave	131,189 (0.020, 0.890)	131,750	82,095 [75,533, 89,711]	0.218	0.313	0.612	-2.01
V1	C	0.20	no	leave	239,604 (0.087, 0.657)	263,500	141,416 [126,925, 155,842]	0.427	0.531	0.789	-1.10
V1	C	0.30	no	leave	327,165 (0.213, 0.440)	395,250	179,884 [159,716, 201,108]	0.590	0.672	0.867	-0.46
V1	C	0.50	no	contain	422,810 (0.510, 0.137)	482,448	177,341 [156,249, 198,208]	0.798	0.861	0.950	0.56
V1	SM	0.05	no	leave	46,683 (0.010, 0.427)	65,875	22,136 [17,220, 28,033]	0.139	0.270	0.845	-5.15
V1	SM	0.10	no	leave	87,150 (0.023, 0.340)	131,750	38,056 [29,976, 47,183]	0.332	0.649	0.967	-4.25
V1	SM	0.20	no	leave	163,788 (0.080, 0.213)	263,500	65,600 [52,600, 79,864]	0.806	0.950	0.992	-3.06
V1	SM	0.30	no	leave	229,516 (0.113, 0.180)	395,250	82,235 [66,228, 97,963]	0.950	0.984	0.996	-2.03
V1	SM	0.50	no	contain	335,349 (0.170, 0.120)	482,448	89,880 [71,745, 108,415]	0.990	0.996	0.999	0.30
Study 4	C	0.05	no	leave	26,200 (0.000, 0.040)	65,875	1,653 [827, 2,617]	0.002	0.030	0.998	-8.81
Study 4	C	0.10	no	leave	50,471 (0.000, 0.017)	131,750	1,378 [276, 2,755]	0.061	0.989	1.000	-7.86
Study 4	C	0.20	no	leave	98,739 (0.000, 0.003)	263,500	551 [0, 1,653]	0.995	0.999	1.000	-6.54
Study 4	C	0.30	no	leave	148,108 (0.000, 0.003)	395,250	827 [0, 2,480]	0.999	1.000	1.000	-5.54
Study 4	C	0.50	no	contain	247,636 (0.003, 0.003)	482,448	2,168 [0, 5,713]	1.000	1.000	1.000	-0.67
Study 4	SM	0.05	no	leave	37,496 (0.000, 0.313)	65,875	12,949 [10,745, 15,154]	0.081	0.227	0.953	-5.88
Study 4	SM	0.10	no	leave	76,316 (0.017, 0.243)	131,750	27,222 [20,432, 34,795]	0.309	0.854	0.990	-4.86
Study 4	SM	0.20	no	leave	143,596 (0.050, 0.160)	263,500	45,408 [33,806, 56,849]	0.929	0.986	0.999	-3.57
Study 4	SM	0.30	no	leave	203,020 (0.073, 0.127)	395,250	55,739 [42,223, 69,546]	0.983	0.995	0.999	-2.39
Study 4	SM	0.50	no	contain	310,533 (0.147, 0.073)	482,448	65,064 [49,266, 80,848]	0.997	0.999	1.000	0.52
V2	C	0.05	yes	leave	66,963 (0.003, 0.990)	65,875	42,416 [40,502, 45,693]	0.096	0.125	0.206	-2.74
V2	C	0.10	no	leave	127,937 (0.003, 0.937)	131,750	78,843 [75,493, 82,789]	0.188	0.241	0.355	-1.96
V2	C	0.20	no	leave	242,328 (0.063, 0.727)	263,500	144,141 [130,884, 157,624]	0.376	0.452	0.629	-1.08
V2	C	0.30	no	leave	331,007 (0.200, 0.473)	395,250	183,726 [162,748, 203,895]	0.528	0.620	0.782	-0.48
V2	C	0.50	no	contain	439,196 (0.573, 0.140)	482,448	193,727 [171,831, 214,636]	0.765	0.820	0.895	0.50
V2	SM	0.05	yes	leave	65,875 (0.000, 1.000)	65,875	41,328 [41,328, 41,328]	0.066	0.075	0.100	-2.44
V2	SM	0.10	yes	leave	131,750 (0.000, 1.000)	131,750	82,656 [82,656, 82,656]	0.129	0.145	0.191	-1.69
V2	SM	0.20	yes	leave	266,190 (0.010, 0.993)	263,500	168,002 [164,201, 173,058]	0.250	0.278	0.346	-0.88
V2	SM	0.30	yes	leave	397,833 (0.090, 0.890)	395,250	250,552 [237,286, 264,376]	0.364	0.398	0.476	-0.33
V2	SM	0.50	no	contain	481,253 (0.983, 0.007)	482,448	235,785 [231,247, 241,112]	0.572	0.613	0.679	0.52
V3	C	0.05	no	leave	26,200 (0.000, 0.040)	65,875	1,653 [827, 2,617]	0.002	0.110	1.000	-9.10
V3	C	0.10	no	leave	50,471 (0.000, 0.017)	131,750	1,378 [276, 2,480]	0.208	0.995	1.000	-8.11
V3	C	0.20	no	leave	101,656 (0.003, 0.013)	263,500	3,468 [551, 7,098]	0.998	1.000	1.000	-6.83
V3	C	0.30	no	leave	148,387 (0.003, 0.000)	395,250	1,106 [0, 3,318]	1.000	1.000	1.000	-5.86
V3	C	0.50	no	contain	246,259 (0.003, 0.000)	482,448	790 [0, 2,370]	1.000	1.000	1.000	0.89
V3	SM	0.05	no	leave	42,992 (0.003, 0.410)	65,875	18,445 [15,291, 22,412]	0.074	0.211	0.946	-5.62
V3	SM	0.10	no	leave	82,653 (0.017, 0.320)	131,750	33,559 [26,583, 41,452]	0.271	0.774	0.991	-4.74
V3	SM	0.20	no	leave	154,293 (0.043, 0.240)	263,500	56,106 [45,053, 67,839]	0.851	0.978	0.998	-3.54
V3	SM	0.30	no	leave	214,860 (0.067, 0.183)	395,250	67,579 [53,772, 82,748]	0.974	0.995	0.999	-2.42
V3	SM	0.50	no	contain	325,282 (0.133, 0.117)	482,448	79,813 [63,263, 98,334]	0.997	0.999	1.000	0.18
V4	C	0.05	no	leave	63,780 (0.007, 0.877)	65,875	39,233 [35,680, 44,562]	0.121	0.199	0.370	-2.91
V4	C	0.10	no	leave	122,781 (0.030, 0.737)	131,750	73,687 [64,649, 83,465]	0.245	0.376	0.655	-2.07
V4	C	0.20	no	leave	222,714 (0.113, 0.493)	263,500	124,526 [108,809, 141,090]	0.513	0.673	0.852	-1.16
V4	C	0.30	no	leave	309,551 (0.230, 0.347)	395,250	162,270 [142,118, 183,259]	0.688	0.804	0.916	-0.48
V4	C	0.50	no	contain	411,770 (0.487, 0.123)	482,448	166,301 [145,576, 187,573]	0.887	0.933	0.973	0.66
V4	SM	0.05	no	leave	24,547 (0.000, 0.000)	65,875	0 [0, 0]	0.001	0.007	0.989	-10.58
V4	SM	0.10	no	leave	49,094 (0.000, 0.000)	131,750	0 [0, 0]	0.014	0.983	0.998	-9.64
V4	SM	0.20	no	leave	98,188 (0.000, 0.000)	263,500	0 [0, 0]	0.992	0.997	1.000	-8.28
V4	SM	0.30	no	leave	147,281 (0.000, 0.000)	395,250	0 [0, 0]	0.997	0.999	1.000	-7.07
V4	SM	0.50	no	contain	245,469 (0.000, 0.000)	482,448	0 [0, 0]	0.999	1.000	1.000	-1.51

Table 6: Verification mass, saving and recovered share per cell over the plausible grid points: number of points with a non-empty interval; m_c ; number of points at which the realised saving $\widehat{D}_c$ is significantly positive / significantly negative / not distinguishable from zero; medians of $\widehat{D}_c$ and of the plug-in saving $\widetilde{D}_c$; number of points with $\widetilde{D}_c$ significantly positive; ρ_c (n.r.: not reported, cost of ambiguity indistinguishable from zero). Descriptive; decides no claim. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	Non-empty	m_c median [min, max]	$\widehat{D}_c$: > 0 / < 0 / covers 0	$\widehat{D}_c$ median	$\widetilde{D}_c$ median	$\widetilde{D}_c > 0$	ρ_c median [min, max]
V1	C	0.05	yes	461	0.042 [0.000, 0.638]	201 / 0 / 327	2.875	1.732	453	0.065 [-0.019, 0.629]
V1	C	0.10	no	461	0.143 [0.000, 0.878]	306 / 0 / 222	7.521	6.934	461	0.092 [-0.044, 0.767]
V1	C	0.20	no	461	0.411 [0.000, 0.988]	334 / 0 / 194	21.918	23.246	461	0.155 [-0.016, 0.857]
V1	C	0.30	no	461	0.572 [0.000, 0.992]	398 / 0 / 130	37.200	34.915	461	0.207 [-0.011, 0.885]
V1	C	0.50	no	461	0.522 [0.000, 0.993]	307 / 0 / 221	26.033	31.820	461	0.147 [-0.043, 0.875]
V1	SM	0.05	no	461	0.052 [0.000, 0.222]	84 / 0 / 444	1.545	2.877	461	0.070 [-0.037, 0.590]
V1	SM	0.10	no	461	0.100 [0.000, 0.331]	162 / 5 / 361	2.683	6.118	461	0.070 [-0.070, 0.661]
V1	SM	0.20	no	461	0.133 [0.000, 0.459]	318 / 0 / 210	9.723	7.295	461	0.148 [-0.025, 0.745]
V1	SM	0.30	no	461	0.136 [0.000, 0.537]	382 / 0 / 146	12.348	7.282	461	0.150 [0.000, 0.772]
V1	SM	0.50	no	461	0.147 [0.000, 0.575]	277 / 0 / 251	9.975	8.381	461	0.111 [-0.009, 0.778]
Study 4	C	0.05	no	461	0.003 [0.000, 0.026]	71 / 0 / 457	370	171	454	0.224 [-0.114, 0.788]
Study 4	C	0.10	no	461	0.004 [0.000, 0.044]	0 / 1 / 527	0	234	409	0.000 [-0.414, 0.574]
Study 4	C	0.20	no	461	0.003 [0.000, 0.061]	0 / 315 / 213	-743	211	333	n.r.
Study 4	C	0.30	no	461	0.004 [0.000, 0.064]	0 / 175 / 353	-502	97	227	n.r.
Study 4	C	0.50	no	461	0.003 [0.000, 0.080]	0 / 0 / 528	503	256	206	n.r.
Study 4	SM	0.05	no	461	0.035 [0.000, 0.188]	33 / 4 / 491	-681	1.895	460	-0.053 [-0.178, 0.500]
Study 4	SM	0.10	no	461	0.066 [0.000, 0.270]	228 / 0 / 300	3.760	4.239	461	0.138 [-0.020, 0.663]
Study 4	SM	0.20	no	461	0.108 [0.000, 0.368]	266 / 0 / 262	6.784	5.888	461	0.149 [0.000, 0.746]
Study 4	SM	0.30	no	461	0.124 [0.000, 0.416]	277 / 0 / 251	7.539	7.171	461	0.135 [-0.008, 0.754]
Study 4	SM	0.50	no	461	0.121 [0.000, 0.438]	282 / 0 / 246	8.918	6.932	461	0.137 [-0.004, 0.793]
V2	C	0.05	yes	461	0.015 [0.000, 0.691]	383 / 0 / 145	1.725	332	436	0.041 [-0.001, 0.570]
V2	C	0.10	no	461	0.113 [0.000, 0.927]	245 / 0 / 283	4.124	4.986	461	0.052 [-0.014, 0.749]
V2	C	0.20	no	461	0.425 [0.000, 0.992]	369 / 0 / 159	22.825	23.695	461	0.158 [-0.011, 0.850]
V2	C	0.30	no	461	0.641 [0.000, 0.998]	350 / 0 / 178	37.162	39.694	461	0.202 [-0.004, 0.881]
V2	C	0.50	no	461	0.578 [0.000, 0.998]	331 / 0 / 197	33.588	36.927	461	0.173 [-0.036, 0.885]
V2	SM	0.05	yes	461	0.000 [0.000, 1.000]	31 / 0 / 497	0	0	141	0.000 [-0.026, 0.517]
V2	SM	0.10	yes	461	0.010 [0.000, 1.000]	123 / 0 / 405	0	303	336	0.000 [-0.025, 0.756]
V2	SM	0.20	yes	461	0.619 [0.000, 1.000]	341 / 0 / 187	20.831	17.007	453	0.124 [-0.004, 0.878]
V2	SM	0.30	yes	461	0.990 [0.000, 1.000]	443 / 0 / 85	82.796	70.951	461	0.330 [-0.012, 0.917]
V2	SM	0.50	no	461	0.879 [0.000, 1.000]	359 / 0 / 169	35.558	47.744	461	0.151 [0.000, 0.909]
V3	C	0.05	no	461	0.003 [0.000, 0.023]	21 / 0 / 507	178	177	438	0.107 [-0.038, 0.806]
V3	C	0.10	no	461	0.007 [0.000, 0.035]	0 / 8 / 520	-177	270	410	-0.128 [-0.681, 0.699]
V3	C	0.20	no	461	0.006 [0.000, 0.055]	259 / 0 / 269	2.049	600	445	0.591 [0.000, 0.825]
V3	C	0.30	no	461	0.006 [0.000, 0.065]	0 / 15 / 513	-257	372	345	n.r.
V3	C	0.50	no	461	0.003 [0.000, 0.080]	0 / 64 / 464	-258	51	186	n.r.
V3	SM	0.05	no	461	0.030 [0.000, 0.188]	95 / 0 / 433	0	1.428	461	0.000 [-0.072, 0.568]
V3	SM	0.10	no	461	0.062 [0.000, 0.278]	225 / 0 / 303	3.849	4.096	461	0.115 [-0.033, 0.681]
V3	SM	0.20	no	461	0.114 [0.000, 0.385]	288 / 0 / 240	8.430	6.174	461	0.150 [-0.022, 0.726]
V3	SM	0.30	no	461	0.134 [0.000, 0.448]	255 / 0 / 273	7.282	8.154	461	0.108 [-0.009, 0.770]
V3	SM	0.50	no	461	0.148 [0.000, 0.520]	337 / 0 / 191	14.026	8.236	461	0.176 [-0.013, 0.799]
V4	C	0.05	no	461	0.058 [0.000, 0.552]	211 / 0 / 317	2.119	3.400	461	0.054 [-0.052, 0.619]
V4	C	0.10	no	461	0.152 [0.000, 0.787]	293 / 0 / 235	8.750	8.187	461	0.119 [-0.026, 0.750]
V4	C	0.20	no	461	0.370 [0.000, 0.925]	347 / 0 / 181	20.055	21.561	461	0.161 [0.000, 0.843]
V4	C	0.30	no	461	0.478 [0.000, 0.968]	459 / 0 / 69	36.047	29.120	461	0.222 [0.000, 0.875]
V4	C	0.50	no	461	0.429 [0.000, 0.973]	383 / 0 / 145	28.192	24.189	461	0.170 [-0.011, 0.875]
V4	SM	0.05	no	461	0.000 [0.000, 0.035]	0 / 215 / 313	-117	33	215	n.r.
V4	SM	0.10	no	461	0.000 [0.000, 0.045]	0 / 89 / 439	-68	2	89	n.r.
V4	SM	0.20	no	461	0.000 [0.000, 0.039]	0 / 31 / 497	0	0	31	n.r.
V4	SM	0.30	no	461	0.000 [0.000, 0.031]	0 / 28 / 500	0	0	28	n.r.
V4	SM	0.50	no	461	0.000 [0.000, 0.028]	0 / 68 / 460	0	0	68	n.r.

Table 7: Critical delay $r_K\tau^*/a$ (Theorem 2) per sensitivity, specificity and cost of a check, in units of a/r_K (–: the interval is empty at every delay), and the number of (x, y) points with a non-empty interval. Closed forms agree with the numerical search at every point.

s	t	$x = 0.001$	$x = 0.01$	$x = 0.05$	$x = 0.1$	$x = 0.2$	$x = 0.5$	Non-empty of 36
0.7	0.7	0.399	0.390	0.350	0.300	0.200	–	24
0.7	0.8	0.499	0.490	0.450	0.400	0.300	–	25
0.7	0.9	0.599	0.590	0.550	0.500	0.400	0.100	31
0.7	0.99	0.689	0.680	0.640	0.590	0.490	0.190	33
0.8	0.7	0.499	0.490	0.450	0.400	0.300	–	25
0.8	0.8	0.599	0.590	0.550	0.500	0.400	0.100	31
0.8	0.9	0.699	0.690	0.650	0.600	0.500	0.200	33
0.8	0.99	0.789	0.780	0.740	0.690	0.590	0.290	35
0.9	0.7	0.599	0.590	0.550	0.500	0.400	0.100	31
0.9	0.8	0.699	0.690	0.650	0.600	0.500	0.200	33
0.9	0.9	0.799	0.790	0.750	0.700	0.600	0.300	35
0.9	0.99	0.889	0.880	0.840	0.790	0.690	0.390	35
0.99	0.7	0.689	0.680	0.640	0.590	0.490	0.190	33
0.99	0.8	0.789	0.780	0.740	0.690	0.590	0.290	35
0.99	0.9	0.889	0.880	0.840	0.790	0.690	0.390	35
0.99	0.99	0.979	0.970	0.930	0.880	0.780	0.480	35

Table 8: Test instances of each cause (n_K compromises, n_F faults; the split is by seed and every test seed contributes one episode of each cause) and convergence of the logistic fit, per design and regime. The fit is Paper 1’s, reproduced exactly by the run. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	n_K	n_F	Converged	Iterations
V1	C	300	300	yes	14
V1	SM	300	300	yes	18
Study 4	C	300	300	yes	17
Study 4	SM	300	300	yes	27
V2	C	300	300	yes	15
V2	SM	300	300	yes	12
V3	C	300	300	yes	12
V3	SM	300	300	yes	15
V4	C	300	300	yes	18
V4	SM	300	300	yes	10

Table 9: Expected cost per instance of every policy per cell, with Paper 1’s cost matrix: the constant policies, the passive rule and the oracle (which knows the cause) do not depend on the grid point; targeted verification and verify-everything are given as median [min, max] over the plausible grid points. Costs in Paper 1’s monetary units. Descriptive; decides no claim. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	Always contain	Always leave	Passive	Oracle	Targeted median [min, max]	Verify all median [min, max]
V1	C	0.05	yes	474,807	65,875	68,724	24,547	65,849 [40,919, 69,549]	152,470 [44,524, 343,303]
V1	C	0.10	no	475,656	131,750	131,189	49,094	123,668 [68,202, 134,805]	185,825 [69,247, 393,803]
V1	C	0.20	no	477,354	263,500	239,604	98,188	217,686 [118,451, 241,936]	250,445 [118,694, 494,803]
V1	C	0.30	no	479,052	395,250	327,165	147,281	289,965 [167,978, 329,125]	315,095 [168,140, 595,803]
V1	C	0.50	no	482,448	658,750	422,810	245,469	396,777 [267,595, 430,354]	443,559 [267,033, 797,803]
V1	SM	0.05	no	474,807	65,875	46,683	24,547	45,138 [33,623, 47,504]	152,470 [44,524, 343,303]
V1	SM	0.10	no	475,656	131,750	87,150	49,094	84,467 [61,980, 89,815]	185,825 [69,247, 393,803]
V1	SM	0.20	no	477,354	263,500	163,788	98,188	154,065 [114,947, 165,402]	250,445 [118,694, 494,803]
V1	SM	0.30	no	479,052	395,250	229,516	147,281	217,168 [166,038, 229,516]	315,095 [168,140, 595,803]
V1	SM	0.50	no	482,448	658,750	335,349	245,469	325,374 [265,452, 336,190]	443,559 [267,033, 797,803]
Study 4	C	0.05	no	474,807	65,875	26,200	24,547	25,830 [24,897, 26,388]	152,470 [44,524, 343,303]
Study 4	C	0.10	no	475,656	131,750	50,471	49,094	50,471 [49,681, 51,042]	185,825 [69,247, 393,803]
Study 4	C	0.20	no	477,354	263,500	98,739	98,188	99,481 [98,739, 100,268]	250,445 [118,694, 494,803]
Study 4	C	0.30	no	479,052	395,250	148,108	147,281	148,610 [147,856, 149,506]	315,095 [168,140, 595,803]
Study 4	C	0.50	no	482,448	658,750	247,636	245,469	247,134 [246,080, 247,950]	443,559 [267,033, 797,803]
Study 4	SM	0.05	no	474,807	65,875	37,496	24,547	38,177 [31,022, 39,798]	152,470 [44,524, 343,303]
Study 4	SM	0.10	no	475,656	131,750	76,316	49,094	72,556 [58,262, 76,859]	185,825 [69,247, 393,803]
Study 4	SM	0.20	no	477,354	263,500	143,596	98,188	136,811 [109,708, 143,596]	250,445 [118,694, 494,803]
Study 4	SM	0.30	no	479,052	395,250	203,020	147,281	195,482 [160,982, 203,473]	315,095 [168,140, 595,803]
Study 4	SM	0.50	no	482,448	658,750	310,533	245,469	301,615 [258,939, 310,820]	443,559 [267,033, 797,803]
V2	C	0.05	yes	474,807	65,875	66,963	24,547	65,237 [42,789, 67,024]	152,470 [44,524, 343,303]
V2	C	0.10	no	475,656	131,750	127,937	49,094	123,813 [68,893, 129,008]	185,825 [69,247, 393,803]
V2	C	0.20	no	477,354	263,500	242,328	98,188	219,503 [119,799, 243,870]	250,445 [118,694, 494,803]
V2	C	0.30	no	479,052	395,250	331,007	147,281	293,846 [169,200, 331,682]	315,095 [168,140, 595,803]
V2	C	0.50	no	482,448	658,750	439,196	245,469	405,608 [267,751, 446,159]	443,559 [267,033, 797,803]
V2	SM	0.05	yes	474,807	65,875	65,875	24,547	65,875 [44,524, 66,944]	152,470 [44,524, 343,303]
V2	SM	0.10	yes	475,656	131,750	131,750	49,094	131,750 [69,247, 133,833]	185,825 [69,247, 393,803]
V2	SM	0.20	yes	477,354	263,500	266,190	98,188	245,359 [118,694, 266,866]	250,445 [118,694, 494,803]
V2	SM	0.30	yes	479,052	395,250	397,833	147,281	315,037 [168,140, 400,867]	315,095 [168,140, 595,803]
V2	SM	0.50	no	482,448	658,750	481,253	245,469	445,695 [267,033, 481,253]	443,559 [267,033, 797,803]
V3	C	0.05	no	474,807	65,875	26,200	24,547	26,022 [24,868, 26,263]	152,470 [44,524, 343,303]
V3	C	0.10	no	475,656	131,750	50,471	49,094	50,648 [49,508, 51,409]	185,825 [69,247, 393,803]
V3	C	0.20	no	477,354	263,500	101,656	98,188	99,607 [98,796, 101,656]	250,445 [118,694, 494,803]
V3	C	0.30	no	479,052	395,250	148,387	147,281	148,644 [147,983, 150,085]	315,095 [168,140, 595,803]
V3	C	0.50	no	482,448	658,750	246,259	245,469	246,517 [246,259, 247,817]	443,559 [267,033, 797,803]
V3	SM	0.05	no	474,807	65,875	42,992	24,547	42,992 [32,521, 44,324]	152,470 [44,524, 343,303]
V3	SM	0.10	no	475,656	131,750	82,653	49,094	78,804 [59,790, 83,772]	185,825 [69,247, 393,803]
V3	SM	0.20	no	477,354	263,500	154,293	98,188	145,863 [113,580, 155,544]	250,445 [118,694, 494,803]
V3	SM	0.30	no	479,052	395,250	214,860	147,281	207,579 [162,810, 215,479]	315,095 [168,140, 595,803]
V3	SM	0.50	no	482,448	658,750	325,282	245,469	311,256 [261,523, 326,307]	443,559 [267,033, 797,803]
V4	C	0.05	no	474,807	65,875	63,780	24,547	61,660 [39,494, 65,801]	152,470 [44,524, 343,303]
V4	C	0.10	no	475,656	131,750	122,781	49,094	114,030 [67,499, 124,681]	185,825 [69,247, 393,803]
V4	C	0.20	no	477,354	263,500	222,714	98,188	202,658 [117,757, 222,714]	250,445 [118,694, 494,803]
V4	C	0.30	no	479,052	395,250	309,551	147,281	273,504 [167,492, 309,551]	315,095 [168,140, 595,803]
V4	C	0.50	no	482,448	658,750	411,770	245,469	383,578 [266,228, 413,623]	443,559 [267,033, 797,803]
V4	SM	0.05	no	474,807	65,875	24,547	24,547	24,663 [24,547, 25,026]	152,470 [44,524, 343,303]
V4	SM	0.10	no	475,656	131,750	49,094	49,094	49,162 [49,094, 49,511]	185,825 [69,247, 393,803]
V4	SM	0.20	no	477,354	263,500	98,188	98,188	98,188 [98,188, 98,567]	250,445 [118,694, 494,803]
V4	SM	0.30	no	479,052	395,250	147,281	147,281	147,281 [147,281, 147,722]	315,095 [168,140, 595,803]
V4	SM	0.50	no	482,448	658,750	245,469	245,469	245,469 [245,469, 246,074]	443,559 [267,033, 797,803]

Table 10: Sensitivity analysis: passive threshold with the measured cost of leaving a fault, and, over the plausible grid points, the median verification mass, the number of points with a significantly positive realised saving, and the number at which targeted verification is significantly cheaper than the best constant policy, under each threshold. Decides nothing. Designs as in Paper 1: Original design (V1), Study 4 (V1, seven channels), Study 5 (V2), Third design (V3), Fourth design (V4); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	q^*	m_c median		$\widehat{D}_c > 0$		Targeted < best constant	
					measured	primary	measured	primary	measured	primary
V1	C	0.05	yes		0.348	0.042	0.049	201	199	150
V1	C	0.10	no		0.348	0.143	0.160	306	305	286
V1	C	0.20	no		0.348	0.411	0.425	334	338	528
V1	C	0.30	no		0.348	0.572	0.575	398	366	528
V1	C	0.50	no		0.348	0.522	0.508	307	287	528
V1	SM	0.05	no		0.309	0.052	0.063	84	114	528
V1	SM	0.10	no		0.309	0.100	0.100	162	263	528
V1	SM	0.20	no		0.309	0.133	0.126	318	322	528
V1	SM	0.30	no		0.309	0.136	0.130	382	299	528
V1	SM	0.50	no		0.309	0.147	0.142	277	267	528
Study 4	C	0.05	no		0.348	0.003	0.002	71	116	528
Study 4	C	0.10	no		0.348	0.004	0.004	0	2	528
Study 4	C	0.20	no		0.348	0.003	0.003	0	0	528
Study 4	C	0.30	no		0.348	0.004	0.004	0	0	528
Study 4	C	0.50	no		0.348	0.003	0.003	0	0	528
Study 4	SM	0.05	no		0.309	0.035	0.043	33	63	528
Study 4	SM	0.10	no		0.309	0.066	0.071	228	241	528
Study 4	SM	0.20	no		0.309	0.108	0.110	266	158	528
Study 4	SM	0.30	no		0.309	0.124	0.122	277	307	528
Study 4	SM	0.50	no		0.309	0.121	0.115	282	273	528
V2	C	0.05	yes		0.364	0.015	0.015	383	382	109
V2	C	0.10	no		0.364	0.113	0.113	245	249	371
V2	C	0.20	no		0.364	0.425	0.426	369	368	528
V2	C	0.30	no		0.364	0.641	0.642	350	352	528
V2	C	0.50	no		0.364	0.578	0.578	331	329	528
V2	SM	0.05	yes		0.364	0.000	0.000	31	31	31
V2	SM	0.10	yes		0.364	0.010	0.010	123	123	123
V2	SM	0.20	yes		0.364	0.619	0.622	341	341	330
V2	SM	0.30	yes		0.364	0.990	0.990	443	448	425
V2	SM	0.50	no		0.364	0.879	0.877	359	358	367
V3	C	0.05	no		0.364	0.003	0.003	21	21	528
V3	C	0.10	no		0.364	0.007	0.007	0	0	528
V3	C	0.20	no		0.364	0.006	0.006	259	5	528
V3	C	0.30	no		0.364	0.006	0.006	0	0	528
V3	C	0.50	no		0.364	0.003	0.003	0	0	528
V3	SM	0.05	no		0.362	0.030	0.031	95	95	528
V3	SM	0.10	no		0.362	0.062	0.062	225	233	528
V3	SM	0.20	no		0.362	0.114	0.113	288	336	528
V3	SM	0.30	no		0.362	0.134	0.134	255	256	528
V3	SM	0.50	no		0.362	0.148	0.148	337	339	528
V4	C	0.05	no		0.362	0.058	0.058	211	212	314
V4	C	0.10	no		0.362	0.152	0.152	293	278	443
V4	C	0.20	no		0.362	0.370	0.373	347	326	528
V4	C	0.30	no		0.362	0.478	0.481	459	445	528
V4	C	0.50	no		0.362	0.429	0.423	383	376	528
V4	SM	0.05	no		0.344	0.000	0.000	0	0	528
V4	SM	0.10	no		0.344	0.000	0.000	0	0	528
V4	SM	0.20	no		0.344	0.000	0.000	0	0	528
V4	SM	0.30	no		0.344	0.000	0.000	0	0	528
V4	SM	0.50	no		0.344	0.000	0.000	0	0	528